BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054501
Article Type:	Original research
Date Submitted by the Author:	14-Jun-2021
Complete List of Authors:	Nnaji, Chukwudi; University of Cape Town Faculty of Health Sciences, Women's Health Research Unit, School of Public Health and Family Medicine; University of Cape Town Faculty of Health Sciences, Cancer Research Initiative Kuodi, Paul; Lira University, Department of Public Health, Faculty of Health Sciences Walter, Fiona; University of Cambridge, Dept of Public Health and Primary Care; Queen Mary University of London Barts and The London School of Medicine and Dentistry, Wolfson Institute of Population Health Moodley, Jennifer; University of Cape Town Faculty of Health Sciences, Women's Health Research Unit, School of Public Health and Family Medicine; University of Cape Town Faculty of Health Sciences, 2Cancer Research Initiative
Keywords:	ONCOLOGY, Breast tumours < ONCOLOGY, PUBLIC HEALTH, Organisation of health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review

Chukwudi A. Nnaji[1,2], Paul Kuodi[3], Fiona M. Walter[4,5], Jennifer Moodley[1,2,6]

Authors affiliations:

[1]Women's Health Research Unit, School of Public Health and Family Medicine, Faculty of Health Sciences, University of Cape Town; Anzio Road. Observatory, 7925, Cape, Town, South Africa

[2]Cancer Research Initiative, Faculty of Health Sciences, University of Cape Town; Anzio Road, Observatory, 7925, Cape, Town, South Africa

[3] Department of Public Health, Faculty of Health Sciences, Lira University, Lira, Uganda

[4] The Primary Care Unit, Department of Public Health and Primary Care, University of Cambridge, United Kingdom

[5] Wolfson Institute of Population Health, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, UK

[6] SAMRC Gynaecology Cancer Research Centre, University of Cape Town, Cape Town, South Africa

Correspondence to:

Dr Chukwudi A. Nnaji; nnjchu001@myuct.ac.za

For peer review only

Abstract

Objectives: To systematically synthesise available evidence on the nature and effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-come countries (LMICs).

Design: A systematic review of published evidence. The review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA).

Data sources: A comprehensive search of published literature was conducted. In addition, relevant grey literature sources and bibliographical references of included studies were searched for potentially eligible evidence.

Study selection: Studies published between January 2010 and November 2020 were eligible for inclusion; they had to report on interventions/strategies for the timely diagnosis of breast and/or cervical cancers in LMIC settings.

Data extraction and synthesis: Literature search, screening, study selection, data extraction and quality appraisal were conducted by two independent reviewers. Evidence was synthesised and reported using a global taxonomy framework for early cancer diagnosis.

Results: From the total of 10593 records identified, 21 studies conducted across 20 LMICs were included in this review. Most included studies (16/21) primarily focused on interventions addressing breast cancers; two focused on cervical cancer while the rest examined multiple cancer types. Reported interventions targeted health care workers (12); women and adolescent girls (7); and both women and health care workers (3). Eight studies reported on interventions addressing access delays; another eight focused on interventions addressing diagnostic delays; one reported on an intervention targeted at addressing both access and diagnostic delays, and five studies assessed interventions addressing access, diagnostic and treatment delays. While most interventions were demonstrated to be feasible and effective, many of the reported outcome measures are of limited clinical relevance to diagnostic timeliness.

Conclusions: Though limited, evidence suggests that interventions aimed at addressing barriers to timely diagnosis of breast and cervical cancer are feasible in resource-limited contexts. Future interventions need to address clinically relevant measures to better assess efficacy of interventions.

Keywords: Cancer, breast, cervical, diagnosis, timeliness, delay, LMICs.

PROSPERO registration number: CRD42020177232.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- This review was designed in accordance with standard systematic review protocol guidelines.
- Literature searching was comprehensive, covering both peer-reviewed and relevant grey literature.
- No language restrictions were applied in the search.
- While most interventions were demonstrated to be feasible and effective, reported outcome measures are of limited clinical relevance to diagnostic timeliness.
- The overall strength of the synthesised evidence and applicability of findings depend on the quality of included studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Breast and cervical cancers both represent a significant burden of disease globally, with a disproportionately high burden of morbidity and mortality in low- and middle-income countries (LMICs). [1-3] The majority (53%) of new breast cancer cases occur in women living in LMICs, accounting for nearly a third of all cancers in women, and with an age-standardised incidence rate (ASIR) of 31 per 100,000 women in these settings.[4,5] Cervical cancer constitutes 16% of women's cancers in LMICs, with an ASIR of 16 per 100,000 women.[2] The increasing burdens of both cancers has been attributed to factors such as population growth, ageing societies and exposures to oncogenic risk factors.[1,6] While the incidence of breast and cervical cancers increases in LMICs, many cases continue to go unreported and undiagnosed; the majority of these women present with late-stage disease, the treatment of which is often costlier and less effective.[7,8] These challenges have been aggravated by the COVID-19 pandemic, which immensely disrupted cancer services globally, particularly in low-resource settings.[9]

The timeliness of diagnosis is an important determining factor of breast and cervical cancer survival.[10-13] Yet, in many LMICs, breast and cervical screening and early diagnosis programmes are often opportunistic and not well organised.[7,14-16] One of the greatest barriers to timely cancer diagnosis in LMICs is lack of geographic access to health facilities offering screening and diagnostic services, as well as the lack of well-trained care providers.[6,17] In addition to these health system factors are the underlying psychosocial and cultural barriers to timely cancer help-seeking and diagnosis.[6-8,18] These include perceptions such as beliefs that breast cancers are contagious or punitive consequences of sins.[15,19]

To conceptualise and understand the pathway to cancer diagnosis, distinct events and processes from symptom awareness to diagnosis and treatment have been described. These include patients' detection of bodily changes and awareness of symptoms; decision to seek help from a healthcare provider; access to clinical evaluation, diagnosis, and staging; access to treatment; and follow-up.[17,20-22] Barriers encountered by patients in any of these phases may pose an impediment to timely cancer diagnosis and delay patients' access to care.[10,23] In an effort to address these barriers and provide global standards for early cancer diagnosis, the World Health Organization (WHO) published the *WHO Guide to Cancer Early Diagnosis* in 2017.[24] It provides an operational framework for cancer control programmes around the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

world to systematically address barriers that may impede timely cancer diagnosis, treatment and care. Furthermore, it specifies three essential steps of early cancer diagnosis: Step 1: awareness of cancer symptoms and accessing care; Step 2: clinical evaluation, diagnosis and staging; and Step 3: access to treatment, including pain relief. These steps were emphasised to be essential for setting cancer control priorities in the 2020 WHO Report on Cancer.[25]

Interventions aimed at effectively addressing barriers to timely cancer diagnosis have been studied in various settings within the different phases of the cancer care continuum.[6,11] It is evident that such interventions can improve the timeliness of cancer diagnosis, enable early initiation of treatment, reduce morbidity and increase the chances of survival.[20,26-30] However, much of this evidence is from high-income countries (HICs). Given the substantial differences between HICs and LMICs regarding health resources, environment, infrastructure, technology, and medical personnel, improving time to diagnosis for breast and cervical cancer in LMIC settings may require different approaches.[3]

While interventions to improve the timely diagnosis of these cancers are increasingly being implemented in LMICs, there is uncertainty about their role and effectiveness. We have identified two previous reviews on this topic within the LMIC context.[6,11] A scoping review by Dalton and colleagues synthesised the evidence on patient navigation strategies for cancer care in LMICs, but focused broadly on the entire cancer detection, treatment and care continuum.[6] A previous systematic review by Qu and colleagues assessed interventions specifically aimed at addressing barriers to timely cancer diagnosis in LMICs.[11] Neither specifically focused on breast and/or cervical cancer.

Therefore, the purpose of this review was to systematically synthesise the evidence on the nature and effectiveness of interventions aimed at improving timely diagnosis of symptomatic breast and cervical cancers in LMICs, by targeting the 3 key steps outlined by the WHO framework.[24] We sought to provide a user-friendly evidence summary for health policy-makers, cancer programme managers, oncologists and cancer programme implementers, for informing further efforts to assist women in LMICs in overcoming socioeconomic, cultural, and structural barriers to timely diagnosis and appropriate care, while identifying gaps for future research.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

METHODS

Study design

This systematic review of evidence was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[31] The review has been registered on the International Prospective Register of Systematic Reviews (PROSPERO, registration number: CRD42020177232).[32] The protocol, which was designed in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) guidelines[33] has been published elsewhere.[34]

Data sources

The following databases were searched: MEDLINE (via PubMed), Cochrane Library (including the Cochrane Central Register of Controlled Trials (CENTRAL) and the Database of Abstracts of Reviews of Effects (DARE)), Scopus, CINAHL, Web of Science and the International Clinical Trials Registry Platform (ICTRP). Relevant grey literature sources; including the publication database of the WHO's International Agency for Research on Cancer (IARC), the Cancer Atlas of the Union for International Cancer Control (UICC) and the Global Cancer Project Map, were also searched for potentially eligible publications. The search strategy (see Appendix 1) was developed with guidance from a health sciences subject librarian. More details of the search strategies used are described in the published review protocol.[34] Only articles published over the last 10 years (from 1 January 2010 to the date of search) were considered eligible, with no language restriction. In addition, the bibliographical references of included studies were also searched. Search was conducted from 22 June 2020 through 30 November 2020.

Eligibility criteria

The study inclusion criteria were guided by the research question: *'What interventions targeting the key steps outlined by the WHO operational framework for early diagnosis have been used for improving timely diagnosis of symptomatic breast and cervical cancers in LMICs, and how effective are they?'*. Eligible studies had to report on interventions or strategies for timely breast and cervical cancer diagnosis, targeting women, the general public or healthcare workers in LMICs. The definition of LMICs was based on the World Bank's current classification using per capita gross national income.[35]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Such interventions or strategies had to be aimed at influencing the timeliness of symptomatic breast or cervical cancer diagnosis, whether as a single focus intervention or as multi-focus intervention targeting more than one cancer type. Studies with interventions focused primarily on screening of asymptomatic individuals were excluded. More detailed information about the eligibility criteria can be found in the published review protocol.[34]

Study selection

The review consisted of two levels of screening: a title and abstract screening to identify potentially eligible publications, and review of full-texts to select those to be included in the review based on pre-defined inclusion/exclusion criteria. For the first level of screening, two reviewers (CAN and PK) independently screened the titles and abstracts of all retrieved records from the search output. Articles that were considered relevant by either or both of the reviewers were included in the full-text review. In the second step, the two reviewers (CAN and PK) independently assess the full-texts to determine if they met the inclusion/exclusion criteria. Any discordance in eligibility assessment was resolved through consensus between the two researchers, with recourse to a third arbitrator (JM) as necessary.

Data extraction

Two reviewers (CAN and PK) independently extracted all relevant data from the included articles using a standardised data extraction tool, adapted from the framework proposed by Carlos and colleagues.[36] Extracted data fell under four domains: (1) study identification details (article title; journal title; authors; country of the study; language; publication year; host institution of the study); (2) methodological characteristics (study design; study objective or research question or hypothesis; sample characteristics (e.g. sample size; sex; age, ethnicity; groups and controls; follow-up duration; validation of measures; statistical analyses); (3) main findings, and (4) conclusions. Study eligibility was re-verified at the start of data extraction. Discrepancies in extraction were resolved through consensus. CAN combined the two spreadsheets of extracted data for analysis, and data were double-checked by PK for completeness and accuracy.

Quality assessment

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Two reviewers (CAN and PK) independently assessed each included study for risk of bias, using the Cochrane risk of bias tool[37] for randomised trials, and the Risk of Bias In Non-randomised Studies of Interventions (ROBINS-I) tool[38] for non-randomised and observational studies. Discrepancies in study quality assessments were resolved by discussion and consensus. Further details of the quality appraisal process are discussed in the published protocol of this review.[34]

Data analysis

A PRISMA flow diagram was used to illustrate the literature search results and study selection process. Outcomes were categorised according to the essential steps of early cancer diagnosis as specified in the WHO Guide to Cancer Early Diagnosis.[24] Plans to conduct a meta-analysis were not feasible as studies were not quantitatively comparable in terms of interventions and reported outcomes. Instead, a thematic narrative synthesis approach was used to aggregate and analyse relevant findings. Plans to assess the certainty of reviewed evidence using the Grades of Recommendation Assessment, Development and Evaluation (GRADE) approach[39] were not feasible for similar reasons. A table summarising findings from each included study was presented. JM and FMW reviewed the analysed data for accuracy and consistency with protocol.

RESULTS

Search results

A total of 10593 records were identified from literature database searches, from which 1244 duplicates were discarded. The remainder, 9349 unique records, had their titles and abstracts screened, from which 9264 clearly ineligible publications were excluded. The full texts of the remaining 85 potentially eligible studies were reviewed against predefined inclusion and exclusion criteria; 21 were included in the review and 64 were excluded for various reasons. **Figure 1** presents the PRISMA flow chart of the study selection process and reasons for exclusion.

Characteristics of included studies

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1 summarises the characteristics of included studies, published between 2015 and 2020. Studies were conducted in 20 LMICs across regions of Africa (8), Europe (4), Latin America (2), Middle-East Asia (1) and South-East Asia (5). Twenty were single-country studies, in which 13 countries were represented: Bangladesh (2)[8,40], Botswana (2)[41,42], Colombia (1)[43], India (1)[44], Indonesia (1)[45], Iran (1)[46], Malaysia (2)[47,48], Nigeria (1)[49], Pakistan (1)[50], Rwanda (1)[51], South Africa (3)[52-54], Tanzania (1)[55] and Zambia (3)[56-58]. One study had a multi-country focus, with participants recruited from nine countries: Bosnia-Herzegovina, Costa Rica, Egypt, India, North Macedonia, Pakistan, Slovenia, Turkey and Uganda.[59] **Figure 2** illustrates the geographical distribution of the studies.

In terms of study design, six studies were randomised controlled trials,[8,40,43,45,46,51] six were quasi-experimental (before and after) studies,[42,44,48,49,52,54] one was a longitudinal (cohort-type) observational study,[55] and four were retrospective studies,[41,47,50,53] while the remaining four were cross-sectional studies.[56-59]

The majority (16/21) of included studies primarily focused on interventions addressing breast cancers; two focused on cervical cancer, whereas the rest examined multiple cancer types (including breast and/or cervical cancers). Of the 21 studies, 12 targeted health care workers (including physicians, nurses, pathologists, non-physician health providers and community health workers); seven targeted women and adolescent girls; and three targeted both women and health care workers. There was an even representation across urban and rural implementation contexts, with reported interventions implemented in 11 urban, 9 rural and one mixed (urban and rural) communities.

When categorised by the three steps of the WHO Guide to Cancer Early Diagnosis[24], eight studies reported on interventions aiming to address barriers in Step 1 (promoting awareness and addressing delays in access to care);[8,40,44-47,49,54] another eight focused on interventions addressing Step 2 (addressing diagnostic delay by optimising clinical evaluation, diagnosis and staging);[42,43,50-52,55,59,60] one study reported on an intervention targeted at both Steps 1 and 2;[53] while five studies assessed interventions targeted at all three steps (addressing access, diagnostic and treatment delays).[48,56-58] **Table 1** describes the categorisation of the interventions by the three steps of the WHO Guide to Cancer Early Diagnosis.[24]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1: Characteristics of included studies and categorisation of interventions according to the essential steps of the WHO Guide to Cancer Early Diagnosis

Step	Study ID	Study Design	Country context	Target population	Cancer type	Setting	Intervention	Duration
Step 1	Gadgil 2017	Quasi-experimental (before and after)	India	Women in an occupational setting.	Breast	Urban	A programme that aimed to increase awareness on breast cancer and access to early detection by emailing awareness brochures annually to a cohort of women in an occupational setting.	3 years
	Ginsburg 2014	RCT	Bangladesh	CHWs	Breast	Rural	An mHealth model to increase clinic attendance for breast symptoms in rural Bangladesh: involving smart phone/applications and CHW training as part of a patient navigation programme to address potential barriers to seeking care.	4 months
	Ifediora 2019	Quasi-experimental (before and after)	Nigeria	High school girls	Cervical	Rural	School-based cervical cancer prevention and early diagnosis education programme - targeting cervical cancer campaigns on teenage high schoolers in resource-limited economies.	6 months
	Chowdhury 2015	RCT	Bangladesh	CHWs	Breast	Rural	Population-based case finding for breast cancer by CHWs in Rural Bangladesh using different data collection methods and approaches.	3 months
	Maimela 2019	Quasi-experimental (before and after)	South Africa	Women	Cervical	Urban	Decentralising colposcopy services from tertiary-level to primary-level care facility to aid early definitive diagnosis of cervical cancer.	6 years
	Setyowibowo 2020	Cluster randomized crossover design	Indonesia	Women	Breast	Both rural and urban	A self-help intervention for reducing time to diagnosis in women with breast cancer symptoms It consisted of health education and psycho-education using a narrative strategy, which involved the use of testimonials and story-telling.	3 months
	Teh 2015		Malaysia	Women	Breast	Urban	Routine (opportunistic) screening mammogram vs targeted (high risk) screening mammogram vs diagnostic mammogram.	NA

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

		Retrospective study						
	Termeh Zonouzy 2019	RCT	Iran	Women	Breast	Urban	An educational intervention based on fear appeals using the extended parallel process model (EPPM) to improve attitudes, intention, and early breast cancer diagnosis.	3 months
	Ali 2018	Retrospective study	Pakistan	Physicians and pathologists	Multiple, including breast cancer	Urban	Adequate clinical information (CI) for improving turnaround time (TAT) in surgical histopathological diagnosis.	4 months
	Aribal 2019	Observational clinical study	Bosnia-Herzegovina, Costa Rica, Egypt, India, North Macedonia, Pakistan, Slovenia, Turkey, Uganda	Oncologists, radiologists, medical physicists and radiographers working on oncology centres	Breast	Urban	A programme aiming to improve early detection of breast cancer by strengthening both the clinical component and quality aspects of imaging practice of oncology centres.	2 years
	Dickerson 2017	Quasi-experimental (before and after)	South Africa	Non-physician providers	Breast	Rural	Breast ultrasound training for non-physician providers with the goal of early diagnosis of breast cancer and down-staging in a rural setting.	3 weeks
	Martei 2018	Retrospective study	Botswana	Pathologists	Breast	Urban	A pathology scale-up programme to optimise breast cancer pathological diagnostic TAT in Botswana.	4 years
	Murillo 2016	RCT	Colombia	Physicians	Breast	Urban	Opportunistic clinic-based breast cancer screening programs to increase breast cancer screening and down-staging in Colombian women. Physicians received a 2-day training course on breast-cancer epidemiology, clinical signs and symptoms, as well as the principles of mammography and BIRADS (Breast Imaging Reporting and Data System) grading.	2 years
Step 2	Ngoma 2014	Prospective cohort study	Tanzania	CHW/village navigators	Multiple, including breast and cervical	Rural	Conventional dispensary self-referral vs. proactive home visits for down-staging cancer in a rural community in Tanzania.	3 years

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Pace 2019	RCT	Rwanda	Health workers	Breast	Rural	Health worker training on diagnosis and staging to facilitate breast cancer early diagnosis. It consisted of instructions for CHWs in symptoms of breast cancer and messaging that community members should come to their local health centres for any breast symptoms.	2 years
	Tapela 2018	Quasi-experimental (before and after)	Botswana	Primary care providers	Multiple, including breast and cervical	Urban	A district-level capacity building training programme targeting primary care providers, aiming to enhance timely diagnosis of cancers.	5 days
Step 1+2	Hadley 2018	Retrospective/post-implementation study	South Africa	Nurses, nursing assistants, and counsellor.	Breast	Rural	An improved and computerized breast cancer patient-tracking system and data base as part of an ultrasound-based breast cancer early detection program in a rural community in South Africa.	NR
	Songiso 2020	Cross-sectional study	Zambia	Women and HCWs	Breast	Urban	Establishment of a district-level breast care specialty clinic for women with breast symptoms. The clinic offers same-day breast self-awareness education, clinical breast examination, breast ultrasound, ultrasound-guided breast biopsy, surgery, referral for treatment and follow-up.	NA
	Pinder 2018	Cross-sectional study	Zambia	Women and HCWs in a rural setting	Breast	Rural	An algorithm that compresses the multi-step breast cancer symptom awareness, diagnostic and treatment pathway into a single visit.	NA
Step 1+2+3	Pinder 2017	Cross-sectional study	Zambia	Women and HCWs in a rural setting	Breast	Rural	An initiative that leverages an existing cervical cancer prevention service platform to build facility-level capacity for breast cancer care by raising awareness among women, creating a resource-appropriate breast cancer care training curricula for mid- and high-level providers, and facility-level capacity building for early detection and treatment capacity.	NA

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Yeoh 2018	Quasi-experimental (before and after)	Malaysia	Women	Breast	urban	A patient navigation programme to Improve timely breast cancer diagnosis and care.	1 year and one month

Abbreviations: CHW: Community Health Workers; HCW: Healthcare Workers; NA: Not applicable; NR: Not reported; RCT: Randomised Controlled Trial; TAT: Turn-around Time

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Only two of the included studies were determined to be of high quality (low risk of bias)[43,49], seven were adjudged to be of moderate quality (moderate risk of bias)[8,40,45,46,51,52,57] while the majority were found to be of low quality (high risk of bias).[41,42,44,47,48,50,53-56,58,59] **Figures 3a and 3b** illustrate the risk of bias assessment outcomes of the included studies.

Interventions targeting Step 1 of the WHO Guide to Cancer Early Diagnosis- promoting awareness and addressing care access delays

Interventions seeking to reduce access delays by promoting awareness, help-seeking and linkage to care were predominantly targeted at women (5),[44-47,54] with a few targeted at adolescent girls (1)[49] and community/lay health workers (2).[8,40]

A cluster randomised trial assessed a self-help intervention for reducing time to diagnosis in Indonesian women with breast cancer symptoms. The intervention consisted of health education and psycho-education using a narrative strategy, which involved the use of testimonials and story-telling.[45] It reduced the time between the first medical consultation and definitive diagnosis among women with breast symptoms by 13.3 days (25.90 in the intervention group vs 39.29 in the control group, P = 0.02). Another trial by Termeh-Zonouzy evaluated the effectiveness of an educational intervention based on fear appeals using the extended parallel process model (EPPM) to improve attitudes, intention, and timely breast cancer diagnosis in Iran. It found a significant improvement in attitude and intention (*P*=0.01 and *P*=0.001, respectively), but no significant improvement was observed for early breast cancer diagnosis (*P*=0.78).[46]

A trial by Gadgil and colleagues investigated a programme that aimed to increase awareness on breast cancer and access to timely detection by emailing awareness brochures annually to a cohort of women in an urban occupational setting in India.[44] They found an increase in the proportion of women with early breast tumours and a concomitant increase in the proportion of women receiving breast conserving surgery increased post-intervention when compared with the pre-intervention period increased from (74% to 81% and 39% to 51% respectively). However, there were no outcomes directly related to the timeliness of the diagnoses.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

A quasi-experimental (before and after) study assessed the effectiveness of decentralising colposcopy services from tertiary-level to primary-level care facility in an urban community in South Africa to aid early definitive diagnosis of cervical cancer.[54] The number of women who had a colposcopy at the primary healthcare facility rose threefold post-decentralisation (from 114 to 350), with an increased proportion of women who had a colposcopy within 3 months of a Pap smear (from 11.8% to 15.4%) and reduced workload at the tertiary facility. A retrospective study compared routine (opportunistic) screening mammogram with targeted (high risk) screening mammogram and diagnostic mammogram among women in Malaysia.[47] It observed cancer detection rates of 0.5 %, 1.25 % and 26 % by opportunistic screening, targeted screening, and diagnostic mammograms, respectively. The proportion of ductal carcinoma in situ was 23.1 % in the two screening groups compared to only 2.5 % in the diagnostic group. However, none of the outcomes were directly related to the timeliness of diagnosis.

Of the interventions targeting lay/community health workers, a trial by Ginsburg et al. assessed the role an mHealth model that used smart phone applications and CHW training as part of a patient navigation programme to address potential barriers to seeking care in a rural community in Bangladesh.[8] The intervention was found to have increased clinic attendance for breast symptoms, improved data quality, higher identification of women with abnormal breast exams and better adherence to diagnostic follow-up. Similarly, another trial of a CHW-driven population-based case finding programme for breast cancer in a rural setting in Bangladesh found the use of mobile technology as feasible and efficient in active breast cancer case finding in rural settings. Manual data collection and reporting had missing data in 80% of cases, and took an average of 5 minutes longer to collect data, versus no missing data when mobile phones were used for data collection and reporting.[40]

A school-based cervical cancer prevention and early diagnosis education programme targeted at rural teenage high school students Nigeria.[49] Notably, no significant improvements in knowledge parameters (Pap smears, risk factors and early symptoms). Nevertheless, participants who engaged more with the intervention showed significant improvements in knowledge across most parameters of knowledge.

Interventions targeting Step 2 of the WHO Guide to Cancer Early Diagnosis-addressing diagnostic delays by optimising clinical evaluation, diagnosis and staging

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

A trial compared diagnostic outcomes between a physician-targeted opportunistic clinic-based breast cancer screening programme and usual care in an urban setting in Colombia. Physicians in the intervention clinics received a 2-day training course on breast-cancer epidemiology, clinical signs and symptoms, as well as the principles of mammography and BIRADS (Breast Imaging Reporting and Data System) grading. They were instructed to perform CBE on all eligible women, record results, and refer women with suspicious findings for further diagnostic procedures. It demonstrated a higher proportion of early breast cancer diagnosed with the intervention; 68.2% of the women with breast cancer in the intervention group had breast conservation surgery , compared with 50.0% in the control group.[43]

Another trial by Pace et al investigated the impact of a health care worker (HCW) training programme on diagnosis and staging to facilitate breast cancer early diagnosis in a rural district of Rwanda.[51] The training consisted of instructions for CHWs in symptoms of breast cancer and messaging that community members should come to their local health centre for any breast symptoms. The intervention led to an increase in diagnostic health facility visits (an increase of 4.7 – 7.9 visits/month in intervention facilities compared with control facilities, P <0.05) and more breast biopsies (36.6 vs 8.9/100,000 person-years, P <0.001) and higher incidence of early-stage breast cancer (3.3 per 100,000 vs 0.7 per 100,000 in control areas, P <.05).

A quasi-experimental study of a breast ultrasound training programme for non-physician providers with the goal of early diagnosis of breast cancer and down-staging in a rural setting in South Africa demonstrated improved diagnostic competencies (with pre- to post-test averages improvement of 68% in total across four competencies (foundational knowledge, descriptive categories, benign vs malignant, and lesion identification).[52] No outcome directly related to diagnostic timeliness was reported, however. Another quasi experimental trial studied district-level capacity building training programme targeting primary care providers in Botswana, aiming to enhance timely diagnosis of cancers. It found an overall performance increase (measured by the immediate impact of training on the acquisition of knowledge, attitude and skills in eight sub-domains: pathophysiology, epidemiology, social context, symptoms, evaluation, treatment, documentation and follow-up) of 16.8% after participation across all subdomains.[42]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

A prospective cohort study by Ngoma et al. assessed the role of village navigators for case-finding and showed significant down-staging of cancers in a Tanzanian village across 3 successive years ($p < 0.001$).[55] An observational clinical study examined the effectiveness of a programme aiming to improve early detection of breast cancer by strengthening both the clinical and quality aspects of the breast imaging practice of oncology centres.[59] The interventions included a shift to image guided core biopsies, implementing multidisciplinary breast meetings, implementing quality assurance procedures, transitioning to digital technologies, facilitating training activities for relevant professionals and improving data collection processes. It was observed that the programme enhanced facility-level breast cancer early diagnostic capacity and increased number of mammography (in 7 out of 10 participating centres) and timely biopsies performed (in 8 out of 10 centres).

A retrospective study assessed the relationship between adequate clinical information (CI) and breast cancer histopathological diagnostic turnaround time (TAT) in Pakistan.[50] It found that breast tissue specimens with deficient clinical information were associated with longer TAT (>80% of the specimens took had TAT of longer than three days vs <50% of specimens with sufficient clinical information). Another retrospective study that assessed a pathology scale-up programme to optimise breast cancer pathological diagnostic TAT in Botswana found that the programme decreased median TAT for biopsy and immunohistochemistry specimens (from 21.5 days to 8 days, before and after the initiation of the programme respectively). However, there was no significant decline in median TAT for surgical specimens.[41]

Intervention targeted at both Steps 1 and 2 of the WHO Guide to Cancer Early Diagnosis-addressing access and diagnostic delays

Hadley and colleagues conducted a retrospective (post-implementation) study to determine the impact of an improved and computerised breast cancer patient-tracking system and database, as part of a nurse-driven and ultrasound-based breast cancer early detection programme in a rural community in South Africa.[53] Most health workers' involved agreed that the intervention can improve early breast cancer detection, patient follow-up, clinical decision making, communication among clinicians, although a majority (71%) of them

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

reported that the system increased visit time, due to factors such as unreliable internet and inadequate computer workstations.

Interventions targeted at all three steps of the WHO Guide to Cancer Early Diagnosis - addressing access, diagnostic and treatment delays

There were interventions targeting all three steps of the WHO Guide. A patient navigation programme to improve timely breast cancer diagnosis, treatment and follow-up for women in Malaysia was found to have increased the proportion of women receiving timely mammography (from 74.4% to 96.4%), biopsy (from 76.1% to 92.5%), and timely communication of diagnostic results (from 58.5% to 80.0%).[48] Other interventions included the establishment of a district-level breast care specialty clinic that offers same-day breast self-awareness education, clinical breast examination, breast ultrasound, ultrasound-guided breast biopsy, surgery, referral for treatment and follow-up for women with breast symptoms in Zambia[58]; an algorithm that compresses the multi-step breast cancer symptom awareness, diagnostic and treatment pathway into a single clinic visit[56]; and an initiative that leverages an existing cervical cancer prevention service platform to build facility-level capacity for breast cancer care by raising awareness among women, creating a resource-appropriate training curricula for mid- and high-level providers for early detection and treatment.[57] Due to the cross-sectional nature of the studies reporting these interventions, they could only demonstrate the feasibility of the interventions within resource-limited contexts, and not their effectiveness.

DISCUSSION

This systematic review highlights the diverse nature and effectiveness of interventions for the early diagnosis of symptomatic breast and cervical cancers, from 21 studies set across 20 LMICs. In spite of the limited number and methodological limitations of the studies included, this review has identified substantial evidence that interventions and strategies targeting the key steps of the WHO operational framework for early diagnosis for improving timely detection of breast and cervical cancers are feasible and can yield desirable outcomes. It is noteworthy that, while most of interventions were reported to be feasible and effective for improving diagnostic processes, many of their reported outcome measures are not directly

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

related to diagnostic timeliness. Although a variety of interventions have been studied, the majority of them were targeted at improving the diagnostic capacity of health care workers (including physicians, nurses, pathologists, non-physician health providers and community health workers) through training, knowledge-acquisition, upskilling and better equipment.

Training interventions to improve clinical knowledge and skills of health care workers were shown to improve diagnostic turnaround time[41] and increase detection rates[43,51] and downstaging.[52] The use of community-based patient navigation programmes driven by trained community- or lay-health workers for breast examination, specimen collection, pathological diagnosis and follow-up demonstrated appreciable improvements in measured outcomes.[8,40] While the reported improvements in early diagnosis knowledge, attitude and skills are generally plausible, the variations in the quality of the studies and their outcome measurement and reporting complicate the aggregation and interpretation of these findings, while limiting the generalisability of the evidence. In addition, only a few of the studies measured intervention outcomes beyond a duration of one year, hence limiting the determination of the long-term impact of the studied interventions. Findings are consistent with those of previous reviews.[6,11]

Similarly, the use of health education and awareness promotion campaigns targeted at schoolgirls and women demonstrated considerable effectiveness in improving knowledge, attitude and help-seeking behaviour. These include culturally-sensitive and context-appropriate strategies such a self-help intervention for reducing time to diagnosis in women with breast cancer symptoms, consisting of health education and psycho-education through narratives, testimonials and story-telling.[45] Nevertheless, it is noteworthy that a school-based cervical cancer prevention and early diagnosis education programme targeting adolescent high schoolers yielded no significant improvements in the participants' overall knowledge.[49]

Interventions addressing structural barriers to access appropriate healthcare, such as proactive home-visits, patient-navigation programmes and decentralisation of diagnostic services from tertiary-level to primary-level care facilities, have shown promising results in reducing diagnostic turn-around time and cancer down-staging.[41,43,54,55] While it is difficult to draw conclusions from these studies owing to their methodological limitations, potential publication bias and heterogeneity, these findings reflect the important role of interventions targeting structural impediments to early cancer diagnosis in LMICs.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The use of varied outcome measures in assessing the effectiveness of interventions across studies complicated the interpretation, aggregation and comparability of the findings. Many outcome measures reported have very limited clinical relevance to the timeliness of breast or cervical cancer diagnosis. Hence, it is imperative for future studies to measure outcomes that are clinically relevant, such as reduced help-seeking intervals, improved and timely access to diagnostic services and down-staging of disease at diagnosis. However, these will require resources for more rigorous study designs, longer patient follow-up and adequate availability of diagnostic services.

Although generally reported to be feasible and beneficial, interventions for facilitating the early detection of breast or cervical cancer were not without their challenges. This review demonstrates that innovative interventions targeted at barriers to early cancer diagnosis can themselves be constrained by impediments such as non-engagement of the target population[49], as well as structural constraints such as unreliable internet service and inadequate computer workstations, which may lead to increased patient visit or consultation time.[53]

While the COVID-19 pandemic has had a considerable impact on cancer diagnosis and care services globally, the findings of this review have important implications for mitigating the pandemic's further disruption of cancer early diagnosis services in LMICs. The implications of these disruptions are grim in resource-limited settings, where cancer programmes are typically under-resourced.[9] As the pandemic evolves and movement restrictions are eased, proactive measures and interventions are needed for the timely and safe restoration of cancer screening and early diagnosis programmes.

The limited number of studies and reported interventions reported points to the dearth of available evidence on the nature and effectiveness of interventions for addressing barriers to early diagnosis of breast and cervical cancer in LMICs. Therefore, our review highlights the need for research across intervention types and study populations to identify effective interventions to overcome barriers to cancer early diagnosis in LMICs, particularly studies that evaluate time to diagnosis. Our findings also underscore the need for studies to investigate long-term post-intervention outcomes. In addition, the finding of heterogeneous measurement of outcomes necessitate the study of interventions using standardised methods, outcome measures, and control of potential biases.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Overall, given the rising burden of breast and cervical cancer in LMICs, the role of strategies and interventions for improving early diagnosis is relatively underexplored. While more interventions are needed, those that target both the general public and health care workers and the whole continuum of care may be more cost-effective, particularly for resource-limited settings within LMICs. Given the variability in cancer burden, health system capacity and contextual norms, it is also imperative for interventions to be contextually appropriate and sustainable within the broader health system. Implementation research has been proposed as an approach to fostering the adoption and expansion of evidence-based cancer control strategies.[61] Hence, it is also imperative to embed implemention research activities in timely diagnosis interventions to evaluate their practicality and effectiveness within local contexts. These context-specific studies are important to inform health policy making and cancer programmes as to which early detection approaches might be more suitable within individual LMIC contexts.

Limitations

Our review is not without its limitations. First, most of the studies assessed intervention outcomes using observational, non-experimental or quasi-experimental study designs, making them prone to selection bias, uncontrolled confounding, and limited external validity. Secondly, as with other reviews of the published literature, our study may be susceptible to publication bias in the reporting of study findings, in which positive findings have a higher likelihood of being reported. This may have manifested in our finding of very limited evidence of non-effectiveness of interventions. Thirdly, the short duration of the majority of the interventions undermines poses additional limitations to the applicability of the evidence. Furthermore, although our search strategy was comprehensive enough, it is possible that we missed some relevant studies. Lastly, our study excluded interventions that focused primarily on screening of asymptomatic individuals, due to the possible lead time and length bias associated with cancer screening programmes and intervention. This may have limited our ability to study the possible effectiveness of screening interventions on early breast and cervical cancer diagnosis in LMICs.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

CONCLUSION

Interventions aimed at improving timely diagnosis and cancer outcomes are feasible and can be effective for improving timely diagnosis of breast and cervical cancers in resource-limited LMIC contexts. Our review highlights the pressing need for future studies to address clinically relevant standardised measure of time to diagnosis to better assess the effectiveness of interventions to overcome barriers to cancer early diagnosis. It is imperative that early diagnosis interventions to adopt contextually appropriate and cost-effective approaches that are well embedded within the broader health system.

Patient and public involvement: This is a review of publicly available literature. Patients and the public were not directly involved in the design, conduct, reporting or dissemination this study.

Authors' contributions: Conceptualisation: JM. Protocol development : CAN, PK, FMW and JM. Duplicate screening, study selection, quality assessment and data collection: CAN and PK. Data analysis and interpretation: CAN, PK, FMW and JM. First draft preparation: CAN. Provision of critical insights and refinement of the manuscript: FMW and JM. All authors have read and approved the final version of the manuscript.

Funding: This research was jointly supported by the Cancer Association of South Africa (CANSA), the University of Cape Town and, the SA Medical Research Council with funds received from the SA National Department of Health, GlaxoSmithKline Africa Non-Communicable Disease Open Lab (via a supporting grant Project Number: 023), the UK Medical Research Council (via the Newton Fund). Authors retained control of the final content of the publication. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Acknowledgments: The authors wish to thank Namhla Madini of the Faculty of Health Sciences Library, University of Cape Town, for the support and guidance in developing the literature search strategy. FMW acknowledges the support of the CanTest Collaborative (funded by Cancer Research UK C8640/A23385) as its Director.

Patient consent for publication: Not required.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Ethics approval: This is a systematic review of publicly available literature, with no primary data collection. Hence, it did not require ethical approval.

Competing interests: None declared.

Provenance and peer review: Not commissioned; externally peer reviewed.

References

1. Global Burden of Disease Cancer Collaborators. Global, Regional, and National Cancer Incidence, Mortality, Years of Life Lost, Years Lived With Disability, and Disability-Adjusted Life-Years for 29 Cancer Groups, 1990 to 2017: A Systematic Analysis for the Global Burden of Disease Study. JAMA oncology 2019; 5:1749-68.
2. de Sanjose S, Tsu VD: Prevention of cervical and breast cancer mortality in low- and middle-income countries: a window of opportunity. International journal of women's health 2019; 11:381-6.
3. Demment MM, Peters K, Dykens JA, et al.: Developing the Evidence Base to Inform Best Practice: A Scoping Study of Breast and Cervical Cancer Reviews in Low- and Middle-Income Countries. PloS one 2015; 10:e0134618.
4. Bray F, Ferlay J, Soerjomataram I, et al.: Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: a cancer journal for clinicians 2018; 68:394-424.
5. GLOBOCAN. Cancer Today 2018. http://gco.iarc.fr/today/home. Accessed 26 March 2020.
6. Dalton M, Holzman E, Erwin E, et al.: Patient navigation services for cancer care in low-and middle-income countries: A scoping review. PloS one 2019; 14:e0223537-e.
7. Tegegne TK, Chojenta C, Loxton D, et al.: The impact of geographic access on institutional delivery care use in low and middle-income countries: Systematic review and meta-analysis. PloS one 2018; 13:e0203130.
8. Ginsburg OM, Chowdhury M, Wu W, et al.: An mHealth Model to Increase Clinic Attendance for Breast Symptoms in Rural Bangladesh: Can Bridging the Digital Divide Help Close the Cancer Divide? The Oncologist 2014; 19:177-85.
9. Nnaji CA, Moodley J: Impact of the COVID-19 pandemic on cancer diagnosis, treatment and research in African health systems: a review of current evidence and contextual perspectives. Ecancermedicalscience 2021; 15:1170.
10. Brand NR, Qu LG, Chao A, et al.: Delays and Barriers to Cancer Care in Low- and Middle-Income Countries: A Systematic Review. Oncologist 2019; 24:e1371-e80.
11. Qu LG, Brand NR, Chao A, et al.: Interventions Addressing Barriers to Delayed Cancer Diagnosis in Low- and Middle-Income Countries: A Systematic Review. Oncologist 2020.
12. Richards MA, Westcombe AM, Love SB, et al.: Influence of delay on survival in patients with breast cancer: a systematic review. Lancet 1999; 353:1119-26.
13. Jedy-Agba E, McCormack V, Adebamowo C, et al.: Stage at diagnosis of breast cancer in sub-Saharan Africa: a systematic review and meta-analysis. The Lancet Global health 2016; 4:e923-e35.
14. Islam RM, Billah B, Hossain MN, et al.: Barriers to Cervical Cancer and Breast Cancer Screening Uptake in Low-Income and Middle-Income Countries: A Systematic Review. Asian Pacific journal of cancer prevention : APJCP 2017; 18:1751-63.
15. Cazap E, Magrath I, Kingham TP, et al.: Structural Barriers to Diagnosis and Treatment of Cancer in Low- and Middle-Income Countries: The Urgent Need for Scaling Up. J Clin Oncol 2016; 34:14-9.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. Moodley J, Walter FM, Scott SE, et al.: Towards timely diagnosis of symptomatic breast and cervical cancer in South Africa. South African medical journal = Suid-Afrikaanse tydskrif vir geneeskunde 2018; 108:803-4.
17. Moodley J, Cairncross L, Naiker T, et al.: Understanding pathways to breast cancer diagnosis among women in the Western Cape Province, South Africa: a qualitative study. BMJ Open 2016; 6:e009905.
18. Poum A, Promthet S, Duffy SW, et al.: Factors associated with delayed diagnosis of breast cancer in northeast Thailand. Journal of epidemiology 2014; 24:102-8.
19. Tetteh DA, Faulkner SL: Sociocultural factors and breast cancer in sub-Saharan Africa: implications for diagnosis and management. Womens Health (Lond) 2016; 12:147-56.
20. Moodley J, Cairncross L, Naiker T, et al.: From symptom discovery to treatment - women's pathways to breast cancer care: a cross-sectional study. BMC cancer 2018; 18:312.
21. Scott SE, Walter FM, Webster A, et al.: The model of pathways to treatment: conceptualization and integration with existing theory. British journal of health psychology 2013; 18:45-65.
22. Walter F, Webster A, Scott S, et al.: The Andersen Model of Total Patient Delay: A Systematic Review of Its Application in Cancer Diagnosis. J Health Serv Res Policy 2012; 17:110-8.
23. Weller D, Vedsted P, Rubin G, et al.: The Aarhus statement: improving design and reporting of studies on early cancer diagnosis. Br J Cancer 2012; 106:1262-7.
24. World Health Organization (WHO). Guide To Cancer Early Diagnosis. Geneva, Switzerland: World Health Organization, 2017. Available from: http://apps.who.int/iris/bitstream/10665/254500/1/9789241511940-eng.pdf?ua=1. Accessed 10 March, 2020. .
25. World Health Organization (WHO). WHO report on cancer: setting priorities, investing wisely and providing care for all. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.
26. Wells KJ, Battaglia TA, Dudley DJ, et al.: Patient navigation: State of the art or is it science? Cancer 2008; 113:1999-2010.
27. Valaitis RK, Carter N, Lam A, et al.: Implementation and maintenance of patient navigation programs linking primary care with community-based health and social services: a scoping literature review. BMC Health Serv Res 2017; 17:116.
28. Battaglia TA, Roloff K, Posner MA, et al.: Improving follow-up to abnormal breast cancer screening in an urban population. A patient navigation intervention. Cancer 2007; 109:359-67.
29. Drake BF, Tannan S, Anwuri VV, et al.: A Community-Based Partnership to Successfully Implement and Maintain a Breast Health Navigation Program. Journal of community health 2015; 40:1216-23.
30. Neal RD, Tharmanathan P, France B, et al.: Is increased time to diagnosis and treatment in symptomatic cancer associated with poorer outcomes? Systematic review. Br J Cancer 2015; 112 Suppl 1:S92-S107.
31. Liberati A, Altman DG, Tetzlaff J, et al.: The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. Annals of internal medicine 2009; 151:W65-94.
32. Booth A, Clarke M, Dooley G, et al.: The nuts and bolts of PROSPERO: an international prospective register of systematic reviews. Syst Rev 2012; 1:2.
33. Shamseer L, Moher D, Clarke M, et al.: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ (Clinical research ed) 2015; 350:g7647.
34. Nnaji CA, Kuodi P, Walter FM, et al.: Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low and middle-income countries: a systematic review protocol. BMJ Open 2020; 10:e042788.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

35. Fantom N, Serajuddin U. The World Bank's Classification of Countries by Income (English). Policy Research Working Paper no. WPS 7528. Washington, DC: World Bank Group, 2016.
36. Carlos LL-J, Cruz LAPd, Leopoldo VC, et al.: Effectiveness of Traditional Chinese Acupuncture versus Sham Acupuncture: a Systematic Review. Rev Lat Am Enfermagem 2016; 24:e2762-e.
37. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.0 (updated July 2019). Cochrane, 2019. Available from https://training.cochrane.org/handbook/current/chapter-08. Accessed 27 March 2020.
38. Sterne JA, Hernan MA, Reeves BC, et al.: ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ (Clinical research ed) 2016; 355:i4919.
39. Guyatt GH, Oxman AD, Kunz R, et al.: What is "quality of evidence" and why is it important to clinicians? BMJ (Clinical research ed) 2008; 336:995-8.
40. Chowdhury TI, Love RR, Chowdhury MT, et al.: Feasibility Study of Case-Finding for Breast Cancer by Community Health Workers in Rural Bangladesh. Asian Pacific journal of cancer prevention : APJCP 2015; 16:7853-7.
41. Martei YM, Narasimhamurthy M, Prabhakar P, et al.: Breast Cancer Pathology Turnaround Time in Botswana. Journal of global oncology 2018; 4:1-7.
42. Tapela NM, Peluso MJ, Kohler RE, et al.: A Step Toward Timely Referral and Early Diagnosis of Cancer: Implementation and Impact on Knowledge of a Primary Care-Based Training Program in Botswana. Frontiers in oncology 2018; 8:187.
43. Murillo R, Díaz S, Perry F, et al.: Increased breast cancer screening and downstaging in Colombian women: A randomized trial of opportunistic breast-screening. International journal of cancer 2016; 138:705-13.
44. Gadgil A, Sauvaget C, Roy N, et al.: Cancer early detection program based on awareness and clinical breast examination: Interim results from an urban community in Mumbai, India. Breast (Edinburgh, Scotland) 2017; 31:85-9.
45. Setyowibowo H, Hunfeld JAM, Iskandarsyah A, et al.: A self-help intervention for reducing time to diagnosis in Indonesian women with breast cancer symptoms. Psycho-oncology 2020; 29:696-702.
46. Termeh Zonouzy V, Niknami S, Ghofranipour F, et al.: An educational intervention based on the extended parallel process model to improve attitude, behavioral intention, and early breast cancer diagnosis: a randomized trial. International journal of women's health 2019; 11:1-10.
47. Teh YC, Tan GH, Taib NA, et al.: Opportunistic mammography screening provides effective detection rates in a limited resource healthcare system. BMC cancer 2015; 15:405.
48. Yeoh ZY, Jaganathan M, Rajaram N, et al.: Feasibility of Patient Navigation to Improve Breast Cancer Care in Malaysia. Journal of global oncology 2018; 4:1-13.
49. Ifediora CO, Azuike EC: Targeting cervical cancer campaigns on teenage high schoolers in resource-limited economies: lessons from an intervention study of Nigerian senior secondary school girls. Family practice 2019; 36:600-6.
50. Ali SMH, Kathia UM, Gondal MUM, et al.: Impact of Clinical Information on the Turnaround Time in Surgical Histopathology: A Retrospective Study. Cureus 2018; 10:e2596.
51. Pace LE, Dusengimana JMV, Shulman LN, et al.: Cluster Randomized Trial to Facilitate Breast Cancer Early Diagnosis in a Rural District of Rwanda. Journal of global oncology 2019; 5:1-13.
52. Dickerson LK, Rositch AF, Lucas S, et al.: Pilot Educational Intervention and Feasibility Assessment of Breast Ultrasound in Rural South Africa. Journal of global oncology 2017; 3:502-8.
53. Hadley M, Mullen LA, Dickerson L, et al.: Assessment and Improvement Strategies for a Breast Cancer Early Detection Program in Rural South Africa. Journal of global oncology 2018; 4:1-12.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

54. Maimela G, Nene X, Mvundla N, et al.: The impact of decentralising colposcopy services from tertiary-level to primary-level care in inner-city Johannesburg, South Africa: a before and after study. BMJ Open 2019; 9:e024726.
55. Ngoma T, Mandeli J, Holland JF: Downstaging cancer in rural Africa. International journal of cancer 2015; 136:2875-9.
56. Pinder LF, Nzayisenga JB, Shibemba A, et al.: Demonstration of an algorithm to overcome health system-related barriers to timely diagnosis of breast diseases in rural Zambia. PloS one 2018; 13:e0196985.
57. Pinder LF, Henry-Tillman R, Linyama D, et al.: Leverage of an Existing Cervical Cancer Prevention Service Platform to Initiate Breast Cancer Control Services in Zambia: Experiences and Early Outcomes. Journal of global oncology 2018; 4:1-8.
58. Songiso M, Pinder LF, Munalula J, et al.: Minimizing Delays in the Breast Cancer Pathway by Integrating Breast Specialty Care Services at the Primary Health Care Level in Zambia. JCO global oncology 2020; 6:859-65.
59. Aribal E, Mora P, Chaturvedi AK, et al.: Improvement of early detection of breast cancer through collaborative multi-country efforts: Observational clinical study. European journal of radiology 2019; 115:31-8.
60. Martei YM, Pace LE, Brock JE, et al.: Breast Cancer in Low- and Middle-Income Countries: Why We Need Pathology Capability to Solve This Challenge. Clinics in laboratory medicine 2018; 38:161-73.
61. Murillo R, Robles C: Research Needs for Implementing Cancer Prevention and Early Detection in Developing Countries: From Scientists' to Implementers' Perspectives. Biomed Res Int 2019; 2019:9607803.

Figure 1: PRISMA flow diagram of study selection process

Figure 2: Map of geographical distribution of included studies

Figures 3a and 3b: Risk of bias assessment outcomes geographical distribution of the studies

Appendix 1: Search strategies

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

FIGURE 1

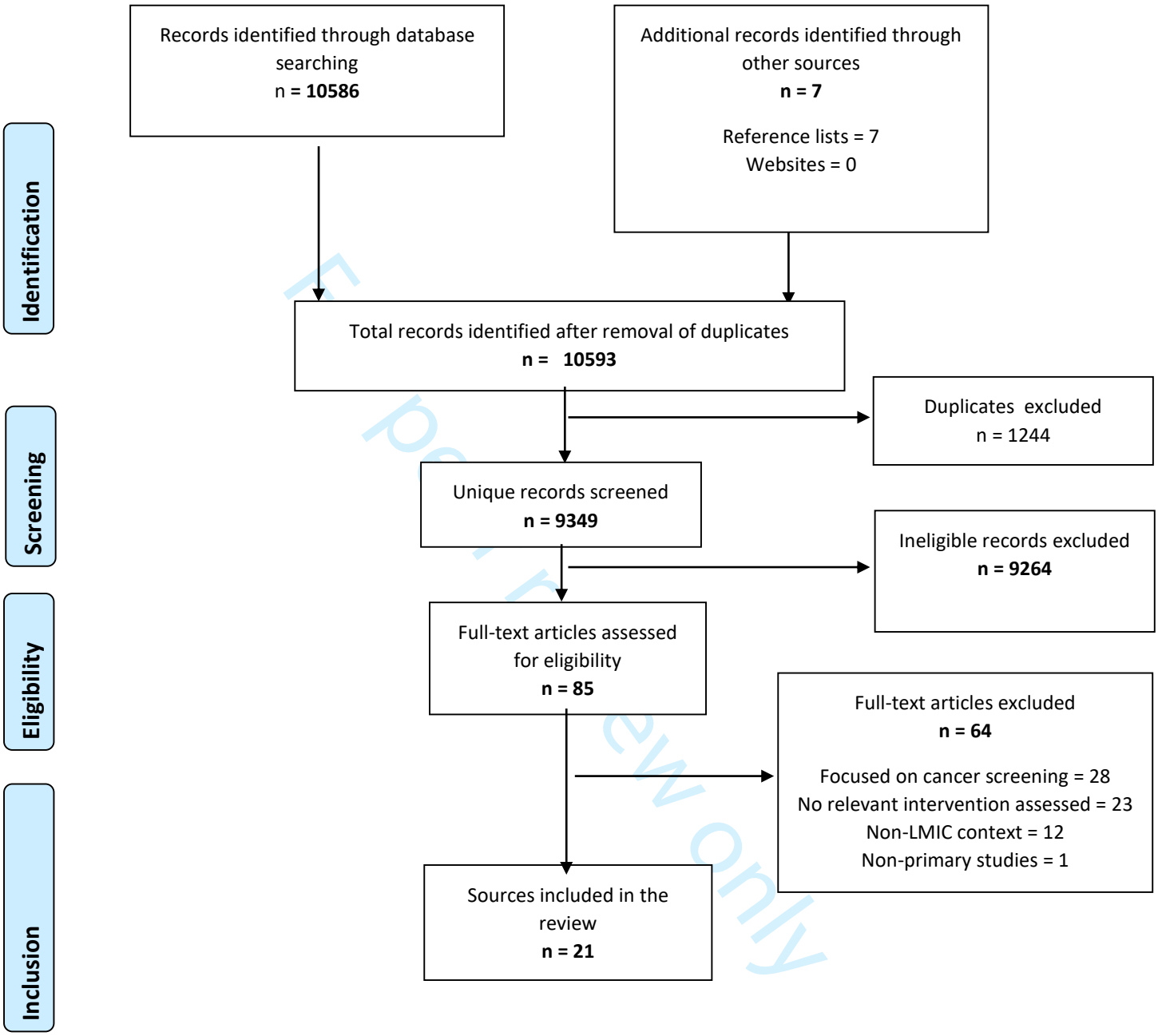


Figure 1: PRISMA flow diagram showing study selection process

For peer review only

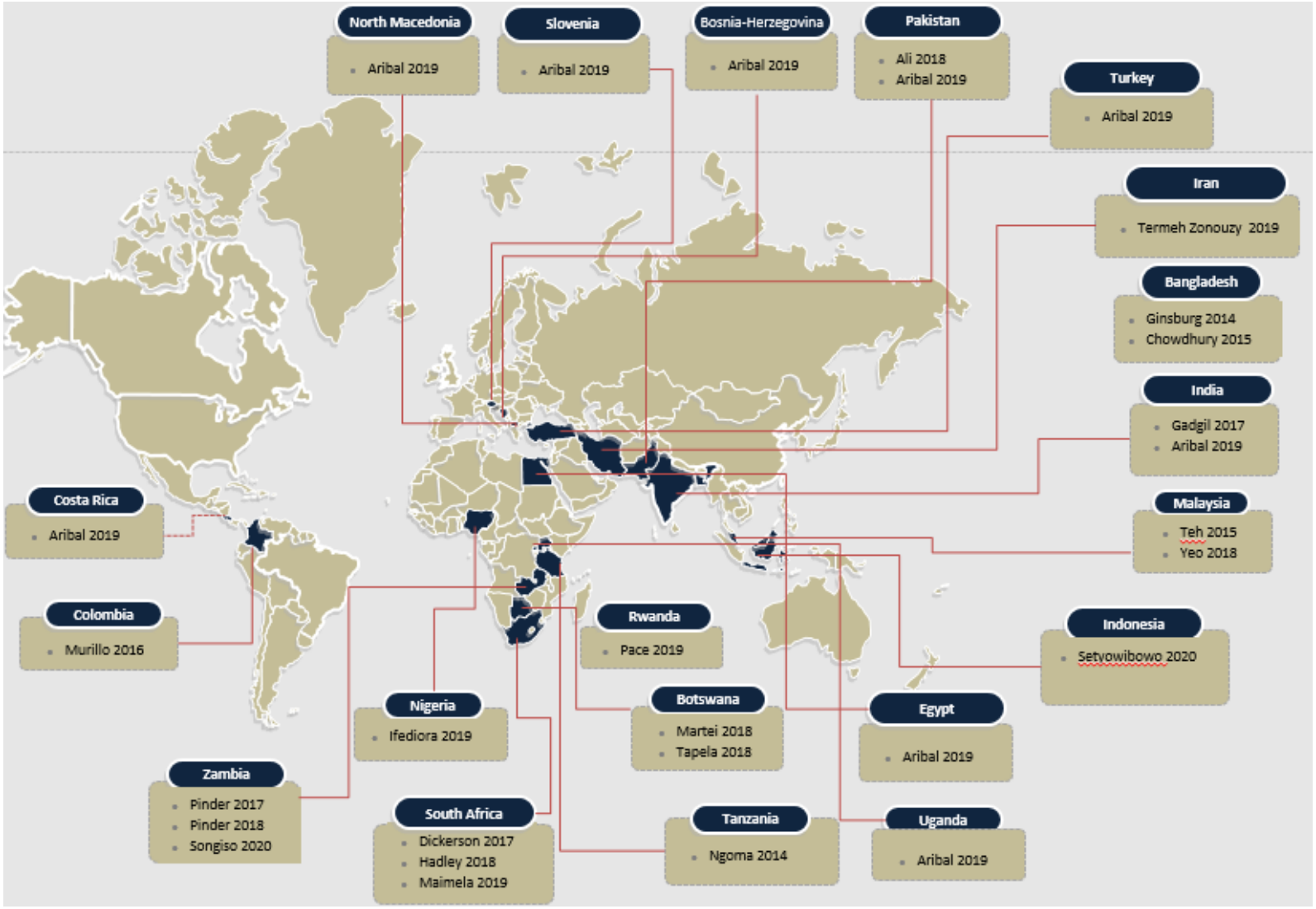


Figure 2: Geographical distribution of included studies (map created by authors using an open-source template sourced from https://yourfreetemplates.com/)

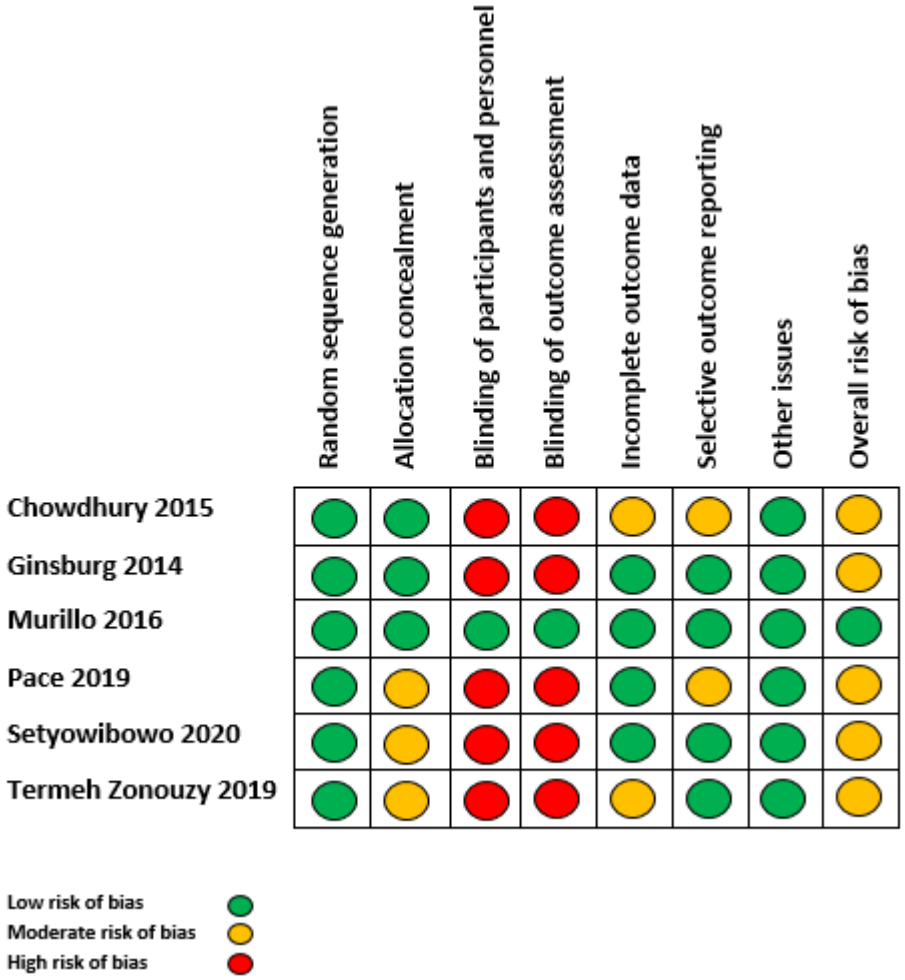


Figure 3a: Risk of bias assessment of randomized controlled trials

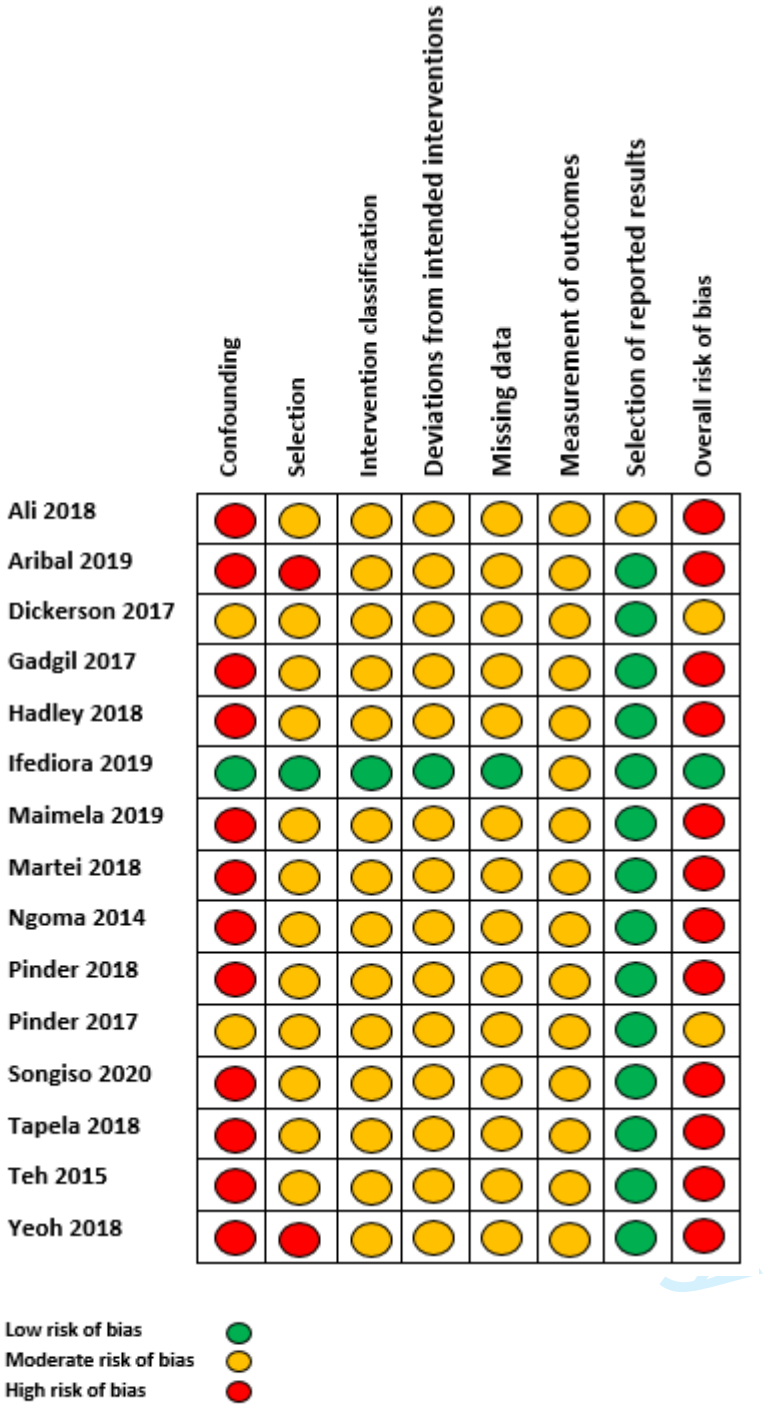


Figure 3b: Risk of bias assessment of observational studies

Appendix: Search strategies

PubMed/MEDLINE search strategy

Search #	Search Texts and Syntaxes
#1	Breast Neoplasms [Mesh] OR Breast cancer[Title/Abstract] OR breast[Title/Abstract] OR cervical cancer[Title/Abstract] OR cervix[Title/Abstract] OR cancer[Title/Abstract] OR malignant[Title/Abstract] OR neoplasm[Title/Abstract] OR neoplasia[Title/Abstract] OR malignancy[Title/Abstract] OR tumour[Title/Abstract]
#2	Diagnosis [Mesh] OR diagnostic[Title/Abstract] OR diagnosis[Title/Abstract] OR diagnosis[Title/Abstract] OR detection[Title/Abstract] OR discovery[Title/Abstract] OR Patient navigation[Title/Abstract] OR Patient pathway[Title/Abstract] OR care continuum[Title/Abstract]
#3	early[Title/Abstract] OR timely[Title/Abstract] OR time[Title/Abstract] OR late[Title/Abstract] OR delay[Title/Abstract]
#4	#2 AND #3
#5	Developing Countries OR Developing Country OR Developing Economies OR Developing Economy OR Developing Nation OR Developing Nations OR Developing Population OR Developing Populations OR Developing World OR LAMI Countries OR LAMI Country OR Less Developed Countries OR Less Developed Country OR Less Developed Economies OR Less Developed Nation OR Less Developed Nations OR Less Developed World OR Lesser Developed Countries OR Lesser Developed Nations OR LMIC OR LMICS OR Low GDP OR Low GNP OR Low Gross Domestic OR Low Gross National OR Low Income OR Lower GDP OR lower gross domestic OR Lower Income OR Middle Income OR Poor Countries OR Poor Country OR Poor Economies OR Poor Economy OR Poor Nation OR Poor Nations OR Poor Population OR Poor Populations OR poor world OR Poorer Countries OR Poorer Economies OR Poorer Economy OR Poorer Nations OR Poorer Population OR Poorer Populations OR Third World OR Transitional Countries OR Transitional Country OR Transitional Economies OR Transitional Economy OR Under Developed Countries OR Under Developed Country OR under developed nations OR Under Developed World OR Under Served Population OR Under Served Populations OR Underdeveloped Countries OR Underdeveloped Country OR underdeveloped economies OR underdeveloped nations OR underdeveloped population OR Underdeveloped World OR Underserved Countries OR Underserved Nations OR Underserved Population OR Underserved Populations OR Afghanistan OR Albania OR Algeria OR American Samoa OR Angola OR Armenia OR Azerbaijan OR Bangladesh OR Belarus OR Byelarus OR Belorussia OR Belize OR Benin OR Bhutan OR Bolivia OR Bosnia OR Botswana OR Brazil OR Bulgaria OR Burma OR Burkina Faso OR Burundi OR Cabo Verde OR Cape Verde OR Cambodia OR

For peer review only

	Cameroon OR Central African Republic OR Chad OR China OR Colombia OR Comoros OR Comores OR Comoro OR Congo OR Costa Rica OR Côte d'Ivoire OR Cuba OR Djibouti OR Dominica OR Dominican Republic OR Ecuador OR Egypt OR El Salvador OR Equatorial Guinea OR Eritrea OR Ethiopia OR Fiji OR Gabon OR Gambia OR Gaza OR Georgia OR Georgia Republic OR Ghana OR Grenada OR Grenadines OR Guatemala OR Guinea OR Guinea- Bissau OR Guyana OR Haiti OR Herzegovina OR Hercegovina OR Honduras OR India OR Indonesia OR Iran OR Iraq OR Ivory Coast OR Jamaica OR Jordan OR Kazakhstan OR Kenya OR Kiribati OR Democratic People's Republic of Korea OR Kosovo OR Kyrgyz OR Kirghizia OR Kirghiz OR Kyrgyzstan OR Lao PDR OR Laos OR Lebanon OR Lesotho OR Liberia OR Libya OR Macedonia OR Madagascar OR Malawi OR Malay OR Malaya OR Malaysia OR Maldives OR Mali OR Marshall Islands OR Mauritania OR Mauritius OR Mexico OR Micronesia OR Moldova OR Mongolia OR Montenegro OR Morocco OR Mozambique OR Myanmar OR Namibia OR Nepal OR Nicaragua OR Niger OR Nigeria OR Pakistan OR Palau OR Papua New Guinea OR Paraguay OR Peru OR Philippines OR Principe OR Romania OR Ruanda OR Rwanda OR Samoa OR Sao Tome OR Senegal OR Serbia OR Sierra Leone OR Solomon Islands OR Somalia OR South Africa OR South Sudan OR Sri Lanka OR St Lucia OR St Vincent OR Sudan OR Surinam OR Suriname OR Swaziland OR Syria OR Syrian Arab Republic OR Tajikistan OR Tadzhikistan OR Tajikistan OR Tadzhik OR Tanzania OR Thailand OR Timor OR Togo OR Tonga OR Tunisia OR Turkey OR Turkmen OR Turkmenistan OR Tuvalu OR Uganda OR Ukraine OR Uzbek OR Uzbekistan OR Vanuatu OR Venezuela OR Vietnam OR West Bank OR Yemen OR Zambia OR Zimbabwe
#6	#1 AND #4 AND #5

For peer review only

Scopus

Search #	Search Texts and Syntaxes
#1	TITLE-ABS-KEY (Breast cancer) OR breast OR (cervical cancer) OR cervix OR cancer OR malignant OR neoplasm OR neoplasia OR malignancy OR tumour
#2	TITLE-ABS-KEY Diagnosis OR diagnostic OR diagnosis OR diagnosis OR detection OR discovery OR (Patient navigation) OR (Patient pathway) OR (care continuum)
#3	Early OR timely OR time OR late OR delay
#4	#2 AND #3
#5	TITLE-ABS-KEY “Deprived Countries” OR “Deprived Population” OR “Deprived Populations” OR “Developing Countries” OR “Developing Country” OR “Developing Economies” OR “Developing Economy” OR “Developing Nation” OR “Developing Nations” OR “Developing Population” OR “Developing Populations” OR “Developing World” OR “LAMI Countries” OR “LAMI Country” OR “Less Developed Countries” OR “Less Developed Country” OR “Less Developed Economies” OR “Less Developed Nation” OR “Less Developed Nations” OR “Less Developed World” OR “Lesser Developed Countries” OR “Lesser Developed Nations” OR LMIC OR LMICS OR “Low GDP” OR “Low GNP” OR “Low Gross Domestic” OR “Low Gross National” OR “Low Income” OR “Lower income” OR “Lower GDP” OR “Lower Gross Domestic” OR “Middle Income” OR “Poor Countries” OR “Poor Country” OR “Poor Economies” OR “Poor Economy” OR “Poor Nation” OR “Poor Nations” OR “Poor Population” OR “Poor Populations” OR “poor world” OR “Poorer Countries” OR “Poorer Economies” OR “Poorer Economy” OR “Poorer Nations” OR “Poorer Population” OR “Poorer Populations” OR “Third World” OR “Transitional Countries” OR “Transitional Country” OR “Transitional Economies” OR “Transitional Economy” OR “Under Developed” OR “Under Served” OR “Underdeveloped Countries” OR “Underdeveloped Country” OR “underdeveloped economies” OR “underdeveloped nations” OR “underdeveloped population” OR “Underdeveloped World” OR “Underserved Countries” OR “Underserved Nations” OR “Underserved Population” OR “Underserved Populations” OR Afghanistan OR Albania OR Algeria OR “American Samoa” OR Angola OR Armenia OR Azerbaijan OR Bangladesh OR Belarus OR Byelarus OR Belorussia OR Belize OR Benin OR Bhutan OR Bolivia OR Bosnia OR Botswana OR Brazil OR Bulgaria OR Burma

For peer review only

	OR “Burkina Faso” OR Burundi OR “Cabo Verde” OR “Cape Verde” OR Cambodia OR Cameroon OR “Central African Republic” OR Chad OR China OR Colombia OR Comoros OR Comores OR Comoro OR Congo OR “Costa Rica” OR “Côte d'Ivoire” OR Cuba OR “Democratic People’s Republic of Korea” OR Djibouti OR Dominica OR “Dominican Republic” OR Ecuador OR Egypt OR “El Salvador” OR Eritrea OR Ethiopia OR “Equatorial Guinea” OR Fiji OR Gabon OR Gambia OR Gaza OR “Georgia Republic” OR Georgia OR Ghana OR Grenada OR Grenadines OR Guatemala OR Guinea OR “Guinea Bissau” OR Guyana OR Haiti OR Herzegovina OR Hercegovina OR Honduras OR India OR Indonesia OR Iran OR Iraq OR “Ivory Coast” OR Jamaica OR Jordan OR Kazakhstan OR Kenya OR Kiribati OR Korea OR Kosovo OR Kyrgyz OR Kirghizia OR Kirghiz OR Kyrgyzstan OR “Lao PDR” OR Laos OR Lebanon OR Lesotho OR Liberia OR Libya OR Macedonia OR Madagascar OR Malawi OR Malay OR Malaya OR Malaysia OR Maldives OR Mali OR “Marshall Islands” OR Mauritania OR Mauritius OR Mexico OR Micronesia OR Moldova OR Mongolia OR Montenegro OR Morocco OR Mozambique OR Myanmar OR Namibia OR Nepal OR Nicaragua OR Niger OR Nigeria OR Pakistan OR Palau OR Papua New Guinea OR Paraguay OR Peru OR Philippines OR Principe OR Romania OR Rwanda OR Ruanda OR Samoa OR “Sao Tome” OR Senegal OR Serbia OR “Sierra Leone” OR “Solomon Islands” OR Somalia OR “South Africa” OR “South Sudan” OR “Sri Lanka” OR “St Lucia” OR “St Vincent” OR Sudan OR Surinam OR Suriname OR Swaziland OR Syria OR “Syrian Arab Republic” OR Tajikistan OR Tadzhikistan OR Tajikistan OR Tadzhik OR Tanzania OR Thailand OR Timor OR Togo OR Tonga OR Tunisia OR Turkey OR Turkmen OR Turkmenistan OR Tuvalu OR Uganda OR Ukraine OR Uzbek OR Uzbekistan OR Vanuatu OR Venezuela OR Vietnam OR “West Bank” OR Yemen OR Zambia OR Zimbabwe
#6	#1 AND #4 AND #5

For peer review only

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7-8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7-8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	8

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	9
Study characteristics	17	Cite each included study and present its characteristics.	9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	10
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	10-18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	10-18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	18
	23b	Discuss any limitations of the evidence included in the review.	21
	23c	Discuss any limitations of the review processes used.	21
	23d	Discuss implications of the results for practice, policy, and future research.	21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	22
Competing interests	26	Declare any competing interests of review authors.	22
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	7

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi:



PRISMA 2020 Checklist

10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only

BMJ Open

Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054501.R1
Article Type:	Original research
Date Submitted by the Author:	14-Feb-2022
Complete List of Authors:	Nnaji, Chukwudi; University of Cape Town Faculty of Health Sciences, Women's Health Research Unit, School of Public Health and Family Medicine; University of Cape Town Faculty of Health Sciences, Cancer Research Initiative Kuodi, Paul; Lira University, Department of Public Health, Faculty of Health Sciences Walter, Fiona; University of Cambridge, Dept of Public Health and Primary Care; Queen Mary University of London Barts and The London School of Medicine and Dentistry, Wolfson Institute of Population Health Moodley, Jennifer; University of Cape Town Faculty of Health Sciences, Women's Health Research Unit, School of Public Health and Family Medicine; University of Cape Town Faculty of Health Sciences, 2Cancer Research Initiative
Primary Subject Heading:	Public health
Secondary Subject Heading:	Oncology, Health services research, Global health, Radiology and imaging
Keywords:	ONCOLOGY, Breast tumours < ONCOLOGY, PUBLIC HEALTH, Organisation of health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review

Chukwudi A. Nnaji[1,2], Paul Kuodi[3], Fiona M. Walter[4,5], Jennifer Moodley[1,2,6]

Authors affiliations:

[1]Women's Health Research Unit, School of Public Health and Family Medicine, Faculty of Health Sciences, University of Cape Town; Anzio Road. Observatory, 7925, Cape, Town, South Africa

[2]Cancer Research Initiative, Faculty of Health Sciences, University of Cape Town; Anzio Road, Observatory, 7925, Cape, Town, South Africa

[3] Department of Public Health, Faculty of Health Sciences, Lira University, Lira, Uganda

[4] The Primary Care Unit, Department of Public Health and Primary Care, University of Cambridge, United Kingdom

[5] Wolfson Institute of Population Health, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, UK

[6] SAMRC Gynaecology Cancer Research Centre, University of Cape Town, Cape Town, South Africa

Correspondence to:

Dr Chukwudi A. Nnaji; nnjchu001@myuct.ac.za

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abstract

Objectives: To systematically synthesise available evidence on the nature and effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-come countries (LMICs).

Design: A systematic review of published evidence. The review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA).

Data sources: A comprehensive search of published literature was conducted. In addition, relevant grey literature sources and bibliographical references of included studies were searched for potentially eligible evidence.

Study selection: Studies published between January 2010 and November 2020 were eligible for inclusion; they had to report on interventions/strategies for the timely diagnosis of breast and/or cervical cancers in LMIC settings.

Data extraction and synthesis: Literature search, screening, study selection, data extraction and quality appraisal were conducted by two independent reviewers. Evidence was synthesised and reported using a global taxonomy framework for early cancer diagnosis.

Results: From the total of 10593 records identified, 21 studies conducted across 20 LMICs were included in this review. Most included studies (16/21) primarily focused on interventions addressing breast cancers; two focused on cervical cancer while the rest examined multiple cancer types. Reported interventions targeted health care workers (12); women and adolescent girls (7); and both women and health care workers (3). Eight studies reported on interventions addressing access delays; another eight focused on interventions addressing diagnostic delays; one reported on an intervention targeted at addressing both access and diagnostic delays, and five studies assessed interventions addressing access, diagnostic and treatment delays. While most interventions were demonstrated to be feasible and effective, many of the reported outcome measures are of limited clinical relevance to diagnostic timeliness.

Conclusions: Though limited, evidence suggests that interventions aimed at addressing barriers to timely diagnosis of breast and cervical cancer are feasible in resource-limited contexts. Future interventions need to address clinically relevant measures to better assess efficacy of interventions.

Keywords: Cancer, breast, cervical, diagnosis, timeliness, delay, LMICs.

PROSPERO registration number: CRD42020177232.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- This review was designed in accordance with standard systematic review protocol guidelines.
- Literature searching was comprehensive, covering both peer-reviewed and relevant grey literature.
- No language restrictions were applied in the search.
- While most interventions were demonstrated to be feasible and effective, reported outcome measures are of limited clinical relevance to diagnostic timeliness.
- The overall strength of the synthesised evidence and applicability of findings depend on the quality of included studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Breast and cervical cancers both represent a significant burden of disease globally, with a disproportionately high burden of morbidity and mortality in low- and middle-income countries (LMICs). [1-3] The majority (53%) of new breast cancer cases occur in women living in LMICs, accounting for nearly a third of all cancers in women, and with an age-standardised incidence rate (ASIR) of 31 per 100,000 women in these settings.[4,5] Cervical cancer constitutes 16% of women's cancers in LMICs, with an ASIR of 16 per 100,000 women.[2] The increasing burdens of both cancers has been attributed to factors such as population growth, ageing societies and exposures to oncogenic risk factors.[1,6] While the incidence of breast and cervical cancers increases in LMICs, many cases continue to go unreported and undiagnosed; the majority of these women present with late-stage disease, the treatment of which is often costlier and less effective.[7,8] These challenges have been aggravated by the COVID-19 pandemic, which immensely disrupted cancer services globally, particularly in low-resource settings.[9]

The timeliness of diagnosis is an important determining factor of breast and cervical cancer survival.[10-13] Yet, in many LMICs, breast and cervical screening and early diagnosis programmes are often opportunistic and not well organised.[7,14-16] One of the greatest barriers to timely cancer diagnosis in LMICs is lack of geographic access to health facilities offering screening and diagnostic services, as well as the lack of well-trained care providers.[6,17] In addition to these health system factors are the underlying psychosocial and cultural barriers to timely cancer help-seeking and diagnosis.[6-8,18] These include perceptions such as beliefs that breast cancers are contagious or punitive consequences of sins.[15,19]

To conceptualise and understand the pathway to cancer diagnosis, distinct events and processes from symptom awareness to diagnosis and treatment have been described. These include patients' detection of bodily changes and awareness of symptoms; decision to seek help from a healthcare provider; access to clinical evaluation, diagnosis, and staging; access to treatment; and follow-up.[17,20-22] Barriers encountered by patients in any of these phases may pose an impediment to timely cancer diagnosis and delay patients' access to care.[10,23] In an effort to address these barriers and provide global standards for early cancer diagnosis, the World Health Organization (WHO) published the *WHO Guide to Cancer Early Diagnosis* in 2017.[24] It provides an operational framework for cancer control programmes around the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

world to systematically address barriers that may impede timely cancer diagnosis, treatment and care. Furthermore, it specifies three essential steps of early cancer diagnosis: Step 1: awareness of cancer symptoms and accessing care; Step 2: clinical evaluation, diagnosis and staging; and Step 3: access to treatment, including pain relief. These steps were emphasised to be essential for setting cancer control priorities in the 2020 WHO Report on Cancer.[25]

Interventions aimed at effectively addressing barriers to timely cancer diagnosis have been studied in various settings within the different phases of the cancer care continuum.[6,11] It is evident that such interventions can improve the timeliness of cancer diagnosis, enable early initiation of treatment, reduce morbidity and increase the chances of survival.[20,26-30] However, much of this evidence is from high-income countries (HICs). Given the substantial differences between HICs and LMICs regarding health resources, environment, infrastructure, technology, and medical personnel, improving time to diagnosis for breast and cervical cancer in LMIC settings may require different approaches.[3]

While interventions to improve the timely diagnosis of these cancers are increasingly being implemented in LMICs, there is uncertainty about their role and effectiveness. We have identified two previous reviews on this topic within the LMIC context.[6,11] A scoping review by Dalton and colleagues synthesised the evidence on patient navigation strategies for cancer care in LMICs, but focused broadly on the entire cancer detection, treatment and care continuum.[6] A previous systematic review by Qu and colleagues assessed interventions specifically aimed at addressing barriers to timely cancer diagnosis in LMICs.[11] Neither specifically focused on breast and/or cervical cancer.

Therefore, the purpose of this review was to systematically synthesise the evidence on the nature and effectiveness of interventions aimed at improving timely diagnosis of symptomatic breast and cervical cancers in LMICs, by targeting the 3 key steps outlined by the WHO framework.[24] Due to the lead time and length bias associated with cancer screening programmes, this review focused on interventions targeted at improving early diagnosis of symptomatic breast and cervical cancer. Moreover, evidence demonstrates that diagnostic delays among symptomatic women are so long in LMICs.[15,19] Hence, interventions to reduce these delays among people with symptoms would likely have substantial impact on outcomes and are also a prerequisite for screening programs. We sought to provide a user-friendly evidence summary for health policy-makers, cancer programme managers, oncologists and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cancer programme implementers, for informing further efforts to assist women in LMICs in overcoming socioeconomic, cultural, and structural barriers to timely diagnosis and appropriate care, while identifying gaps for future research.

METHODS

Study design

This systematic review of evidence was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[31] The review has been registered on the International Prospective Register of Systematic Reviews (PROSPERO, registration number: CRD42020177232).[32] The protocol, which was designed in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) guidelines[33] has been published elsewhere.[34]

Data sources

The following databases were searched: MEDLINE (via PubMed), Cochrane Library (including the Cochrane Central Register of Controlled Trials (CENTRAL) and the Database of Abstracts of Reviews of Effects (DARE)), Scopus, CINAHL, Web of Science and the International Clinical Trials Registry Platform (ICTRP). Relevant grey literature sources; including the publication database of the WHO's International Agency for Research on Cancer (IARC), the Cancer Atlas of the Union for International Cancer Control (UICC) and the Global Cancer Project Map, were also searched for potentially eligible publications. The search strategy (see Appendix 1) was developed with guidance from a health sciences subject librarian. More details of the search strategies used are described in the published review protocol.[34] Only articles published over the last 10 years (from 1 January 2010 to the date of search) were considered eligible, with no language restriction. In addition, the bibliographical references of included studies were also searched. Search was conducted from 22 June 2020 through 30 November 2020.

Eligibility criteria

The study inclusion criteria were guided by the research question: *'What interventions targeting the key steps outlined by the WHO operational framework for early diagnosis have been used for improving timely diagnosis of symptomatic breast and cervical cancers in LMICs, and how effective are they?'*. Eligible studies had to report on interventions or strategies for

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

timely breast and cervical cancer diagnosis, targeting women, the general public or healthcare workers in LMICs. The definition of LMICs was based on the World Bank's current classification using per capita gross national income.[35]

Such interventions or strategies had to be aimed at influencing the timeliness of symptomatic breast or cervical cancer diagnosis along any or a combination of the three steps of the WHO framework for early diagnosis, whether as a single focus intervention or as a multi-focus intervention targeting more than one cancer type. Studies with interventions focused primarily on screening of asymptomatic individuals were excluded. More detailed information about the eligibility criteria can be found in the published review protocol.[34]

Study selection

The review consisted of two levels of screening: a title and abstract screening to identify potentially eligible publications, and review of full-texts to select those to be included in the review based on pre-defined inclusion/exclusion criteria. For the first level of screening, two reviewers (CAN and PK) independently screened the titles and abstracts of all retrieved records from the search output. Articles that were considered relevant by either or both of the reviewers were included in the full-text review. In the second step, the two reviewers (CAN and PK) independently assess the full-texts to determine if they met the inclusion/exclusion criteria. Any discordance in eligibility assessment was resolved through consensus between the two researchers, with recourse to a third arbitrator (JM) as necessary.

Data extraction

Two reviewers (CAN and PK) independently extracted all relevant data from the included articles using a standardised data extraction tool, adapted from the framework proposed by Carlos and colleagues.[36] Extracted data fell under four domains: (1) study identification details (article title; journal title; authors; country of the study; language; publication year; host institution of the study); (2) methodological characteristics (study design; study objective or research question or hypothesis; sample characteristics (e.g. sample size; sex; age, ethnicity; groups and controls; follow-up duration; validation of measures; statistical analyses); (3) main findings, and (4) conclusions. Study eligibility was re-verified at the start of data extraction. Discrepancies in extraction were resolved through consensus. CAN combined the two

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

spreadsheets of extracted data for analysis, and data were double-checked by PK for completeness and accuracy.

Quality assessment

Two reviewers (CAN and PK) independently assessed each included study for risk of bias, using the Cochrane risk of bias tool[37] for randomised clinical trials, and the Risk of Bias In Non-randomised Studies of Interventions (ROBINS-I) tool[38] for non-randomised and observational studies. Discrepancies in study quality assessments were resolved by discussion and consensus. Further details of the quality appraisal process are discussed in the published protocol of this review.[34]

Data analysis

A PRISMA flow diagram was used to illustrate the literature search results and study selection process. Outcomes were categorised according to the essential steps of early cancer diagnosis as specified in the WHO Guide to Cancer Early Diagnosis.[24] Plans to conduct a meta-analysis were not feasible as studies were not quantitatively comparable in terms of interventions and reported outcomes. Instead, a thematic narrative synthesis approach was used to aggregate and analyse relevant findings. Plans to assess the certainty of reviewed evidence using the Grades of Recommendation Assessment, Development and Evaluation (GRADE) approach[39] were not feasible for similar reasons. A table summarising findings from each included study was presented. JM and FMW reviewed the analysed data for accuracy and consistency with protocol.

RESULTS

Search results

A total of 10593 records were identified from literature database searches, from which 1244 duplicates were discarded. The remainder, 9349 unique records, had their titles and abstracts screened, from which 9264 clearly ineligible publications were excluded. The full texts of the remaining 85 potentially eligible studies were reviewed against predefined inclusion and exclusion criteria; 21 were included in the review and 64 were excluded for various reasons. **Figure 1** presents the PRISMA flow chart of the study selection process and reasons for exclusion.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Characteristics of included studies

Table 1 summarises the characteristics of included studies, published between 2015 and 2020. Studies were conducted in 20 LMICs across regions of Africa (8), Europe (4), Latin America (2), Middle-East Asia (1) and South-East Asia (5). Twenty were single-country studies, in which 13 countries were represented: Bangladesh (2)[8,40], Botswana (2)[41,42], Colombia (1)[43], India (1)[44], Indonesia (1)[45], Iran (1)[46], Malaysia (2)[47,48], Nigeria (1)[49], Pakistan (1)[50], Rwanda (1)[51], South Africa (3)[52-54], Tanzania (1)[55] and Zambia (3)[56-58]. One study had a multi-country focus, with participants recruited from nine countries: Bosnia-Herzegovina, Costa Rica, Egypt, India, North Macedonia, Pakistan, Slovenia, Turkey and Uganda.[59] **Figure 2** illustrates the geographical distribution of the studies.

In terms of study design, six studies were randomised clinical trials (RCT),[8,40,43,45,46,51] six were before-and-after studies,[42,44,48,49,52,54] one was a longitudinal (cohort-type) observational study,[55] four were retrospective cohort-type studies,[41,47,50,53] while the remaining four were cross-sectional studies.[56-59]

The majority (16/21) of included studies primarily focused on interventions addressing breast cancers; two focused on cervical cancer, whereas the rest examined multiple cancer types (including breast and/or cervical cancers). Of the 21 studies, 12 targeted health care workers (including physicians, nurses, pathologists, non-physician health providers and community health workers); seven targeted women and adolescent girls; and three targeted both women and health care workers. There was an even representation across urban and rural implementation contexts, with reported interventions implemented in 11 urban, 9 rural and one mixed (urban and rural) communities.

When categorised by the three steps of the WHO Guide to Cancer Early Diagnosis[24], eight studies reported on interventions aiming to address barriers in Step 1 (promoting awareness and addressing delays in access to care);[8,40,44-47,49,54] another eight focused on interventions addressing Step 2 (addressing diagnostic delay by optimising clinical evaluation, diagnosis and staging);[42,43,50-52,55,59,60] one study reported on an intervention targeted at both Steps 1 and 2;[53] while five studies assessed interventions targeted at all three steps (addressing access, diagnostic and treatment delays).[48,56-58] **Table 1** describes the categorisation of the interventions by the three steps of the WHO Guide to Cancer Early Diagnosis.[24]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1: Characteristics of included studies and categorisation of interventions according to the essential steps of the WHO Guide to Cancer Early Diagnosis

Step	Study ID	Study Design	Country	Target population	Cancer type	Setting	Intervention	Primary outcome	Duration
Step 1	Gadgil 2017	Before-and-after study	India	Women in an occupational setting.	Breast	Urban	A programme that aimed to increase awareness on breast cancer and access to early detection by emailing awareness brochures annually to a cohort of women in an occupational setting.	Proportion of women with early tumours	3 years
	Ginsburg 2014	RCT	Bangladesh	CHWs	Breast	Rural	An mHealth model to increase clinic attendance for breast symptoms in rural Bangladesh: involving smart phone/applications and CHW training as part of a patient navigation programme to address potential barriers to seeking care.	Number of women with breast symptoms identified	4 months
	Ifediora 2019	Before-and-after study	Nigeria	High school girls	Cervical	Rural	School-based cervical cancer prevention and early diagnosis education programme - targeting cervical cancer campaigns on teenage high schoolers in resource-limited economies.	Improvement in cervical cancer awareness and help-seeking attitude	6 months
	Chowdhury 2015	RCT	Bangladesh	CHWs	Breast	Rural	Population-based case finding for breast cancer by CHWs in Rural Bangladesh using different data collection methods and approaches.	Accuracy of cancer case-finding data	3 months
	Maimela 2019	Before-and-after study	South Africa	Women	Cervical	Urban	Decentralising colposcopy services from tertiary-level to primary-level care facility to aid early definitive diagnosis of cervical cancer.	Time to colposcopy and number of women who had colposcopy following Pap smear	6 years
	Setyowibowo 2020		Indonesia	Women	Breast	Both rural and urban	A self-help intervention for reducing time to diagnosis in women with breast cancer symptoms It consisted of health education and psycho-education using a narrative	Time between the first medical consultation and	3 months

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

		Cluster randomized crossover trial					strategy, which involved the use of testimonials and story-telling.	the definitive diagnosis	
	Teh 2015	Retrospective study	Malaysia	Women	Breast	Urban	Routine (opportunistic) screening mammogram vs targeted (high risk) screening mammogram vs diagnostic mammogram.	Proportion of women with early tumours	NA
	Termeh Zonouzy 2019	RCT	Iran	Women	Breast	Urban	An educational intervention based on fear appeals using the extended parallel process model (EPPM) to improve attitudes, intention, and early breast cancer diagnosis.	Changes in attitude, behavioral intention towards early diagnosis	3 months
	Ali 2018	Retrospective study	Pakistan	Physicians and pathologists	Multiple, including breast cancer	Urban	Adequate clinical information (CI) for improving turnaround time (TAT) in surgical histopathological diagnosis.	Pathology results TAT	4 months
	Aribal 2019	Observational clinical study	Bosnia-Herzegovina, Costa Rica, Egypt, India, North Macedonia, Pakistan, Slovenia, Turkey, Uganda	Oncologists, radiologists, medical physicists and radiographers working on oncology centres	Breast	Urban	A programme aiming to improve early detection of breast cancer by strengthening both the clinical component and quality aspects of imaging practice of oncology centres.	Changes in mammography practice and image guided interventions	2 years
	Dickerson 2017	Before-and-after study	South Africa	Non-physician providers	Breast	Rural	Breast ultrasound training for non-physician providers with the goal of early diagnosis of breast cancer and down-staging in a rural setting.	Diagnostic ultrasound competence scores	3 weeks
	Martei 2018	Retrospective study	Botswana	Pathologists	Breast	Urban	A pathology scale-up programme to optimise breast cancer pathological diagnostic TAT in Botswana.	Breast biopsy TAT	4 years
Step 2	Murillo 2016		Colombia	Physicians	Breast	Urban			2 years

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

		RCT					Opportunistic clinic-based breast cancer screening programs to increase breast cancer screening and down-staging in Colombian women. Physicians received a 2-day training course on mammography and BIRADS (Breast Imaging Reporting and Data System) grading.	Proportion of women with early tumours	
	Ngoma 2014	Prospective cohort study	Tanzania	CHW/village navigators	Multiple, including breast and cervical	Rural	Conventional dispensary self-referral vs. proactive home visits for down-staging cancer in a rural community in Tanzania.	Proportion of women with early tumours	3 years
	Pace 2019	RCT	Rwanda	Health workers	Breast	Rural	Health worker training on diagnosis and staging to facilitate breast cancer early diagnosis. It consisted of instructions for CHWs in symptoms of breast cancer and messaging that community members should come to their local health centres for any breast symptoms.	Number of breast biopsies and proportion of women with early tumours	2 years
	Tapela 2018	Before-and-after study	Botswana	Primary care providers	Multiple, including breast and cervical	Urban	A district-level capacity building training programme targeting primary care providers, aiming to enhance timely diagnosis of cancers.	Participants' early diagnosis competence scores	5 days
Step 1+2	Hadley 2018	Retrospective/post-implementation study	South Africa	Nurses, nursing assistants, and counsellor.	Breast	Rural	An improved and computerised breast cancer patient-tracking system and data base as part of an ultrasound-based breast cancer early detection program in a rural community in South Africa.	Improvement in quality of care, follow-up and clinical decision-making	NR
Step 1+2+3	Songiso 2020	Cross-sectional study	Zambia	Women and HCWs	Breast	Urban	Establishment of a district-level breast care specialty clinic for women with breast symptoms. The clinic offers same-day breast self-awareness education, clinical breast examination, breast ultrasound, ultrasound-guided breast biopsy, surgery, referral for treatment and follow-up.	Number of single-visit breats biopsy and cytology performed	NA

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Pinder 2018	Cross-sectional study	Zambia	Women and HCWs in a rural setting	Breast	Rural	An algorithm that compresses the multi-step breast cancer symptom awareness, diagnostic and treatment pathway into a single visit.	Number of single-visit clinical breast examination, ultrasound, biopsy and cytology performed	NA
	Pinder 2017	Cross-sectional study	Zambia	Women and HCWs in a rural setting	Breast	Rural	An initiative that leverages an existing cervical cancer prevention service platform to build facility-level capacity for breast cancer care by raising awareness among women, creating a resource-appropriate breast cancer care training curricula for mid- and high-level providers, and facility-level capacity building for early detection and treatment capacity.	Number of breast biopsies performed	NA
	Yeoh 2018	Before-and-after study	Malaysia	Women	Breast	urban	A patient navigation programme to Improve timely breast cancer diagnosis and care.	Breast biopsy TAT and proportion of women with early tumours	1 year and one month

Abbreviations: CHW: Community Health Workers; HCW: Healthcare Workers; NA: Not applicable; NR: Not reported; RCT: Randomised Clinical Trial; TAT: Turn-around Time

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Only two of the included studies were determined to be of high quality (low risk of bias)[43,49], seven were adjudged to be of moderate quality (moderate risk of bias)[8,40,45,46,51,52,57] while the majority were found to be of low quality (high risk of bias).[41,42,44,47,48,50,53-56,58,59] **Figures 3a and 3b** illustrate the risk of bias assessment outcomes of the included studies.

Interventions targeting Step 1 of the WHO Guide to Cancer Early Diagnosis- promoting awareness and addressing care access delays

Interventions seeking to reduce access delays by promoting awareness, help-seeking and linkage to care were predominantly targeted at women (5),[44-47,54] with a few targeted at adolescent girls (1)[49] and community/lay health workers (2).[8,40]

A cluster randomised trial assessed a self-help intervention for reducing time to diagnosis in Indonesian women with breast cancer symptoms. The intervention consisted of health education and psycho-education using a narrative strategy, which involved the use of testimonials and story-telling.[45] It reduced the time between the first medical consultation and definitive diagnosis among women with breast symptoms by 13.3 days (25.90 in the intervention group vs 39.29 in the control group, P = 0.02). Another RCT by Termeh-Zonouzy evaluated the effectiveness of an educational intervention based on fear appeals using the extended parallel process model (EPPM) to improve attitudes, intention, and timely breast cancer diagnosis in Iran. It found a significant improvement in attitude and intention (*P*=0.01 and *P*=0.001, respectively), but no significant improvement was observed for early breast cancer diagnosis (*P*=0.78).[46]

An RCT by Gadgil and colleagues investigated a programme that aimed to increase awareness on breast cancer and access to timely detection by emailing awareness brochures annually to a cohort of women in an urban occupational setting in India.[44] They found an increase in the proportion of women with early breast tumours and a concomitant increase in the proportion of women receiving breast conserving surgery increased post-intervention when compared with the pre-intervention period increased from (74% to 81% and 39% to 51% respectively). However, there were no outcomes directly related to the timeliness of the diagnoses.

A before-and-after study assessed the effectiveness of decentralising colposcopy services from tertiary-level to primary-level care facility in an urban community in South Africa to aid early definitive diagnosis of cervical cancer.[54] The number of women who had a colposcopy

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

at the primary healthcare facility rose threefold post-decentralisation (from 114 to 350), with an increased proportion of women who had a colposcopy within 3 months of a Pap smear (from 11.8% to 15.4%) and reduced workload at the tertiary facility. A retrospective study compared routine (opportunistic) screening mammogram with targeted (high risk) screening mammogram and diagnostic mammogram among women in Malaysia.[47] It observed cancer detection rates of 0.5 %, 1.25 % and 26 % by opportunistic screening, targeted screening, and diagnostic mammograms, respectively. The proportion of non-invasive cancer (ductal carcinoma in situ) was higher (23.1 %) in the two screening groups compared with only 2.5 % in the diagnostic group, while the proportion of invasive cancer was higher in the diagnostic group. None of these outcomes were directly related to the timeliness of diagnosis.

Of the interventions targeting lay/community health workers, an RCT by Ginsburg et al. assessed the role an mHealth model that used smart phone applications and CHW training as part of a patient navigation programme to address potential barriers to seeking care in a rural community in Bangladesh.[8] The intervention was found to have increased clinic attendance for breast symptoms, improved data quality, higher identification of women with abnormal breast exams and better adherence to diagnostic follow-up. Similarly, another RCT of a CHW-driven population-based case finding programme for breast cancer in a rural setting in Bangladesh found the use of mobile technology as feasible and efficient in active breast cancer case finding in rural settings. Manual data collection and reporting had missing data in 80% of cases, and took an average of 5 minutes longer to collect data, versus no missing data when mobile phones were used for data collection and reporting.[40]

A school-based cervical cancer prevention and early diagnosis education programme targeted at rural teenage high school students Nigeria.[49] Notably, no significant improvements in knowledge parameters (Pap smears, risk factors and early symptoms). Nevertheless, participants who engaged more with the intervention showed significant improvements in knowledge across most parameters of knowledge.

Interventions targeting Step 2 of the WHO Guide to Cancer Early Diagnosis-addressing diagnostic delays by optimising clinical evaluation, diagnosis and staging

An RCT compared diagnostic outcomes between a physician-targeted opportunistic clinic-based breast cancer screening programme and usual care in an urban setting in Colombia. Physicians in the intervention clinics received a 2-day training course on breast-

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cancer epidemiology, clinical signs and symptoms, as well as the principles of mammography and BIRADS (Breast Imaging Reporting and Data System) grading. They were instructed to perform CBE on all eligible women, record results, and refer women with suspicious findings for further diagnostic procedures. It demonstrated a higher proportion of early breast cancer diagnosed with the intervention; 68.2% of the women with breast cancer in the intervention group had breast conservation surgery , compared with 50.0% in the control group.[43]

Another RCT by Pace et al investigated the impact of a health care worker (HCW) training programme on diagnosis and staging to facilitate breast cancer early diagnosis in a rural district of Rwanda.[51] The training consisted of instructions for CHWs in symptoms of breast cancer and messaging that community members should come to their local health centre for any breast symptoms. The intervention led to an increase in diagnostic health facility visits (an increase of 4.7 – 7.9 visits/month in intervention facilities compared with control facilities, P <0.05) and more breast biopsies (36.6 vs 8.9/100,000 person-years, P <0.001) and higher incidence of early-stage breast cancer (3.3 per 100,000 vs 0.7 per 100,000 in control areas, P <.05).

A before-and-after study of a breast ultrasound training programme for non-physician providers with the goal of early diagnosis of breast cancer and down-staging in a rural setting in South Africa demonstrated improved diagnostic competencies (with pre- to post-test averages improvement of 68% in total across four competencies (foundational knowledge, descriptive categories, benign vs malignant, and lesion identification).[52] No outcome directly related to diagnostic timeliness was reported, however. Another before-and-after study studied district-level capacity building training programme targeting primary care providers in Botswana, aiming to enhance timely diagnosis of cancers. It found an overall performance increase (measured by the immediate impact of training on the acquisition of knowledge, attitude and skills in eight sub-domains: pathophysiology, epidemiology, social context, symptoms, evaluation, treatment, documentation and follow-up) of 16.8% after participation across all subdomains.[42]

A prospective cohort study by Ngoma et al. assessed the role of village navigators for case-finding and showed significant down-staging of cancers in a Tanzanian village across 3 successive years ($p < 0.001$).[55] An observational clinical study examined the effectiveness of a programme aiming to improve early detection of breast cancer by strengthening both the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

clinical and quality aspects of the breast imaging practice of oncology centres.[59] The interventions included a shift to image guided core biopsies, implementing multidisciplinary breast meetings, implementing quality assurance procedures, transitioning to digital technologies, facilitating training activities for relevant professionals and improving data collection processes. It was observed that the programme enhanced facility-level breast cancer early diagnostic capacity and increased number of mammography (in 7 out of 10 participating centres) and timely biopsies performed (in 8 out of 10 centres).

A retrospective study assessed the relationship between adequate clinical information (CI) and breast cancer histopathological diagnostic turnaround time (TAT) in Pakistan.[50] It found that breast tissue specimens with deficient clinical information were associated with longer TAT (>80% of the specimens took had TAT of longer than three days vs <50% of specimens with sufficient clinical information). Another retrospective study that assessed a pathology scale-up programme to optimise breast cancer pathological diagnostic TAT in Botswana found that the programme decreased median TAT for biopsy and immunohistochemistry specimens (from 21.5 days to 8 days, before and after the initiation of the programme respectively). However, there was no significant decline in median TAT for surgical specimens.[41]

Intervention targeted at both Steps 1 and 2 of the WHO Guide to Cancer Early Diagnosis- addressing access and diagnostic delays

Hadley and colleagues conducted a retrospective (post-implementation) study to determine the impact of an improved and computerised breast cancer patient-tracking system and database, as part of a nurse-driven and ultrasound-based breast cancer early detection programme in a rural community in South Africa.[53] Most health workers' involved agreed that the intervention can improve early breast cancer detection, patient follow-up, clinical decision making, communication among clinicians. However, a majority (71%) of them reported that the system increased visit time, due to factors such as unreliable internet and inadequate computer workstations.

Interventions targeted at all three steps of the WHO Guide to Cancer Early Diagnosis - addressing access, diagnostic and treatment delays

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

There were interventions targeting all three steps of the WHO Guide. A patient navigation programme to improve timely breast cancer diagnosis, treatment and follow-up for women in Malaysia was found to have increased the proportion of women receiving timely mammography (from 74.4% to 96.4%), biopsy (from 76.1% to 92.5%), and timely communication of diagnostic results (from 58.5% to 80.0%).[48] Other interventions included the establishment of a district-level breast care specialty clinic that offers same-day breast self-awareness education, clinical breast examination, breast ultrasound, ultrasound-guided breast biopsy, surgery, referral for treatment and follow-up for women with breast symptoms in Zambia[58]; an algorithm that compresses the multi-step breast cancer symptom awareness, diagnostic and treatment pathway into a single clinic visit[56]; and an initiative that leverages an existing cervical cancer prevention service platform to build facility-level capacity for breast cancer care by raising awareness among women, creating a resource-appropriate training curricula for mid- and high-level providers for early detection and treatment.[57] Due to the cross-sectional nature of the studies reporting these interventions, they could only demonstrate the feasibility of the interventions within resource-limited contexts, and not their effectiveness.

DISCUSSION

This systematic review highlights the diverse nature and effectiveness of interventions for the early diagnosis of symptomatic breast and cervical cancers, from 21 studies set across 20 LMICs. In spite of the limited number and methodological limitations of the studies included, this review has identified a substantial body of evidence suggesting that interventions and strategies targeting the three essential steps of the WHO operational framework for early cancer diagnosis may be feasible and effective in the context of breast and cervical cancer in LMICs. It is noteworthy that, while most of interventions were reported to be feasible and effective for improving diagnostic processes, many of their reported outcome measures are not directly related to diagnostic timeliness. Although a variety of interventions have been studied, the majority of them were targeted at improving the diagnostic capacity of health care workers (including physicians, nurses, pathologists, non-physician health providers and community health workers) through training, knowledge-acquisition, upskilling and better equipment.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Training interventions aimed at improving diagnostic knowledge and skills of health care workers were shown to improve diagnostic turnaround time[41] and increase detection rates[43,51] and downstaging.[52] The use of community-based patient navigation programmes driven by trained community- or lay-health workers for breast examination, specimen collection, pathological diagnosis and follow-up was also shown to improve early diagnostic outcomes.[8,40] Findings are consistent with those of previous reviews.[6,11]

Similarly, evidence suggests that the use of health education and awareness promotion campaigns targeted at women may be effective in improving breast cancer knowledge, attitude and help-seeking behaviour. These could include culturally-sensitive and context-appropriate strategies such a self-help intervention for reducing time to diagnosis in women with breast cancer symptoms, consisting of health education and psycho-education through narratives, testimonials and story-telling.[45] Nevertheless, a school-based cervical cancer prevention and early diagnosis education programme targeting adolescent high schoolers yielded no significant improvements in the participants' overall knowledge.[49] This may reflect the vital influence of intervention-related and contextual factors on the effectiveness of education and awareness promotion interventions that need to be taken into account when designing such interventions.

Interventions addressing structural barriers to diagnostic services and appropriate healthcare, such as proactive home-visits, patient-navigation programmes and decentralisation of diagnostic services from tertiary-level to primary-level care facilities, have shown promising results in reducing diagnostic turn-around time and cancer down-staging.[41,43,54,55] While it is difficult to draw conclusions from these studies owing to their methodological limitations and potential publication bias, these findings reflect the important role of interventions targeting health system structural impediments to early cancer diagnosis in LMICs.

Evidence suggests that the use of technological interventions, such as computerised breast cancer patient-tracking systems for improving early breast cancer detection, patient follow-up and clinical decision making, may be feasible even in rural settings.[53] This demonstrates the potential for the adoption and scale up of innovative early cancer diagnosis interventions in low-resource settings. However, the effectiveness of such interventions depends largely on an adequate supply of the needed infrastructure and operational readiness, without which

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

interventions may be ineffectiveness or counter-productive. An indication of this is the report of unreliable internet service and inadequate computer workstations as factors constraining the use of a computerised breast cancer patient-tracking system and leading to increased patient visit or consultation time.[53]

Our review shows a dearth of studies evaluating the effectiveness of cervical cancer early diagnosis interventions for symptomatic women, despite the substantial burden of cervical cancer and late-stage diagnosis in LMICs.[1-5] This trend may be as a result of the lingering stigmatisation of cervical cancer in many LMIC settings or may reflect the fact that women with cervical cancer are likely to be at an advanced stage at the time of diagnosis in such settings.[61] Increased investments are needed to address current cervical cancer programme evaluation and research gaps, such as through improved research funding and research capacity building; and the integration of research activities within cancer programme implementation plans. This also underscores the need for greater public health education and community awareness efforts, to mitigate existing stigma and misconceptions associated with cervical cancer that may pose barriers to cervical cancer diagnostic service delivery and research.

The use of varied outcome measures in assessing the effectiveness of interventions across studies complicated the interpretation, aggregation and comparability of the findings. Many outcome measures reported have very limited clinical relevance to the timeliness of breast or cervical cancer diagnosis. While the most important outcome of early cancer diagnosis programmes would be to reduce cancer mortality, evaluating this outcome does require more rigourous study designs such as randomised clinical trials with long follow-up periods, the cost of which may be prohibitive in low-resource settings. Hence, it is imperative for future studies in LMICs to explore the assessment of more clinically relevant proxy outcomes such as reduced time to diagnosis, improved pathology TAT and down-staging. The Anderson model, Model of Pathways to Treatment and Aarhus Statement are some of the useful tools for assessing diagnostic time-related outcomes of early cancer diagnosis interventions in standardised ways.[21,22]

Overall, given the rising burden of breast and cervical cancer in LMICs, the role of strategies and interventions for improving early diagnosis is relatively underexplored. While more interventions are needed, those that target both the general public and health care workers

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

and the whole continuum of care may be more cost-effective, particularly for resource-limited settings. To enhance the feasibility and usefulness of intervention studies for generating contextually-relevant evidence for improving early diagnosis, it is imperative that outcome evaluation activities are embedded as an integral component of early cancer diagnosis programmes. This can be achieved though mixed-methods implementation research strategies to evaluate and support the application of research findings into policy and practice.[62] It is noteworthy that the evidence included in this review relates to the period before the COVID-19 pandemic which has disrupted essential health services, including early cancer diagnosis programmes across the world.[9] For this reason, it is necessary for future research to investigate the short- and long-term impacts of the pandemic on the effectiveness of early diagnosis interventions, as movement restrictions are eased and cancer programmes return to pre-pandemic levels of operation.

Limitations

Our review has some limitations worthy of note. First, most of the included studies assessed intervention outcomes using observational, non-experimental and non-standardised methods, making them prone to recall bias, selection bias, uncontrolled confounding, and limited external validity. Secondly, as with other reviews, our study may be susceptible to publication bias in the reporting of study findings, in which positive findings have a higher likelihood of being reported. This may have been refleced in our finding of very limited evidence of non-effectiveness of interventions. Thirdly, only a few of the studies measured intervention outcomes beyond a duration of one year, hence limiting the determination of the long-term impact of reviewed interventions. Furthermore, although we searched a wide array of literature sources, it is possible that we missed some relevant studies. Lastly, our study excluded interventions that focused primarily on screening of asymptomatic individuals for reasons earlier alluded to. Hence, it is possible that some the screening studies excluded had early diagnosis components.

CONCLUSION

Interventions aimed at improving timely diagnosis and cancer outcomes are feasible and can be effective for improving timely diagnosis of breast and cervical cancers in resource-limited LMIC contexts. To better generate translatable evidence for improving early cancer diagnosis,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

our review underscores the pressing need for future studies to assess interventions using standardised methods, while reporting clinically relevant diagnostic timeliness outcomes. It is also imperative that early diagnosis interventions to adopt contextually appropriate and cost-effective approaches that are well embedded within the broader health system.

Patient and public involvement: This is a review of publicly available literature. Patients and the public were not directly involved in the design, conduct, reporting or dissemination this study.

Authors' contributions: Conceptualisation: JM. Protocol development: CAN, PK, FMW and JM. Duplicate screening, study selection, quality assessment and data collection: CAN and PK. Data analysis and interpretation: CAN, PK, FMW and JM. First draft preparation: CAN. Provision of critical insights and refinement of the manuscript: FMW and JM. All authors have read and approved the final version of the manuscript.

Funding: This research was jointly supported by the Cancer Association of South Africa (CANSA), the University of Cape Town and, the SA Medical Research Council with funds received from the SA National Department of Health, GlaxoSmithKline Africa Non-Communicable Disease Open Lab (via a supporting grant Project Number: 023), the UK Medical Research Council (via the Newton Fund). Authors retained control of the final content of the publication. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Acknowledgments: The authors wish to thank Namhla Madini of the Faculty of Health Sciences Library, University of Cape Town, for the support and guidance in developing the literature search strategy. FMW acknowledges the support of the CanTest Collaborative (funded by Cancer Research UK C8640/A23385) as its Director.

Patient consent for publication: Not required.

Ethics approval: This is a systematic review of publicly available literature, with no primary data collection. Hence, it did not require ethical approval.

Competing interests: None declared.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Provenance and peer review: Not commissioned; externally peer reviewed.

References

1. Global Burden of Disease Cancer Collaborators. Global, Regional, and National Cancer Incidence, Mortality, Years of Life Lost, Years Lived With Disability, and Disability-Adjusted Life-Years for 29 Cancer Groups, 1990 to 2017: A Systematic Analysis for the Global Burden of Disease Study. JAMA oncology 2019; 5:1749-68.
2. de Sanjose S, Tsu VD: Prevention of cervical and breast cancer mortality in low- and middle-income countries: a window of opportunity. International journal of women's health 2019; 11:381-6.
3. Demment MM, Peters K, Dykens JA, et al.: Developing the Evidence Base to Inform Best Practice: A Scoping Study of Breast and Cervical Cancer Reviews in Low- and Middle-Income Countries. PloS one 2015; 10:e0134618.
4. Bray F, Ferlay J, Soerjomataram I, et al.: Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: a cancer journal for clinicians 2018; 68:394-424.
5. GLOBOCAN. Cancer Today 2018. http://gco.iarc.fr/today/home. Accessed 26 March 2021.
6. Dalton M, Holzman E, Erwin E, et al.: Patient navigation services for cancer care in low-and middle-income countries: A scoping review. PloS one 2019; 14:e0223537-e.
7. Tegegne TK, Chojenta C, Loxton D, et al.: The impact of geographic access on institutional delivery care use in low and middle-income countries: Systematic review and meta-analysis. PloS one 2018; 13:e0203130.
8. Ginsburg OM, Chowdhury M, Wu W, et al.: An mHealth Model to Increase Clinic Attendance for Breast Symptoms in Rural Bangladesh: Can Bridging the Digital Divide Help Close the Cancer Divide? The Oncologist 2014; 19:177-85.
9. Nnaji CA, Moodley J: Impact of the COVID-19 pandemic on cancer diagnosis, treatment and research in African health systems: a review of current evidence and contextual perspectives. Ecancermedicalscience 2021; 15:1170.
10. Brand NR, Qu LG, Chao A, et al.: Delays and Barriers to Cancer Care in Low- and Middle-Income Countries: A Systematic Review. Oncologist 2019; 24:e1371-e80.
11. Qu LG, Brand NR, Chao A, et al.: Interventions Addressing Barriers to Delayed Cancer Diagnosis in Low- and Middle-Income Countries: A Systematic Review. Oncologist 2020.
12. Richards MA, Westcombe AM, Love SB, et al.: Influence of delay on survival in patients with breast cancer: a systematic review. Lancet 1999; 353:1119-26.
13. Jedy-Agba E, McCormack V, Adebamowo C, et al.: Stage at diagnosis of breast cancer in sub-Saharan Africa: a systematic review and meta-analysis. The Lancet Global health 2016; 4:e923-e35.
14. Islam RM, Billah B, Hossain MN, et al.: Barriers to Cervical Cancer and Breast Cancer Screening Uptake in Low-Income and Middle-Income Countries: A Systematic Review. Asian Pacific journal of cancer prevention : APJCP 2017; 18:1751-63.
15. Cazap E, Magrath I, Kingham TP, et al.: Structural Barriers to Diagnosis and Treatment of Cancer in Low- and Middle-Income Countries: The Urgent Need for Scaling Up. J Clin Oncol 2016; 34:14-9.
16. Moodley J, Walter FM, Scott SE, et al.: Towards timely diagnosis of symptomatic breast and cervical cancer in South Africa. South African medical journal = Suid-Afrikaanse tydskrif vir geneeskunde 2018; 108:803-4.
17. Moodley J, Cairncross L, Naiker T, et al.: Understanding pathways to breast cancer diagnosis among women in the Western Cape Province, South Africa: a qualitative study. BMJ Open 2016; 6:e009905.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

18. Poum A, Promthet S, Duffy SW, et al.: Factors associated with delayed diagnosis of breast cancer in northeast Thailand. Journal of epidemiology 2014; 24:102-8.
19. Tetteh DA, Faulkner SL: Sociocultural factors and breast cancer in sub-Saharan Africa: implications for diagnosis and management. Womens Health (Lond) 2016; 12:147-56.
20. Moodley J, Cairncross L, Naiker T, et al.: From symptom discovery to treatment - women's pathways to breast cancer care: a cross-sectional study. BMC cancer 2018; 18:312.
21. Scott SE, Walter FM, Webster A, et al.: The model of pathways to treatment: conceptualization and integration with existing theory. British journal of health psychology 2013; 18:45-65.
22. Walter F, Webster A, Scott S, et al.: The Andersen Model of Total Patient Delay: A Systematic Review of Its Application in Cancer Diagnosis. J Health Serv Res Policy 2012; 17:110-8.
23. Weller D, Vedsted P, Rubin G, et al.: The Aarhus statement: improving design and reporting of studies on early cancer diagnosis. Br J Cancer 2012; 106:1262-7.
24. World Health Organization (WHO). Guide To Cancer Early Diagnosis. Geneva, Switzerland: World Health Organization, 2017. Available from: http://apps.who.int/iris/bitstream/10665/254500/1/9789241511940-eng.pdf?ua=1. Accessed 10 March, 2021.
25. World Health Organization (WHO). WHO report on cancer: setting priorities, investing wisely and providing care for all. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.
26. Wells KJ, Battaglia TA, Dudley DJ, et al.: Patient navigation: State of the art or is it science? Cancer 2008; 113:1999-2010.
27. Valaitis RK, Carter N, Lam A, et al.: Implementation and maintenance of patient navigation programs linking primary care with community-based health and social services: a scoping literature review. BMC Health Serv Res 2017; 17:116.
28. Battaglia TA, Roloff K, Posner MA, et al.: Improving follow-up to abnormal breast cancer screening in an urban population. A patient navigation intervention. Cancer 2007; 109:359-67.
29. Drake BF, Tannan S, Anwuri VV, et al.: A Community-Based Partnership to Successfully Implement and Maintain a Breast Health Navigation Program. Journal of community health 2015; 40:1216-23.
30. Neal RD, Tharmanathan P, France B, et al.: Is increased time to diagnosis and treatment in symptomatic cancer associated with poorer outcomes? Systematic review. Br J Cancer 2015; 112 Suppl 1:S92-S107.
31. Liberati A, Altman DG, Tetzlaff J, et al.: The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. Annals of internal medicine 2009; 151:W65-94.
32. Booth A, Clarke M, Dooley G, et al.: The nuts and bolts of PROSPERO: an international prospective register of systematic reviews. Syst Rev 2012; 1:2.
33. Shamseer L, Moher D, Clarke M, et al.: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ (Clinical research ed) 2015; 350:g7647.
34. Nnaji CA, Kuodi P, Walter FM, et al.: Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low and middle-income countries: a systematic review protocol. BMJ Open 2020; 10:e042788.
35. Fantom N, Serajuddin U. The World Bank's Classification of Countries by Income (English). Policy Research Working Paper no. WPS 7528. Washington, DC: World Bank Group, 2016.
36. Carlos LL-J, Cruz LAPd, Leopoldo VC, et al.: Effectiveness of Traditional Chinese Acupuncture versus Sham Acupuncture: a Systematic Review. Rev Lat Am Enfermagem 2016; 24:e2762-e.
37. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.0 (updated July 2019). Cochrane,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2019. Available from https://training.cochrane.org/handbook/current/chapter-08. Accessed 27 March 2020.
38. Sterne JA, Hernan MA, Reeves BC, et al.: ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ (Clinical research ed) 2016; 355:i4919.
39. Guyatt GH, Oxman AD, Kunz R, et al.: What is "quality of evidence" and why is it important to clinicians? BMJ (Clinical research ed) 2008; 336:995-8.
40. Chowdhury TI, Love RR, Chowdhury MT, et al.: Feasibility Study of Case-Finding for Breast Cancer by Community Health Workers in Rural Bangladesh. Asian Pacific journal of cancer prevention : APJCP 2015; 16:7853-7.
41. Martei YM, Narasimhamurthy M, Prabhakar P, et al.: Breast Cancer Pathology Turnaround Time in Botswana. Journal of global oncology 2018; 4:1-7.
42. Tapela NM, Peluso MJ, Kohler RE, et al.: A Step Toward Timely Referral and Early Diagnosis of Cancer: Implementation and Impact on Knowledge of a Primary Care-Based Training Program in Botswana. Frontiers in oncology 2018; 8:187.
43. Murillo R, Díaz S, Perry F, et al.: Increased breast cancer screening and downstaging in Colombian women: A randomized trial of opportunistic breast-screening. International journal of cancer 2016; 138:705-13.
44. Gadgil A, Sauvaget C, Roy N, et al.: Cancer early detection program based on awareness and clinical breast examination: Interim results from an urban community in Mumbai, India. Breast (Edinburgh, Scotland) 2017; 31:85-9.
45. Setyowibowo H, Hunfeld JAM, Iskandarsyah A, et al.: A self-help intervention for reducing time to diagnosis in Indonesian women with breast cancer symptoms. Psycho-oncology 2020; 29:696-702.
46. Termeh Zonouzy V, Niknami S, Ghofranipour F, et al.: An educational intervention based on the extended parallel process model to improve attitude, behavioral intention, and early breast cancer diagnosis: a randomized trial. International journal of women's health 2019; 11:1-10.
47. Teh YC, Tan GH, Taib NA, et al.: Opportunistic mammography screening provides effective detection rates in a limited resource healthcare system. BMC cancer 2015; 15:405.
48. Yeoh ZY, Jaganathan M, Rajaram N, et al.: Feasibility of Patient Navigation to Improve Breast Cancer Care in Malaysia. Journal of global oncology 2018; 4:1-13.
49. Ifediora CO, Azuike EC: Targeting cervical cancer campaigns on teenage high schoolers in resource-limited economies: lessons from an intervention study of Nigerian senior secondary school girls. Family practice 2019; 36:600-6.
50. Ali SMH, Kathia UM, Gondal MUM, et al.: Impact of Clinical Information on the Turnaround Time in Surgical Histopathology: A Retrospective Study. Cureus 2018; 10:e2596.
51. Pace LE, Dusengimana JMV, Shulman LN, et al.: Cluster Randomized Trial to Facilitate Breast Cancer Early Diagnosis in a Rural District of Rwanda. Journal of global oncology 2019; 5:1-13.
52. Dickerson LK, Rositch AF, Lucas S, et al.: Pilot Educational Intervention and Feasibility Assessment of Breast Ultrasound in Rural South Africa. Journal of global oncology 2017; 3:502-8.
53. Hadley M, Mullen LA, Dickerson L, et al.: Assessment and Improvement Strategies for a Breast Cancer Early Detection Program in Rural South Africa. Journal of global oncology 2018; 4:1-12.
54. Maimela G, Nene X, Mvundla N, et al.: The impact of decentralising colposcopy services from tertiary-level to primary-level care in inner-city Johannesburg, South Africa: a before and after study. BMJ Open 2019; 9:e024726.
55. Ngoma T, Mandeli J, Holland JF: Downstaging cancer in rural Africa. International journal of cancer 2015; 136:2875-9.
56. Pinder LF, Nzayisenga JB, Shibemba A, et al.: Demonstration of an algorithm to overcome health system-related barriers to timely diagnosis of breast diseases in rural Zambia. PloS one 2018; 13:e0196985.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

57. Pinder LF, Henry-Tillman R, Linyama D, et al.: Leverage of an Existing Cervical Cancer Prevention Service Platform to Initiate Breast Cancer Control Services in Zambia: Experiences and Early Outcomes. Journal of global oncology 2018; 4:1-8.
58. Songiso M, Pinder LF, Munalula J, et al.: Minimizing Delays in the Breast Cancer Pathway by Integrating Breast Specialty Care Services at the Primary Health Care Level in Zambia. JCO global oncology 2020; 6:859-65.
59. Aribal E, Mora P, Chaturvedi AK, et al.: Improvement of early detection of breast cancer through collaborative multi-country efforts: Observational clinical study. European journal of radiology 2019; 115:31-8.
60. Martei YM, Pace LE, Brock JE, et al.: Breast Cancer in Low- and Middle-Income Countries: Why We Need Pathology Capability to Solve This Challenge. Clinics in laboratory medicine 2018; 38:161-73.
61. Pierz AJ, Randall TC, Castle PE, et al.: A scoping review: Facilitators and barriers of cervical cancer screening and early diagnosis of breast cancer in Sub-Saharan African health settings. Gynecologic oncology reports 2020; 33:100605-.
62. Murillo R, Robles C: Research Needs for Implementing Cancer Prevention and Early Detection in Developing Countries: From Scientists' to Implementers' Perspectives. Biomed Res Int 2019; 2019:9607803.

Figure 1: PRISMA flow diagram of study selection process

Figure 2: Map of geographical distribution of included studies

Figures 3a and 3b: Risk of bias assessment outcomes geographical distribution of the studies

Appendix 1: Search strategies

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

FIGURE 1

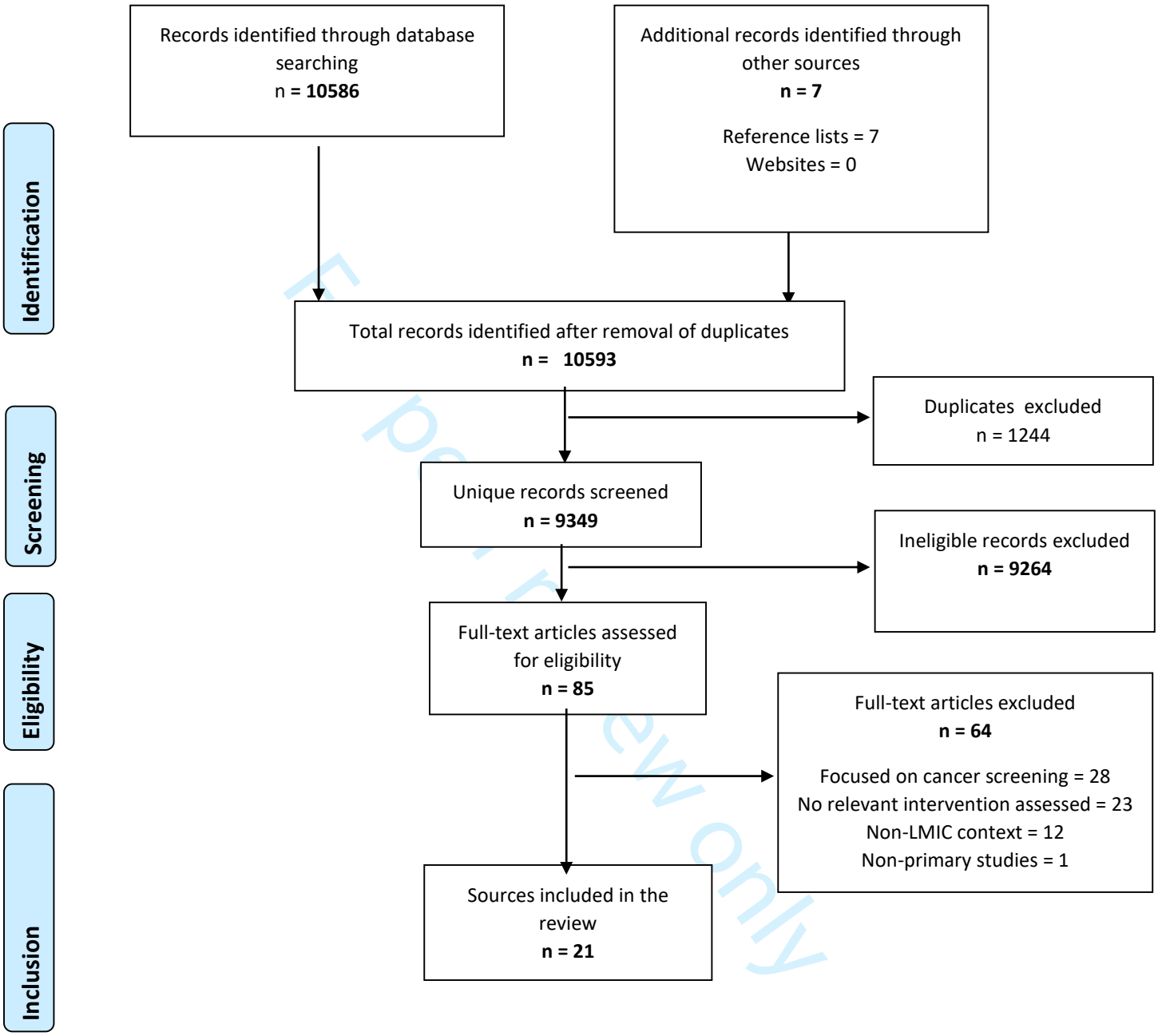


Figure 1: PRISMA flow diagram showing study selection process

For peer review only

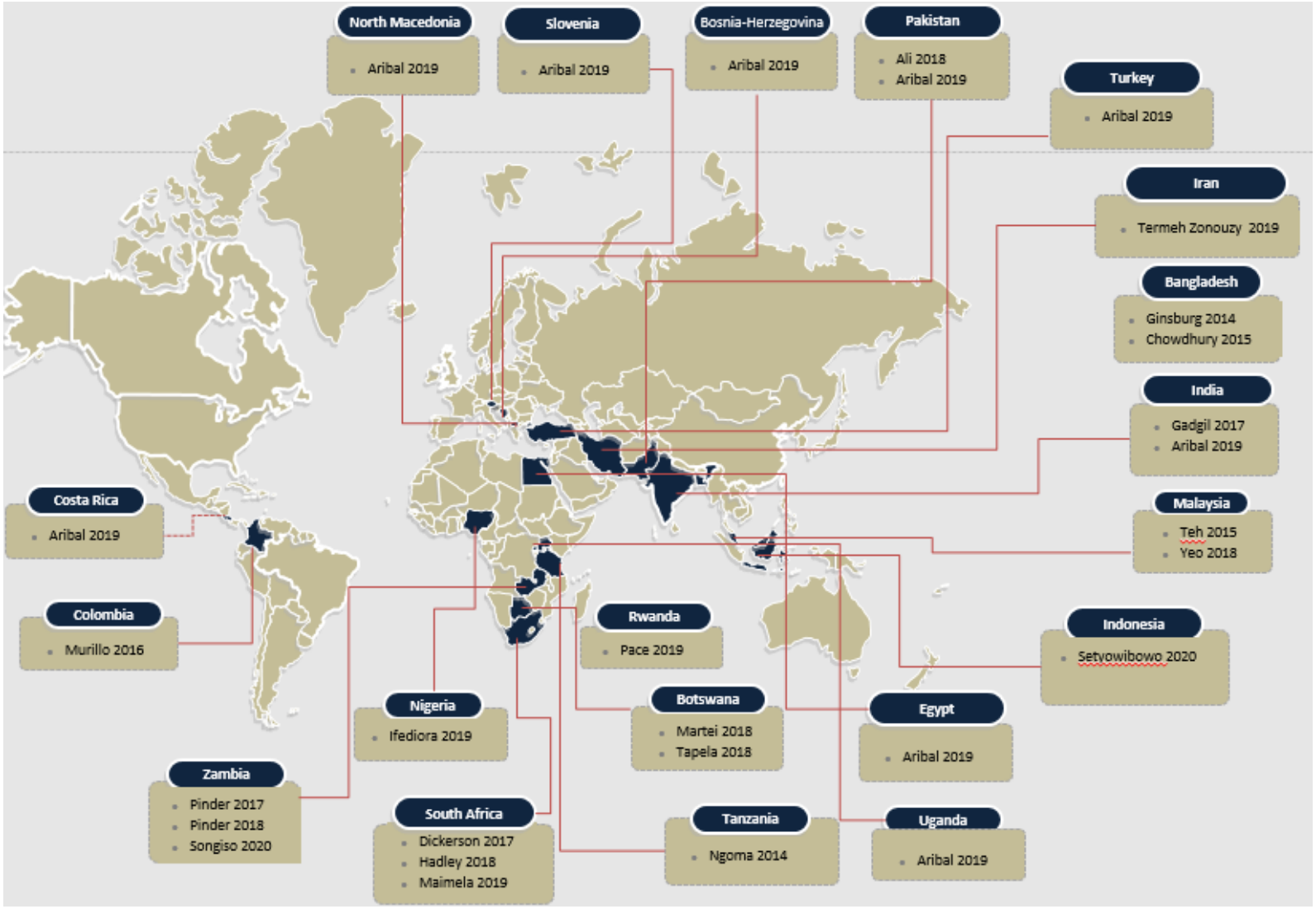


Figure 2: Geographical distribution of included studies (map created by authors using an open-source template sourced from https://yourfreetemplates.com/)

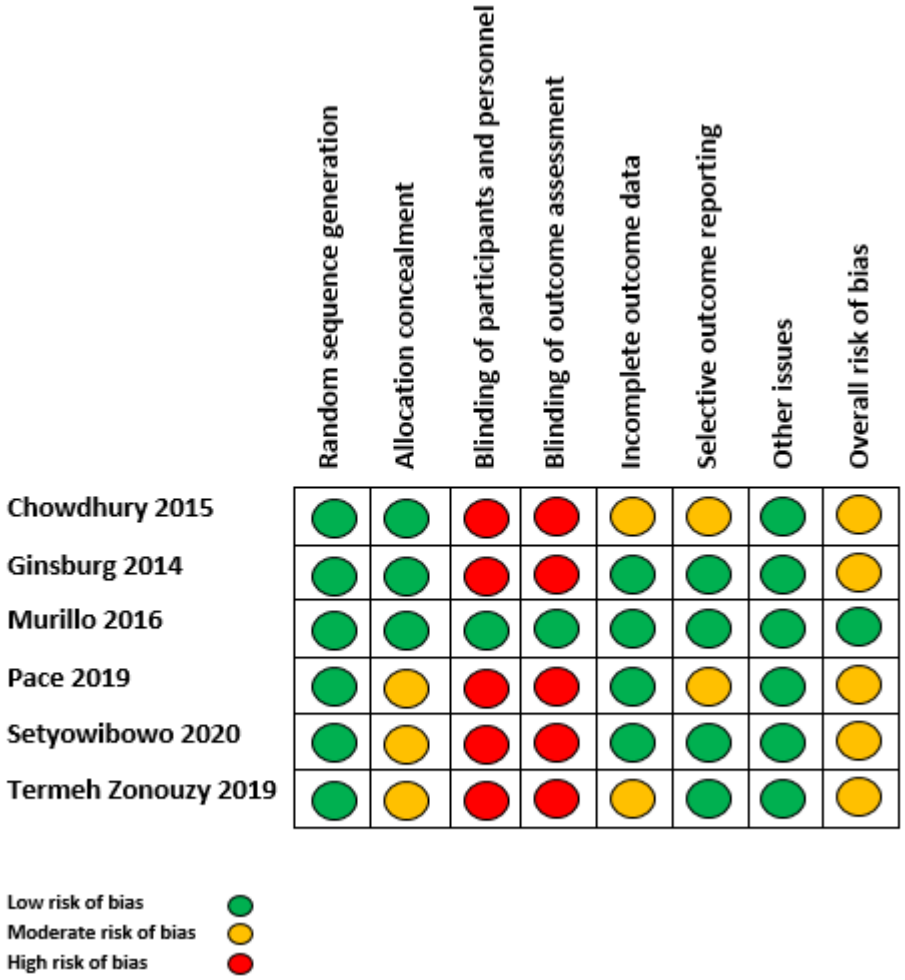


Figure 3a: Risk of bias assessment of randomized controlled trials

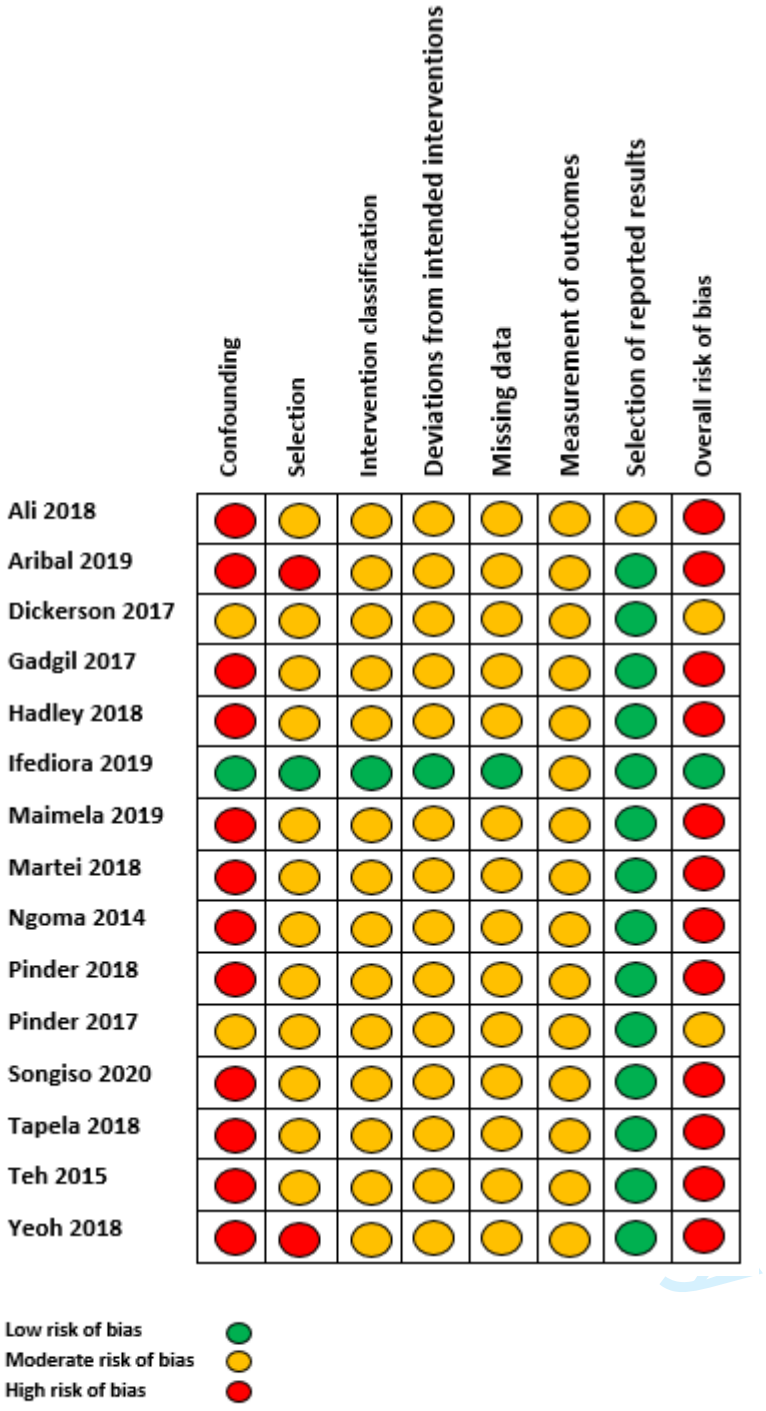


Figure 3b: Risk of bias assessment of observational studies

Appendix: Search strategies

PubMed/MEDLINE search strategy

Search #	Search Texts and Syntaxes
#1	Breast Neoplasms [Mesh] OR Breast cancer[Title/Abstract] OR breast[Title/Abstract] OR cervical cancer[Title/Abstract] OR cervix[Title/Abstract] OR cancer[Title/Abstract] OR malignant[Title/Abstract] OR neoplasm[Title/Abstract] OR neoplasia[Title/Abstract] OR malignancy[Title/Abstract] OR tumour[Title/Abstract]
#2	Diagnosis [Mesh] OR diagnostic[Title/Abstract] OR diagnosis[Title/Abstract] OR diagnosis[Title/Abstract] OR detection[Title/Abstract] OR discovery[Title/Abstract] OR Patient navigation[Title/Abstract] OR Patient pathway[Title/Abstract] OR care continuum[Title/Abstract]
#3	early[Title/Abstract] OR timely[Title/Abstract] OR time[Title/Abstract] OR late[Title/Abstract] OR delay[Title/Abstract]
#4	#2 AND #3
#5	Developing Countries OR Developing Country OR Developing Economies OR Developing Economy OR Developing Nation OR Developing Nations OR Developing Population OR Developing Populations OR Developing World OR LAMI Countries OR LAMI Country OR Less Developed Countries OR Less Developed Country OR Less Developed Economies OR Less Developed Nation OR Less Developed Nations OR Less Developed World OR Lesser Developed Countries OR Lesser Developed Nations OR LMIC OR LMICS OR Low GDP OR Low GNP OR Low Gross Domestic OR Low Gross National OR Low Income OR Lower GDP OR lower gross domestic OR Lower Income OR Middle Income OR Poor Countries OR Poor Country OR Poor Economies OR Poor Economy OR Poor Nation OR Poor Nations OR Poor Population OR Poor Populations OR poor world OR Poorer Countries OR Poorer Economies OR Poorer Economy OR Poorer Nations OR Poorer Population OR Poorer Populations OR Third World OR Transitional Countries OR Transitional Country OR Transitional Economies OR Transitional Economy OR Under Developed Countries OR Under Developed Country OR under developed nations OR Under Developed World OR Under Served Population OR Under Served Populations OR Underdeveloped Countries OR Underdeveloped Country OR underdeveloped economies OR underdeveloped nations OR underdeveloped population OR Underdeveloped World OR Underserved Countries OR Underserved Nations OR Underserved Population OR Underserved Populations OR Afghanistan OR Albania OR Algeria OR American Samoa OR Angola OR Armenia OR Azerbaijan OR Bangladesh OR Belarus OR Byelarus OR Belorussia OR Belize OR Benin OR Bhutan OR Bolivia OR Bosnia OR Botswana OR Brazil OR Bulgaria OR Burma OR Burkina Faso OR Burundi OR Cabo Verde OR Cape Verde OR Cambodia OR

For peer review only

	Cameroon OR Central African Republic OR Chad OR China OR Colombia OR Comoros OR Comores OR Comoro OR Congo OR Costa Rica OR Côte d'Ivoire OR Cuba OR Djibouti OR Dominica OR Dominican Republic OR Ecuador OR Egypt OR El Salvador OR Equatorial Guinea OR Eritrea OR Ethiopia OR Fiji OR Gabon OR Gambia OR Gaza OR Georgia OR Georgia Republic OR Ghana OR Grenada OR Grenadines OR Guatemala OR Guinea OR Guinea- Bissau OR Guyana OR Haiti OR Herzegovina OR Hercegovina OR Honduras OR India OR Indonesia OR Iran OR Iraq OR Ivory Coast OR Jamaica OR Jordan OR Kazakhstan OR Kenya OR Kiribati OR Democratic People's Republic of Korea OR Kosovo OR Kyrgyz OR Kirghizia OR Kirghiz OR Kyrgyzstan OR Lao PDR OR Laos OR Lebanon OR Lesotho OR Liberia OR Libya OR Macedonia OR Madagascar OR Malawi OR Malay OR Malaya OR Malaysia OR Maldives OR Mali OR Marshall Islands OR Mauritania OR Mauritius OR Mexico OR Micronesia OR Moldova OR Mongolia OR Montenegro OR Morocco OR Mozambique OR Myanmar OR Namibia OR Nepal OR Nicaragua OR Niger OR Nigeria OR Pakistan OR Palau OR Papua New Guinea OR Paraguay OR Peru OR Philippines OR Principe OR Romania OR Ruanda OR Rwanda OR Samoa OR Sao Tome OR Senegal OR Serbia OR Sierra Leone OR Solomon Islands OR Somalia OR South Africa OR South Sudan OR Sri Lanka OR St Lucia OR St Vincent OR Sudan OR Surinam OR Suriname OR Swaziland OR Syria OR Syrian Arab Republic OR Tajikistan OR Tadzhikistan OR Tajikistan OR Tadzhik OR Tanzania OR Thailand OR Timor OR Togo OR Tonga OR Tunisia OR Turkey OR Turkmen OR Turkmenistan OR Tuvalu OR Uganda OR Ukraine OR Uzbek OR Uzbekistan OR Vanuatu OR Venezuela OR Vietnam OR West Bank OR Yemen OR Zambia OR Zimbabwe
#6	#1 AND #4 AND #5

For peer review only

Scopus

Search #	Search Texts and Syntaxes
#1	TITLE-ABS-KEY (Breast cancer) OR breast OR (cervical cancer) OR cervix OR cancer OR malignant OR neoplasm OR neoplasia OR malignancy OR tumour
#2	TITLE-ABS-KEY Diagnosis OR diagnostic OR diagnosis OR diagnosis OR detection OR discovery OR (Patient navigation) OR (Patient pathway) OR (care continuum)
#3	Early OR timely OR time OR late OR delay
#4	#2 AND #3
#5	TITLE-ABS-KEY “Deprived Countries” OR “Deprived Population” OR “Deprived Populations” OR “Developing Countries” OR “Developing Country” OR “Developing Economies” OR “Developing Economy” OR “Developing Nation” OR “Developing Nations” OR “Developing Population” OR “Developing Populations” OR “Developing World” OR “LAMI Countries” OR “LAMI Country” OR “Less Developed Countries” OR “Less Developed Country” OR “Less Developed Economies” OR “Less Developed Nation” OR “Less Developed Nations” OR “Less Developed World” OR “Lesser Developed Countries” OR “Lesser Developed Nations” OR LMIC OR LMICS OR “Low GDP” OR “Low GNP” OR “Low Gross Domestic” OR “Low Gross National” OR “Low Income” OR “Lower income” OR “Lower GDP” OR “Lower Gross Domestic” OR “Middle Income” OR “Poor Countries” OR “Poor Country” OR “Poor Economies” OR “Poor Economy” OR “Poor Nation” OR “Poor Nations” OR “Poor Population” OR “Poor Populations” OR “poor world” OR “Poorer Countries” OR “Poorer Economies” OR “Poorer Economy” OR “Poorer Nations” OR “Poorer Population” OR “Poorer Populations” OR “Third World” OR “Transitional Countries” OR “Transitional Country” OR “Transitional Economies” OR “Transitional Economy” OR “Under Developed” OR “Under Served” OR “Underdeveloped Countries” OR “Underdeveloped Country” OR “underdeveloped economies” OR “underdeveloped nations” OR “underdeveloped population” OR “Underdeveloped World” OR “Underserved Countries” OR “Underserved Nations” OR “Underserved Population” OR “Underserved Populations” OR Afghanistan OR Albania OR Algeria OR “American Samoa” OR Angola OR Armenia OR Azerbaijan OR Bangladesh OR Belarus OR Byelarus OR Belorussia OR Belize OR Benin OR Bhutan OR Bolivia OR Bosnia OR Botswana OR Brazil OR Bulgaria OR Burma

For peer review only

	OR “Burkina Faso” OR Burundi OR “Cabo Verde” OR “Cape Verde” OR Cambodia OR Cameroon OR “Central African Republic” OR Chad OR China OR Colombia OR Comoros OR Comores OR Comoro OR Congo OR “Costa Rica” OR “Côte d'Ivoire” OR Cuba OR “Democratic People’s Republic of Korea” OR Djibouti OR Dominica OR “Dominican Republic” OR Ecuador OR Egypt OR “El Salvador” OR Eritrea OR Ethiopia OR “Equatorial Guinea” OR Fiji OR Gabon OR Gambia OR Gaza OR “Georgia Republic” OR Georgia OR Ghana OR Grenada OR Grenadines OR Guatemala OR Guinea OR “Guinea Bissau” OR Guyana OR Haiti OR Herzegovina OR Hercegovina OR Honduras OR India OR Indonesia OR Iran OR Iraq OR “Ivory Coast” OR Jamaica OR Jordan OR Kazakhstan OR Kenya OR Kiribati OR Korea OR Kosovo OR Kyrgyz OR Kirghizia OR Kirghiz OR Kyrgyzstan OR “Lao PDR” OR Laos OR Lebanon OR Lesotho OR Liberia OR Libya OR Macedonia OR Madagascar OR Malawi OR Malay OR Malaya OR Malaysia OR Maldives OR Mali OR “Marshall Islands” OR Mauritania OR Mauritius OR Mexico OR Micronesia OR Moldova OR Mongolia OR Montenegro OR Morocco OR Mozambique OR Myanmar OR Namibia OR Nepal OR Nicaragua OR Niger OR Nigeria OR Pakistan OR Palau OR Papua New Guinea OR Paraguay OR Peru OR Philippines OR Principe OR Romania OR Rwanda OR Ruanda OR Samoa OR “Sao Tome” OR Senegal OR Serbia OR “Sierra Leone” OR “Solomon Islands” OR Somalia OR “South Africa” OR “South Sudan” OR “Sri Lanka” OR “St Lucia” OR “St Vincent” OR Sudan OR Surinam OR Suriname OR Swaziland OR Syria OR “Syrian Arab Republic” OR Tajikistan OR Tadzhikistan OR Tajikistan OR Tadzhik OR Tanzania OR Thailand OR Timor OR Togo OR Tonga OR Tunisia OR Turkey OR Turkmen OR Turkmenistan OR Tuvalu OR Uganda OR Ukraine OR Uzbek OR Uzbekistan OR Vanuatu OR Venezuela OR Vietnam OR “West Bank” OR Yemen OR Zambia OR Zimbabwe
#6	#1 AND #4 AND #5

For peer review only

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7-8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7-8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	8

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	9
Study characteristics	17	Cite each included study and present its characteristics.	9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	10
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	10-18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	10-18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	18
	23b	Discuss any limitations of the evidence included in the review.	21
	23c	Discuss any limitations of the review processes used.	21
	23d	Discuss implications of the results for practice, policy, and future research.	21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	22
Competing interests	26	Declare any competing interests of review authors.	22
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	7

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi:



PRISMA 2020 Checklist

10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only

BMJ Open

Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054501.R2
Article Type:	Original research
Date Submitted by the Author:	24-Mar-2022
Complete List of Authors:	Nnaji, Chukwudi; University of Cape Town Faculty of Health Sciences, Women's Health Research Unit, School of Public Health and Family Medicine; University of Cape Town Faculty of Health Sciences, Cancer Research Initiative Kuodi, Paul; Lira University, Department of Public Health, Faculty of Health Sciences Walter, Fiona; University of Cambridge, Dept of Public Health and Primary Care; Queen Mary University of London Barts and The London School of Medicine and Dentistry, Wolfson Institute of Population Health Moodley, Jennifer; University of Cape Town Faculty of Health Sciences, Women's Health Research Unit, School of Public Health and Family Medicine; University of Cape Town Faculty of Health Sciences, 2Cancer Research Initiative
Primary Subject Heading:	Public health
Secondary Subject Heading:	Oncology, Health services research, Global health, Radiology and imaging
Keywords:	ONCOLOGY, Breast tumours < ONCOLOGY, PUBLIC HEALTH, Organisation of health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-income countries: A systematic review

Chukwudi A. Nnaji[1,2], Paul Kuodi[3], Fiona M. Walter[4,5], Jennifer Moodley[1,2,6]

Authors affiliations:

[1]Women's Health Research Unit, School of Public Health and Family Medicine, Faculty of Health Sciences, University of Cape Town; Anzio Road. Observatory, 7925, Cape, Town, South Africa

[2]Cancer Research Initiative, Faculty of Health Sciences, University of Cape Town; Anzio Road, Observatory, 7925, Cape, Town, South Africa

[3] Department of Public Health, Faculty of Health Sciences, Lira University, Lira, Uganda

[4] The Primary Care Unit, Department of Public Health and Primary Care, University of Cambridge, United Kingdom

[5] Wolfson Institute of Population Health, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, UK

[6] SAMRC Gynaecology Cancer Research Centre, University of Cape Town, Cape Town, South Africa

Correspondence to:

Dr Chukwudi A. Nnaji; nnjchu001@myuct.ac.za

For peer review only

Abstract

Objectives: To systematically synthesise available evidence on the nature and effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low- and middle-come countries (LMICs).

Design: A systematic review of published evidence. The review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA).

Data sources: A comprehensive search of published literature was conducted. In addition, relevant grey literature sources and bibliographical references of included studies were searched for potentially eligible evidence.

Study selection: Studies published between January 2010 and November 2020 were eligible for inclusion. To be eligible, studies had to report on interventions/strategies targeted at women, the general public or healthcare workers, aimed at improving the timely diagnosis of breast and/or cervical cancers in LMIC settings.

Data extraction and synthesis: Literature search, screening, study selection, data extraction and quality appraisal were conducted by two independent reviewers. Evidence was synthesised and reported using a global taxonomy framework for early cancer diagnosis.

Results: From the total of 10593 records identified, 21 studies conducted across 20 LMICs were included in this review. Most of the included studies (16/21) focused primarily on interventions addressing breast cancers; two focused on cervical cancer while the rest examined multiple cancer types. Reported interventions targeted health care workers (12); women and adolescent girls (7); and both women and health care workers (3). Eight studies reported on interventions addressing access delays; seven focused on interventions addressing diagnostic delays; two reported on interventions targeted at addressing both access and diagnostic delays, and four studies assessed interventions addressing access, diagnostic and treatment delays. While most interventions were demonstrated to be feasible and effective, many of the reported outcome measures are of limited clinical relevance to diagnostic timeliness.

Conclusions: Though limited, evidence suggests that interventions aimed at addressing barriers to timely diagnosis of breast and cervical cancer are feasible in resource-limited contexts. Future interventions need to address clinically relevant measures to better assess efficacy of interventions.

Keywords: Cancer, breast, cervical, diagnosis, timeliness, delay, LMICs.

PROSPERO registration number: CRD42020177232.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- This review was designed in accordance with standard systematic review protocol guidelines.
- Literature searching was comprehensive, covering both peer-reviewed and relevant grey literature.
- No language restrictions were applied in the search.
- While most interventions were demonstrated to be feasible and effective, reported outcome measures are of limited clinical relevance to diagnostic timeliness.
- The overall strength of the synthesised evidence and applicability of findings depend on the quality of included studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Breast and cervical cancers both represent a significant burden of disease globally, with a disproportionately high burden of morbidity and mortality in low- and middle-income countries (LMICs). [1-3] The majority (53%) of new breast cancer cases occur in women living in LMICs, accounting for nearly a third of all cancers in women, and with an age-standardised incidence rate (ASIR) of 31 per 100,000 women in these settings.[4,5] Cervical cancer constitutes 16% of women's cancers in LMICs, with an ASIR of 16 per 100,000 women.[2] The increasing burdens of both cancers has been attributed to factors such as population growth, ageing societies and exposures to oncogenic risk factors.[1,6] While the incidence of breast and cervical cancers increases in LMICs, many cases continue to go unreported and undiagnosed; the majority of these women present with late-stage disease, the treatment of which is often costlier and less effective.[7,8] These challenges have been aggravated by the COVID-19 pandemic, which immensely disrupted cancer services globally, particularly in low-resource settings.[9]

The timeliness of diagnosis is an important determining factor of breast and cervical cancer survival.[10-13] Yet, in many LMICs, breast and cervical screening and early diagnosis programmes are often opportunistic and not well organised.[7,14-16] One of the greatest barriers to timely cancer diagnosis in LMICs is lack of geographic access to health facilities offering screening and diagnostic services, as well as the lack of well-trained care providers.[6,17] In addition to these health system factors are the underlying psychosocial and cultural barriers to timely cancer help-seeking and diagnosis.[6-8,18] These include perceptions such as beliefs that breast cancers are contagious or punitive consequences of sins.[15,19]

To conceptualise and understand the pathway to cancer diagnosis, distinct events and processes from symptom awareness to diagnosis and treatment have been described. These include patients' detection of bodily changes and awareness of symptoms; decision to seek help from a healthcare provider; access to clinical evaluation, diagnosis, and staging; access to treatment; and follow-up.[17,20-22] Barriers encountered by patients in any of these phases may pose an impediment to timely cancer diagnosis and delay patients' access to care.[10,23] In an effort to address these barriers and provide global standards for early cancer diagnosis, the World Health Organization (WHO) published the *WHO Guide to Cancer Early Diagnosis* in 2017.[24] It provides an operational framework for cancer control programmes around the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

world to systematically address barriers that may impede timely cancer diagnosis, treatment and care. Furthermore, it specifies three essential steps of early cancer diagnosis: Step 1: awareness of cancer symptoms and accessing care; Step 2: clinical evaluation, diagnosis and staging; and Step 3: access to treatment, including pain relief. These steps were emphasised to be essential for setting cancer control priorities in the 2020 WHO Report on Cancer.[25]

Interventions aimed at effectively addressing barriers to timely cancer diagnosis have been studied in various settings within the different phases of the cancer care continuum.[6,11] It is evident that such interventions can improve the timeliness of cancer diagnosis, enable early initiation of treatment, reduce morbidity and increase the chances of survival.[20,26-30] However, much of this evidence is from high-income countries (HICs). Given the substantial differences between HICs and LMICs regarding health resources, environment, infrastructure, technology, and medical personnel, improving time to diagnosis for breast and cervical cancer in LMIC settings may require different approaches.[3]

While interventions to improve the timely diagnosis of these cancers are increasingly being implemented in LMICs, there is uncertainty about their role and effectiveness. We have identified two previous reviews on this topic within the LMIC context.[6,11] A scoping review by Dalton and colleagues synthesised the evidence on patient navigation strategies for cancer care in LMICs, but focused broadly on the entire cancer detection, treatment and care continuum.[6] A previous systematic review by Qu and colleagues assessed interventions specifically aimed at addressing barriers to timely cancer diagnosis in LMICs.[11] Neither specifically focused on breast and/or cervical cancer.

Therefore, the purpose of this review was to systematically synthesise the evidence on the nature and effectiveness of interventions aimed at improving timely diagnosis of symptomatic breast and cervical cancers in LMICs, by targeting the 3 key steps outlined by the WHO framework.[24] Due to the lead time and length bias associated with cancer screening programmes, this review focused on interventions targeted at improving early diagnosis of symptomatic breast and cervical cancer. Moreover, evidence demonstrates that diagnostic delays among symptomatic women are so long in LMICs.[15,19] Hence, interventions to reduce these delays among people with symptoms would likely have substantial impact on outcomes and are also a prerequisite for screening programs. We sought to provide a user-friendly evidence summary for health policy-makers, cancer programme managers, oncologists and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cancer programme implementers, for informing further efforts to assist women in LMICs in overcoming socioeconomic, cultural, and structural barriers to timely diagnosis and appropriate care, while identifying gaps for future research.

METHODS

Study design

This systematic review of evidence was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[31] The review has been registered on the International Prospective Register of Systematic Reviews (PROSPERO, registration number: CRD42020177232).[32] The protocol, which was designed in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) guidelines[33] has been published elsewhere.[34]

Data sources

The following databases were searched: MEDLINE (via PubMed), Cochrane Library (including the Cochrane Central Register of Controlled Trials (CENTRAL) and the Database of Abstracts of Reviews of Effects (DARE)), Scopus, CINAHL, Web of Science and the International Clinical Trials Registry Platform (ICTRP). Relevant grey literature sources; including the publication database of the WHO's International Agency for Research on Cancer (IARC), the Cancer Atlas of the Union for International Cancer Control (UICC) and the Global Cancer Project Map, were also searched for potentially eligible publications. The search strategy (see Appendix 1) was developed with guidance from a health sciences subject librarian. More details of the search strategies used are described in the published review protocol.[34] Only articles published over the last 10 years (from 1 January 2010 to the date of search) were considered eligible, with no language restriction. In addition, the bibliographical references of included studies were also searched. Search was conducted from 22 June 2020 through 30 November 2020.

Eligibility criteria

The study inclusion criteria were guided by the research question: *'What interventions targeting the key steps outlined by the WHO operational framework for early diagnosis have been used for improving timely diagnosis of symptomatic breast and cervical cancers in LMICs, and how effective are they?'*. Eligibility was determined the PICO (participants; interventions;

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

comparator; outcomes) criteria in line with the research question. Participants included women, general public or healthcare workers in LMICs. The definition of LMICs was based on the World Bank's current classification using per capita gross national income.[35] Interventions included strategies aimed at influencing the timeliness of symptomatic breast or cervical cancer diagnosis along any or a combination of the three steps of the WHO framework for early diagnosis, whether as a single focus intervention or as a multi-focus intervention targeting more than one cancer type. Studies with interventions focused primarily on screening of asymptomatic individuals were excluded. Comparator was the standard of care. Outcomes were those related to any or a combination of the three steps of the WHO framework for early diagnosis, such as improved access to early diagnostic services, reduced diagnostic time and tumour downstaging. More detailed information about the eligibility criteria can be found in the published review protocol.[34]

Study selection

The review consisted of two levels of screening: a title and abstract screening to identify potentially eligible publications, and review of full-texts to select those to be included in the review based on pre-defined inclusion/exclusion criteria. For the first level of screening, two reviewers (CAN and PK) independently screened the titles and abstracts of all retrieved records from the search output. Articles that were considered relevant by either or both of the reviewers were included in the full-text review. In the second step, the two reviewers (CAN and PK) independently assess the full-texts to determine if they met the inclusion/exclusion criteria. Any discordance in eligibility assessment was resolved through consensus between the two researchers, with recourse to a third arbitrator (JM) as necessary.

Data extraction

Two reviewers (CAN and PK) independently extracted all relevant data from the included articles using a standardised data extraction tool, adapted from the framework proposed by Carlos and colleagues.[36] Extracted data fell under four domains: (1) study identification details (article title; journal title; authors; country of the study; language; publication year; host institution of the study); (2) methodological characteristics (study design; study objective or research question or hypothesis; sample characteristics (e.g. sample size; sex; age, ethnicity; groups and controls; follow-up duration; validation of measures; statistical analyses); (3) main

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

findings, and (4) conclusions. Study eligibility was re-verified at the start of data extraction. Discrepancies in extraction were resolved through consensus. CAN combined the two spreadsheets of extracted data for analysis, and data were double-checked by PK for completeness and accuracy.

Quality assessment

Two reviewers (CAN and PK) independently assessed each included study for risk of bias, using the Cochrane risk of bias tool[37] for randomised clinical trials, and the Risk of Bias In Non-randomised Studies of Interventions (ROBINS-I) tool[38] for non-randomised and observational studies. Discrepancies in study quality assessments were resolved by discussion and consensus. Further details of the quality appraisal process are discussed in the published protocol of this review.[34]

Data analysis

A PRISMA flow diagram was used to illustrate the literature search results and study selection process. Outcomes were categorised according to the essential steps of early cancer diagnosis as specified in the WHO Guide to Cancer Early Diagnosis.[24] Plans to conduct a meta-analysis were not feasible as studies were not quantitatively comparable in terms of interventions and reported outcomes. Instead, a thematic narrative synthesis approach was used to aggregate and analyse relevant findings. Plans to assess the certainty of reviewed evidence using the Grades of Recommendation Assessment, Development and Evaluation (GRADE) approach[39] were not feasible for similar reasons. A table summarising findings from each included study was presented. JM and FMW reviewed the analysed data for accuracy and consistency with protocol.

Patient and public involvement

This is a review of publicly available literature. Patients and the public were not directly involved in the design, conduct, reporting or dissemination this study.

RESULTS

Search results

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

A total of 10593 records were identified from literature database searches, from which 1244 duplicates were discarded. The remainder, 9349 unique records, had their titles and abstracts screened, from which 9264 clearly ineligible publications were excluded. The full texts of the remaining 85 potentially eligible studies were reviewed against predefined inclusion and exclusion criteria; 21 were included in the review and 64 were excluded for various reasons. **Figure 1** presents the PRISMA flow chart of the study selection process and reasons for exclusion.

Characteristics of included studies

Tables 1a and 1b summarise the characteristics of included studies, published between 2015 and 2020. Studies were conducted in 20 LMICs across regions of Africa (8), Europe (4), Latin America (2), Middle-East Asia (1) and South-East Asia (5). Twenty were single-country studies, in which 13 countries were represented: Bangladesh (2)[8,40], Botswana (2)[41,42], Colombia (1)[43], India (1)[44], Indonesia (1)[45], Iran (1)[46], Malaysia (2)[47,48], Nigeria (1)[49], Pakistan (1)[50], Rwanda (1)[51], South Africa (3)[52-54], Tanzania (1)[55] and Zambia (3)[56-58]. One study had a multi-country focus, with participants recruited from nine countries: Bosnia-Herzegovina, Costa Rica, Egypt, India, North Macedonia, Pakistan, Slovenia, Turkey and Uganda.[59] **Figure 2** illustrates the geographical distribution of the studies.

In terms of study design, six studies were randomised clinical trials (RCT),[8,40,43,45,46,51] six were before-and-after studies,[42,44,48,49,52,54] one was a longitudinal (cohort-type) observational study,[55] four were retrospective cohort-type studies,[41,47,50,53] while the remaining four were cross-sectional studies.[56-59]

The majority (16/21) of included studies primarily focused on interventions addressing breast cancers; two focused on cervical cancer, whereas the rest examined multiple cancer types (including breast and/or cervical cancers). Of the 21 studies, 12 targeted health care workers (including physicians, nurses, pathologists, non-physician health providers and community health workers); seven targeted women and adolescent girls; and three targeted both women and health care workers (HCWs). There was an even representation across urban and rural implementation contexts, with reported interventions implemented in 11 urban, 9 rural and one mixed (urban and rural) communities.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

When categorised by the three steps of the WHO Guide to Cancer Early Diagnosis[24], eight studies reported on interventions aiming to address barriers in Step 1 (promoting awareness and addressing delays in access to care);[8,40,44-47,49,54] another eight focused on interventions addressing Step 2 (addressing diagnostic delay by optimising clinical evaluation, diagnosis and staging);[42,43,50-52,55,59,60] one study reported on an intervention targeted at both Steps 1 and 2;[53] while five studies assessed interventions targeted at all three steps (addressing access, diagnostic and treatment delays).[48,56-58] **Tables 1a and 1b** describe the categorisation of the interventions by the three steps of the WHO Guide to Cancer Early Diagnosis.[24]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1a: Characteristics of included studies and categorisation of interventions according to the essential steps of the WHO Guide to Cancer Early Diagnosis (single-step interventions)

Step	Study ID	Study Design	Country	Target population	Cancer type	Setting	Intervention	Primary outcome	Duration
Step 1	Gadgil 2017	Before-and-after study	India	Women in an occupational setting.	Breast	Urban	A programme that aimed to increase awareness on breast cancer and access to early detection by emailing awareness brochures annually to a cohort of women in an occupational setting.	Proportion of women with early tumours	3 years
	Ginsburg 2014	RCT	Bangladesh	CHWs	Breast	Rural	An mHealth model to increase clinic attendance for breast symptoms in rural Bangladesh: involving smart phone/applications and CHW training as part of a patient navigation programme to address potential barriers to seeking care.	Number of women with breast symptoms identified	4 months
	Ifediora 2019	Before-and-after study	Nigeria	High school girls	Cervical	Rural	School-based cervical cancer prevention and early diagnosis education programme - targeting cervical cancer campaigns on teenage high schoolers in resource-limited economies.	Improvement in cervical cancer awareness and help-seeking attitude	6 months
	Chowdhury 2015	RCT	Bangladesh	CHWs	Breast	Rural	Population-based case finding for breast cancer by CHWs in Rural Bangladesh using different data collection methods and approaches.	Accuracy of cancer case-finding data	3 months
	Maimela 2019	Before-and-after study	South Africa	Women	Cervical	Urban	Decentralising colposcopy services from tertiary-level to primary-level care facility to aid early definitive diagnosis of cervical cancer.	Time to colposcopy and number of women who had colposcopy following Pap smear	6 years
	Setyowibowo 2020	Cluster randomized crossover trial	Indonesia	Women	Breast	Both rural and urban	A self-help intervention for reducing time to diagnosis in women with breast cancer symptoms It consisted of health education and psycho-education using a narrative strategy, which involved the use of testimonials and story-telling.	Time between the first medical consultation and the definitive diagnosis	3 months
	Teh 2015	Retrospective study	Malaysia	Women	Breast	Urban	Routine (opportunistic) screening mammogram vs targeted (high risk) screening mammogram vs diagnostic mammogram.	Proportion of women with early tumours	NA
	Termeh Zonouzy 2019		Iran	Women	Breast	Urban	An educational intervention based on fear appeals using the extended parallel process model (EPPM) to improve attitudes, intention, and early breast cancer diagnosis.	Changes in attitude, behavioral intention towards early diagnosis	3 months

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

		RCT							
	Ali 2018	Retrospective study	Pakistan	Physicians and pathologists	Multiple, including breast cancer	Urban	Adequate clinical information (CI) for improving turnaround time (TAT) in surgical histopathological diagnosis.	Pathology results TAT	4 months
	Aribal 2019	Observational clinical study	Bosnia-Herzegovina, Costa Rica, Egypt, India, North Macedonia, Pakistan, Slovenia, Turkey, Uganda	Oncologists, radiologists, medical physicists and radiographers working on oncology centres	Breast	Urban	A programme aiming to improve early detection of breast cancer by strengthening both the clinical component and quality aspects of imaging practice of oncology centres.	Changes in mammography practice and image guided interventions	2 years
	Dickerson 2017	Before-and-after study	South Africa	Non-physician providers	Breast	Rural	Breast ultrasound training for non-physician providers with the goal of early diagnosis of breast cancer and down-staging in a rural setting.	Diagnostic ultrasound competence scores	3 weeks
	Martei 2018	Retrospective study	Botswana	Pathologists	Breast	Urban	A pathology scale-up programme to optimise breast cancer pathological diagnostic TAT in Botswana.	Breast biopsy TAT	4 years
	Murillo 2016	RCT	Colombia	Physicians	Breast	Urban	Opportunistic clinic-based breast cancer screening programs to increase breast cancer screening and down-staging in Colombian women. Physicians received a 2-day training course on mammography and BIRADS (Breast Imaging Reporting and Data System) grading.	Proportion of women with early tumours	2 years
	Ngoma 2014	Prospective cohort study	Tanzania	CHW/village navigators	Multiple, including breast and cervical	Rural	Conventional dispensary self-referral vs. proactive home visits for down-staging cancer in a rural community in Tanzania.	Proportion of women with early tumours	3 years
Step 2	Tapela 2018	Before-and-after study	Botswana	Primary care providers	Multiple, including breast and cervical	Urban	A district-level capacity building training programme targeting primary care providers, aiming to enhance timely diagnosis of cancers.	Participants' early diagnosis competence scores	5 days

Abbreviations: CHW: Community Health Workers; HCW: Healthcare Workers; NA: Not applicable; NR: Not reported; RCT: Randomised Clinical Trial; TAT: Turn-around Time

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1b: Characteristics of included studies and categorisation of interventions according to the essential steps of the WHO Guide to Cancer Early Diagnosis (Multi-step interventions)

Step	Study ID	Study Design	Country	Target population	Cancer type	Setting	Intervention	Primary outcome	Duration
Step 1+2	Hadley 2018	Retrospective/post-implementation study	South Africa	Nurses, nursing assistants, and counsellor.	Breast	Rural	An improved and computerised breast cancer patient-tracking system and data base as part of an ultrasound-based breast cancer early detection program in a rural community in South Africa.	Improvement in quality of care, follow-up and clinical decision-making	NR
	Pace 2019	RCT	Rwanda	Health workers	Breast	Rural	It consisted of training of CHWs in how to educate community members about breast cancer symptoms recognition and early help-seeking; training of nurses in symptom recognition, CBE and appropriate referral; training of hospital-based clinicians in diagnostic breast ultrasound and ultrasound-guided core needle biopsy.	Number of breast biopsies and proportion of women with early tumours	2 years
Step 1+2+3	Songiso 2020	Cross-sectional study	Zambia	Women and HCWs	Breast	Urban	Establishment of a district-level breast care specialty clinic for women with breast symptoms. The clinic offers same-day breast self-awareness education, clinical breast examination, breast ultrasound, ultrasound-guided breast biopsy, surgery, referral for treatment and follow-up.	Number of single-visit breats biopsy and cytology performed	NA
	Pinder 2018	Cross-sectional study	Zambia	Women and HCWs in a rural setting	Breast	Rural	An algorithm that compresses the multi-step breast cancer symptom awareness, diagnostic and treatment pathway into a single visit.	Number of single-visit clinical breast examination, ultrasound, biopsy and cytology performed	NA

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Pinder 2017	Cross-sectional study	Zambia	Women and HCWs in a rural setting	Breast	Rural	An initiative that leverages an existing cervical cancer prevention service platform to build facility-level capacity for breast cancer care by raising awareness among women, creating a resource-appropriate breast cancer care training curricula for mid- and high-level providers, and facility-level capacity building for early detection and treatment capacity.	Number of breast biopsies performed	NA
	Yeoh 2018	Before-and-after study	Malaysia	Women	Breast	urban	A patient navigation programme to Improve timely breast cancer diagnosis and care.	Breast biopsy TAT and proportion of women with early tumours	1 year and one month

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Only two of the included studies were determined to be of high quality (low risk of bias)[43,49], seven were adjudged to be of moderate quality (moderate risk of bias)[8,40,45,46,51,52,57] while the majority were found to be of low quality (high risk of bias).[41,42,44,47,48,50,53-56,58,59] **Figures 3a and 3b** illustrate the risk of bias assessment outcomes of the included studies.

Interventions targeting Step 1 of the WHO Guide to Cancer Early Diagnosis- promoting awareness and addressing care access delays

Interventions seeking to reduce access delays by promoting awareness, help-seeking and linkage to care were predominantly targeted at women (5),[44-47,54] with a few targeted at adolescent girls (1)[49] and community/lay health workers (2).[8,40]

A cluster randomised trial assessed a self-help intervention for reducing time to diagnosis in Indonesian women with breast cancer symptoms. The intervention consisted of health education and psycho-education using a narrative strategy, which involved the use of testimonials and story-telling.[45] It reduced the time between the first medical consultation and definitive diagnosis among women with breast symptoms by 13.3 days (25.90 in the intervention group vs 39.29 in the control group, P = 0.02). Another RCT by Termeh-Zonouzy evaluated the effectiveness of an educational intervention based on fear appeals using the extended parallel process model (EPPM) to improve attitudes, intention, and timely breast cancer diagnosis in Iran. It found a significant improvement in attitude and intention (*P*=0.01 and *P*=0.001, respectively), but no significant improvement was observed for early breast cancer diagnosis (*P*=0.78).[46]

An RCT by Gadgil and colleagues investigated a programme that aimed to increase awareness on breast cancer and access to timely detection by emailing awareness brochures annually to a cohort of women in an urban occupational setting in India.[44] They found an increase in the proportion of women with early breast tumours and a concomitant increase in the proportion of women receiving breast conserving surgery increased post-intervention when compared with the pre-intervention period increased from (74% to 81% and 39% to 51% respectively). However, there were no outcomes directly related to the timeliness of the diagnoses.

A before-and-after study assessed the effectiveness of decentralising colposcopy services from tertiary-level to primary-level care facility in an urban community in South Africa to aid early definitive diagnosis of cervical cancer.[54] The number of women who had a colposcopy

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

at the primary healthcare facility rose threefold post-decentralisation (from 114 to 350), with an increased proportion of women who had a colposcopy within 3 months of a Pap smear (from 11.8% to 15.4%) and reduced workload at the tertiary facility. A retrospective study compared routine (opportunistic) screening mammogram with targeted (high risk) screening mammogram and diagnostic mammogram among women in Malaysia.[47] It observed cancer detection rates of 0.5 %, 1.25 % and 26 % by opportunistic screening, targeted screening, and diagnostic mammograms, respectively. The proportion of non-invasive cancer (ductal carcinoma in situ) was higher (23.1 %) in the two screening groups compared with only 2.5 % in the diagnostic group, while the proportion of invasive cancer was higher in the diagnostic group. None of these outcomes were directly related to the timeliness of diagnosis.

Of the interventions targeting lay/community health workers, an RCT by Ginsburg et al. assessed the role an mHealth model that used smart phone applications and CHW training as part of a patient navigation programme to address potential barriers to seeking care in a rural community in Bangladesh.[8] The intervention was found to have increased clinic attendance for breast symptoms, improved data quality, higher identification of women with abnormal breast exams and better adherence to diagnostic follow-up. Similarly, another RCT of a CHW-driven population-based case finding programme for breast cancer in a rural setting in Bangladesh found the use of mobile technology as feasible and efficient in active breast cancer case finding in rural settings. Manual data collection and reporting had missing data in 80% of cases, and took an average of 5 minutes longer to collect data, versus no missing data when mobile phones were used for data collection and reporting.[40]

A school-based cervical cancer prevention and early diagnosis education programme targeted at rural teenage high school students Nigeria.[49] Notably, no significant improvements in knowledge parameters (Pap smears, risk factors and early symptoms). Nevertheless, participants who engaged more with the intervention showed significant improvements in knowledge across most parameters of knowledge.

Interventions targeting Step 2 of the WHO Guide to Cancer Early Diagnosis-addressing diagnostic delays by optimising clinical evaluation, diagnosis and staging

An RCT compared diagnostic outcomes between a physician-targeted opportunistic clinic-based breast cancer screening programme and usual care in an urban setting in Colombia. Physicians in the intervention clinics received a 2-day training course on breast-

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cancer epidemiology, clinical signs and symptoms, as well as the principles of mammography and BIRADS (Breast Imaging Reporting and Data System) grading. They were instructed to perform CBE on all eligible women, record results, and refer women with suspicious findings for further diagnostic procedures. It demonstrated a higher proportion of early breast cancer diagnosed with the intervention; 68.2% of the women with breast cancer in the intervention group had breast conservation surgery , compared with 50.0% in the control group.[43]

A before-and-after study of a breast ultrasound training programme for non-physician providers with the goal of early diagnosis of breast cancer and down-staging in a rural setting in South Africa demonstrated improved diagnostic competencies (with pre- to post-test averages improvement of 68% in total across four competencies (foundational knowledge, descriptive categories, benign vs malignant, and lesion identification).[52] No outcome directly related to diagnostic timeliness was reported, however. Another before-and-after study studied district-level capacity building training programme targeting primary care providers in Botswana, aiming to enhance timely diagnosis of cancers. It found an overall performance increase (measured by the immediate impact of training on the acquisition of knowledge, attitude and skills in eight sub-domains: pathophysiology, epidemiology, social context, symptoms, evaluation, treatment, documentation and follow-up) of 16.8% after participation across all subdomains.[42]

A prospective cohort study by Ngoma et al. assessed the role of village navigators for case-finding and showed significant down-staging of cancers in a Tanzanian village across 3 successive years ($p < 0.001$).[55] An observational clinical study examined the effectiveness of a programme aiming to improve early detection of breast cancer by strengthening both the clinical and quality aspects of the breast imaging practice of oncology centres.[59] The interventions included a shift to image guided core biopsies, implementing multidisciplinary breast meetings, implementing quality assurance procedures, transitioning to digital technologies, facilitating training activities for relevant professionals and improving data collection processes. It was observed that the programme enhanced facility-level breast cancer early diagnostic capacity and increased number of mammography (in 7 out of 10 participating centres) and timely biopsies performed (in 8 out of 10 centres).

A retrospective study assessed the relationship between adequate clinical information (CI) and breast cancer histopathological diagnostic turnaround time (TAT) in Pakistan.[50] It found

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

that breast tissue specimens with deficient clinical information were associated with longer TAT (>80% of the specimens took had TAT of longer than three days vs <50% of specimens with sufficient clinical information). Another retrospective study that assessed a pathology scale-up programme to optimise breast cancer pathological diagnostic TAT in Botswana found that the programme decreased median TAT for biopsy and immunohistochemistry specimens (from 21.5 days to 8 days, before and after the initiation of the programme respectively). However, there was no significant decline in median TAT for surgical specimens.[41]

Intervention targeted at both Steps 1 and 2 of the WHO Guide to Cancer Early Diagnosis- addressing access and diagnostic delays

A cluster-randomised RCT by Pace et al investigated the impact of a HCW training programme on diagnosis and staging to facilitate breast cancer early diagnosis in a rural district of Rwanda.[51] It consisted of training of CHWs in how to educate community members about breast cancer symptoms recognition and early help-seeking; training of nurses in symptom recognition, CBE and appropriate referral; training of hospital-based clinicians in diagnostic breast ultrasound and ultrasound-guided core needle biopsy. The intervention led to an increase in diagnostic health facility visits (an increase of 4.7 – 7.9 visits/month in intervention facilities compared with control facilities, P <0.05) and more breast biopsies (36.6 vs 8.9/100,000 person-years, P <0.001) and higher incidence of early-stage breast cancer (3.3 per 100,000 vs 0.7 per 100,000 in control areas, P <.05).

Hadley and colleagues conducted a retrospective (post-implementation) study to determine the impact of an improved and computerised breast cancer patient-tracking system and database, as part of a nurse-driven and ultrasound-based breast cancer early detection programme in a rural community in South Africa.[53] Most health workers' involved agreed that the intervention can improve early breast cancer detection, patient follow-up, clinical decision making, communication among clinicians. However, a majority (71%) of them reported that the system increased visit time, due to factors such as unreliable internet and inadequate computer workstations.

Interventions targeted at all three steps of the WHO Guide to Cancer Early Diagnosis - addressing access, diagnostic and treatment delays

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

There were interventions targeting all three steps of the WHO Guide. A patient navigation programme to improve timely breast cancer diagnosis, treatment and follow-up for women in Malaysia was found to have increased the proportion of women receiving timely mammography (from 74.4% to 96.4%), biopsy (from 76.1% to 92.5%), and timely communication of diagnostic results (from 58.5% to 80.0%).[48] Other interventions included the establishment of a district-level breast care specialty clinic that offers same-day breast self-awareness education, clinical breast examination, breast ultrasound, ultrasound-guided breast biopsy, surgery, referral for treatment and follow-up for women with breast symptoms in Zambia[58]; an algorithm that compresses the multi-step breast cancer symptom awareness, diagnostic and treatment pathway into a single clinic visit[56]; and an initiative that leverages an existing cervical cancer prevention service platform to build facility-level capacity for breast cancer care by raising awareness among women, creating a resource-appropriate training curricula for mid- and high-level providers for early detection and treatment.[57] Due to the cross-sectional nature of the studies reporting these interventions, they could only demonstrate the feasibility of the interventions within resource-limited contexts, and not their effectiveness.

DISCUSSION

This systematic review highlights the diverse nature and effectiveness of interventions for the early diagnosis of symptomatic breast and cervical cancers, from 21 studies set across 20 LMICs. In spite of the limited number and methodological limitations of the studies included, this review has identified a substantial body of evidence suggesting that interventions and strategies targeting the three essential steps of the WHO operational framework for early cancer diagnosis may be feasible and effective in the context of breast and cervical cancer in LMICs. It is noteworthy that, while most of interventions were reported to be feasible and effective for improving diagnostic processes, many of their reported outcome measures are not directly related to diagnostic timeliness. Although a variety of interventions have been studied, the majority of them were targeted at improving the diagnostic capacity of health care workers (including physicians, nurses, pathologists, non-physician health providers and community health workers) through training, knowledge-acquisition, upskilling and better equipment.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Training interventions aimed at improving diagnostic knowledge and skills of health care workers were shown to improve diagnostic turnaround time[41] and increase detection rates[43,51] and downstaging.[52] The use of community-based patient navigation programmes driven by trained community- or lay-health workers for breast examination, specimen collection, pathological diagnosis and follow-up was also shown to improve early diagnostic outcomes.[8,40] Findings are consistent with those of previous reviews.[6,11]

Similarly, evidence suggests that the use of health education and awareness promotion campaigns targeted at women may be effective in improving breast cancer knowledge, attitude and help-seeking behaviour. These could include culturally-sensitive and context-appropriate strategies such a self-help intervention for reducing time to diagnosis in women with breast cancer symptoms, consisting of health education and psycho-education through narratives, testimonials and story-telling.[45] Nevertheless, a school-based cervical cancer prevention and early diagnosis education programme targeting adolescent high schoolers yielded no significant improvements in the participants' overall knowledge.[49] This may reflect the vital influence of intervention-related and contextual factors on the effectiveness of education and awareness promotion interventions that need to be taken into account when designing such interventions.

Interventions addressing structural barriers to diagnostic services and appropriate healthcare, such as proactive home-visits, patient-navigation programmes and decentralisation of diagnostic services from tertiary-level to primary-level care facilities, have shown promising results in reducing diagnostic turn-around time and cancer down-staging.[41,43,54,55] While it is difficult to draw conclusions from these studies owing to their methodological limitations and potential publication bias, these findings reflect the important role of interventions targeting health system structural impediments to early cancer diagnosis in LMICs.

Evidence suggests that the use of technological interventions, such as computerised breast cancer patient-tracking systems for improving early breast cancer detection, patient follow-up and clinical decision making, may be feasible even in rural settings.[53] This demonstrates the potential for the adoption and scale up of innovative early cancer diagnosis interventions in low-resource settings. However, the effectiveness of such interventions depends largely on an adequate supply of the needed infrastructure and operational readiness, without which

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

interventions may be ineffectiveness or counter-productive. An indication of this is the report of unreliable internet service and inadequate computer workstations as factors constraining the use of a computerised breast cancer patient-tracking system and leading to increased patient visit or consultation time.[53]

Our review shows a dearth of studies evaluating the effectiveness of cervical cancer early diagnosis interventions for symptomatic women, despite the substantial burden of cervical cancer and late-stage diagnosis in LMICs.[1-5] This trend may be as a result of the lingering stigmatisation of cervical cancer in many LMIC settings or may reflect the fact that women with cervical cancer are likely to be at an advanced stage at the time of diagnosis in such settings.[61] Increased investments are needed to address current cervical cancer programme evaluation and research gaps, such as through improved research funding and research capacity building; and the integration of research activities within cancer programme implementation plans. This also underscores the need for greater public health education and community awareness efforts, to mitigate existing stigma and misconceptions associated with cervical cancer that may pose barriers to cervical cancer diagnostic service delivery and research.

The use of varied outcome measures in assessing the effectiveness of interventions across studies complicated the interpretation, aggregation and comparability of the findings. Many outcome measures reported have very limited clinical relevance to the timeliness of breast or cervical cancer diagnosis. While the most important outcome of early cancer diagnosis programmes would be to reduce cancer mortality, evaluating this outcome does require more rigourous study designs such as randomised clinical trials with long follow-up periods, the cost of which may be prohibitive in low-resource settings. Hence, it is imperative for future studies in LMICs to explore the assessment of more clinically relevant proxy outcomes such as reduced time to diagnosis, improved pathology TAT and down-staging. The Anderson model, Model of Pathways to Treatment and Aarhus Statement are some of the useful tools for assessing diagnostic time-related outcomes of early cancer diagnosis interventions in standardised ways.[21,22]

Overall, given the rising burden of breast and cervical cancer in LMICs, the role of strategies and interventions for improving early diagnosis is relatively underexplored. While more interventions are needed, those that target both the general public and health care workers

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

and the whole continuum of care may be more cost-effective, particularly for resource-limited settings. To enhance the feasibility and usefulness of intervention studies for generating contextually-relevant evidence for improving early diagnosis, it is imperative that outcome evaluation activities are embedded as an integral component of early cancer diagnosis programmes. This can be achieved though mixed-methods implementation research strategies to evaluate and support the application of research findings into policy and practice.[62] It is noteworthy that the evidence included in this review relates to the period before the COVID-19 pandemic which has disrupted essential health services, including early cancer diagnosis programmes across the world.[9] For this reason, it is necessary for future research to investigate the short- and long-term impacts of the pandemic on the effectiveness of early diagnosis interventions, as movement restrictions are eased and cancer programmes return to pre-pandemic levels of operation.

Limitations

Our review has some limitations worthy of note. First, most of the included studies assessed intervention outcomes using observational, non-experimental and non-standardised methods, making them prone to recall bias, selection bias, uncontrolled confounding, and limited external validity. Secondly, as with other reviews, our study may be susceptible to publication bias in the reporting of study findings, in which positive findings have a higher likelihood of being reported. This may have been refleced in our finding of very limited evidence of non-effectiveness of interventions. Thirdly, only a few of the studies measured intervention outcomes beyond a duration of one year, hence limiting the determination of the long-term impact of reviewed interventions. Furthermore, although we searched a wide array of literature sources, it is possible that we missed some relevant studies. Lastly, our study excluded interventions that focused primarily on screening of asymptomatic individuals for reasons earlier alluded to. Hence, it is possible that some the screening studies excluded had early diagnosis components.

CONCLUSION

Interventions aimed at improving timely diagnosis and cancer outcomes are feasible and can be effective for improving timely diagnosis of breast and cervical cancers in resource-limited LMIC contexts. To better generate translatable evidence for improving early cancer diagnosis,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

our review underscores the pressing need for future studies to assess interventions using standardised methods, while reporting clinically relevant diagnostic timeliness outcomes. It is also imperative that early diagnosis interventions to adopt contextually appropriate and cost-effective approaches that are well embedded within the broader health system.

Authors' contributions: Conceptualisation: JM. Protocol development: CAN, PK, FMW and JM. Duplicate screening, study selection, quality assessment and data collection: CAN and PK. Data analysis and interpretation: CAN, PK, FMW and JM. First draft preparation: CAN. Provision of critical insights and refinement of the manuscript: FMW and JM. All authors have read and approved the final version of the manuscript.

Funding: This research was jointly supported by the Cancer Association of South Africa (CANSA), the University of Cape Town and, the SA Medical Research Council with funds received from the SA National Department of Health, GlaxoSmithKline Africa Non-Communicable Disease Open Lab (via a supporting grant Project Number: 023), the UK Medical Research Council (via the Newton Fund). Authors retained control of the final content of the publication. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Acknowledgments: The authors wish to thank Namhla Madini of the Faculty of Health Sciences Library, University of Cape Town, for the support and guidance in developing the literature search strategy. FMW acknowledges the support of the CanTest Collaborative (funded by Cancer Research UK C8640/A23385) as its Director.

Patient consent for publication: Not required.

Ethics approval: This is a systematic review of publicly available literature, with no primary data collection. Hence, it did not require ethical approval.

Competing interests: None declared.

Data sharing statement: All data relevant to the study are included in the article or uploaded as supplementary information

Provenance and peer review: Not commissioned; externally peer reviewed.

For peer review only

References

1. Global Burden of Disease Cancer Collaborators. Global, Regional, and National Cancer Incidence, Mortality, Years of Life Lost, Years Lived With Disability, and Disability-Adjusted Life-Years for 29 Cancer Groups, 1990 to 2017: A Systematic Analysis for the Global Burden of Disease Study. JAMA oncology 2019; 5:1749-68.
2. de Sanjose S, Tsu VD: Prevention of cervical and breast cancer mortality in low- and middle-income countries: a window of opportunity. International journal of women's health 2019; 11:381-6.
3. Demment MM, Peters K, Dykens JA, et al.: Developing the Evidence Base to Inform Best Practice: A Scoping Study of Breast and Cervical Cancer Reviews in Low- and Middle-Income Countries. PloS one 2015; 10:e0134618.
4. Bray F, Ferlay J, Soerjomataram I, et al.: Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: a cancer journal for clinicians 2018; 68:394-424.
5. GLOBOCAN. Cancer Today 2018. http://gco.iarc.fr/today/home. Accessed 26 March 2021.
6. Dalton M, Holzman E, Erwin E, et al.: Patient navigation services for cancer care in low-and middle-income countries: A scoping review. PloS one 2019; 14:e0223537-e.
7. Tegegne TK, Chojenta C, Loxton D, et al.: The impact of geographic access on institutional delivery care use in low and middle-income countries: Systematic review and meta-analysis. PloS one 2018; 13:e0203130.
8. Ginsburg OM, Chowdhury M, Wu W, et al.: An mHealth Model to Increase Clinic Attendance for Breast Symptoms in Rural Bangladesh: Can Bridging the Digital Divide Help Close the Cancer Divide? The Oncologist 2014; 19:177-85.
9. Nnaji CA, Moodley J: Impact of the COVID-19 pandemic on cancer diagnosis, treatment and research in African health systems: a review of current evidence and contextual perspectives. Ecancermedicalscience 2021; 15:1170.
10. Brand NR, Qu LG, Chao A, et al.: Delays and Barriers to Cancer Care in Low- and Middle-Income Countries: A Systematic Review. Oncologist 2019; 24:e1371-e80.
11. Qu LG, Brand NR, Chao A, et al.: Interventions Addressing Barriers to Delayed Cancer Diagnosis in Low- and Middle-Income Countries: A Systematic Review. Oncologist 2020.
12. Richards MA, Westcombe AM, Love SB, et al.: Influence of delay on survival in patients with breast cancer: a systematic review. Lancet 1999; 353:1119-26.
13. Jedy-Agba E, McCormack V, Adebamowo C, et al.: Stage at diagnosis of breast cancer in sub-Saharan Africa: a systematic review and meta-analysis. The Lancet Global health 2016; 4:e923-e35.
14. Islam RM, Billah B, Hossain MN, et al.: Barriers to Cervical Cancer and Breast Cancer Screening Uptake in Low-Income and Middle-Income Countries: A Systematic Review. Asian Pacific journal of cancer prevention : APJCP 2017; 18:1751-63.
15. Cazap E, Magrath I, Kingham TP, et al.: Structural Barriers to Diagnosis and Treatment of Cancer in Low- and Middle-Income Countries: The Urgent Need for Scaling Up. J Clin Oncol 2016; 34:14-9.
16. Moodley J, Walter FM, Scott SE, et al.: Towards timely diagnosis of symptomatic breast and cervical cancer in South Africa. South African medical journal = Suid-Afrikaanse tydskrif vir geneeskunde 2018; 108:803-4.
17. Moodley J, Cairncross L, Naiker T, et al.: Understanding pathways to breast cancer diagnosis among women in the Western Cape Province, South Africa: a qualitative study. BMJ Open 2016; 6:e009905.
18. Poum A, Promthet S, Duffy SW, et al.: Factors associated with delayed diagnosis of breast cancer in northeast Thailand. Journal of epidemiology 2014; 24:102-8.
19. Tetteh DA, Faulkner SL: Sociocultural factors and breast cancer in sub-Saharan Africa: implications for diagnosis and management. Womens Health (Lond) 2016; 12:147-56.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Moodley J, Cairncross L, Naiker T, et al.: From symptom discovery to treatment - women's pathways to breast cancer care: a cross-sectional study. BMC cancer 2018; 18:312.
21. Scott SE, Walter FM, Webster A, et al.: The model of pathways to treatment: conceptualization and integration with existing theory. British journal of health psychology 2013; 18:45-65.
22. Walter F, Webster A, Scott S, et al.: The Andersen Model of Total Patient Delay: A Systematic Review of Its Application in Cancer Diagnosis. J Health Serv Res Policy 2012; 17:110-8.
23. Weller D, Vedsted P, Rubin G, et al.: The Aarhus statement: improving design and reporting of studies on early cancer diagnosis. Br J Cancer 2012; 106:1262-7.
24. World Health Organization (WHO). Guide To Cancer Early Diagnosis. Geneva, Switzerland: World Health Organization, 2017. Available from: http://apps.who.int/iris/bitstream/10665/254500/1/9789241511940-eng.pdf?ua=1. Accessed 10 March, 2021.
25. World Health Organization (WHO). WHO report on cancer: setting priorities, investing wisely and providing care for all. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.
26. Wells KJ, Battaglia TA, Dudley DJ, et al.: Patient navigation: State of the art or is it science? Cancer 2008; 113:1999-2010.
27. Valaitis RK, Carter N, Lam A, et al.: Implementation and maintenance of patient navigation programs linking primary care with community-based health and social services: a scoping literature review. BMC Health Serv Res 2017; 17:116.
28. Battaglia TA, Roloff K, Posner MA, et al.: Improving follow-up to abnormal breast cancer screening in an urban population. A patient navigation intervention. Cancer 2007; 109:359-67.
29. Drake BF, Tannan S, Anwuri VV, et al.: A Community-Based Partnership to Successfully Implement and Maintain a Breast Health Navigation Program. Journal of community health 2015; 40:1216-23.
30. Neal RD, Tharmanathan P, France B, et al.: Is increased time to diagnosis and treatment in symptomatic cancer associated with poorer outcomes? Systematic review. Br J Cancer 2015; 112 Suppl 1:S92-S107.
31. Liberati A, Altman DG, Tetzlaff J, et al.: The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. Annals of internal medicine 2009; 151:W65-94.
32. Booth A, Clarke M, Dooley G, et al.: The nuts and bolts of PROSPERO: an international prospective register of systematic reviews. Syst Rev 2012; 1:2.
33. Shamseer L, Moher D, Clarke M, et al.: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ (Clinical research ed) 2015; 350:g7647.
34. Nnaji CA, Kuodi P, Walter FM, et al.: Effectiveness of interventions for improving timely diagnosis of breast and cervical cancers in low and middle-income countries: a systematic review protocol. BMJ Open 2020; 10:e042788.
35. Fantom N, Serajuddin U. The World Bank's Classification of Countries by Income (English). Policy Research Working Paper no. WPS 7528. Washington, DC: World Bank Group, 2016.
36. Carlos LL-J, Cruz LAPd, Leopoldo VC, et al.: Effectiveness of Traditional Chinese Acupuncture versus Sham Acupuncture: a Systematic Review. Rev Lat Am Enfermagem 2016; 24:e2762-e.
37. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.0 (updated July 2019). Cochrane, 2019. Available from https://training.cochrane.org/handbook/current/chapter-08. Accessed 27 March 2020.
38. Sterne JA, Hernan MA, Reeves BC, et al.: ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ (Clinical research ed) 2016; 355:i4919.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

39. Guyatt GH, Oxman AD, Kunz R, et al.: What is "quality of evidence" and why is it important to clinicians? BMJ (Clinical research ed) 2008; 336:995-8.
40. Chowdhury TI, Love RR, Chowdhury MT, et al.: Feasibility Study of Case-Finding for Breast Cancer by Community Health Workers in Rural Bangladesh. Asian Pacific journal of cancer prevention : APJCP 2015; 16:7853-7.
41. Martei YM, Narasimhamurthy M, Prabhakar P, et al.: Breast Cancer Pathology Turnaround Time in Botswana. Journal of global oncology 2018; 4:1-7.
42. Tapela NM, Peluso MJ, Kohler RE, et al.: A Step Toward Timely Referral and Early Diagnosis of Cancer: Implementation and Impact on Knowledge of a Primary Care-Based Training Program in Botswana. Frontiers in oncology 2018; 8:187.
43. Murillo R, Díaz S, Perry F, et al.: Increased breast cancer screening and downstaging in Colombian women: A randomized trial of opportunistic breast-screening. International journal of cancer 2016; 138:705-13.
44. Gadgil A, Sauvaget C, Roy N, et al.: Cancer early detection program based on awareness and clinical breast examination: Interim results from an urban community in Mumbai, India. Breast (Edinburgh, Scotland) 2017; 31:85-9.
45. Setyowibowo H, Hunfeld JAM, Iskandarsyah A, et al.: A self-help intervention for reducing time to diagnosis in Indonesian women with breast cancer symptoms. Psycho-oncology 2020; 29:696-702.
46. Termeh Zonouzy V, Niknami S, Ghofranipour F, et al.: An educational intervention based on the extended parallel process model to improve attitude, behavioral intention, and early breast cancer diagnosis: a randomized trial. International journal of women's health 2019; 11:1-10.
47. Teh YC, Tan GH, Taib NA, et al.: Opportunistic mammography screening provides effective detection rates in a limited resource healthcare system. BMC cancer 2015; 15:405.
48. Yeoh ZY, Jaganathan M, Rajaram N, et al.: Feasibility of Patient Navigation to Improve Breast Cancer Care in Malaysia. Journal of global oncology 2018; 4:1-13.
49. Ifediora CO, Azuike EC: Targeting cervical cancer campaigns on teenage high schoolers in resource-limited economies: lessons from an intervention study of Nigerian senior secondary school girls. Family practice 2019; 36:600-6.
50. Ali SMH, Kathia UM, Gondal MUM, et al.: Impact of Clinical Information on the Turnaround Time in Surgical Histopathology: A Retrospective Study. Cureus 2018; 10:e2596.
51. Pace LE, Dusengimana JMV, Shulman LN, et al.: Cluster Randomized Trial to Facilitate Breast Cancer Early Diagnosis in a Rural District of Rwanda. Journal of global oncology 2019; 5:1-13.
52. Dickerson LK, Rositch AF, Lucas S, et al.: Pilot Educational Intervention and Feasibility Assessment of Breast Ultrasound in Rural South Africa. Journal of global oncology 2017; 3:502-8.
53. Hadley M, Mullen LA, Dickerson L, et al.: Assessment and Improvement Strategies for a Breast Cancer Early Detection Program in Rural South Africa. Journal of global oncology 2018; 4:1-12.
54. Maimela G, Nene X, Mvundla N, et al.: The impact of decentralising colposcopy services from tertiary-level to primary-level care in inner-city Johannesburg, South Africa: a before and after study. BMJ Open 2019; 9:e024726.
55. Ngoma T, Mandeli J, Holland JF: Downstaging cancer in rural Africa. International journal of cancer 2015; 136:2875-9.
56. Pinder LF, Nzayisenga JB, Shibemba A, et al.: Demonstration of an algorithm to overcome health system-related barriers to timely diagnosis of breast diseases in rural Zambia. PloS one 2018; 13:e0196985.
57. Pinder LF, Henry-Tillman R, Linyama D, et al.: Leverage of an Existing Cervical Cancer Prevention Service Platform to Initiate Breast Cancer Control Services in Zambia: Experiences and Early Outcomes. Journal of global oncology 2018; 4:1-8.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

58. Songiso M, Pinder LF, Munalula J, et al.: Minimizing Delays in the Breast Cancer Pathway by Integrating Breast Specialty Care Services at the Primary Health Care Level in Zambia. JCO global oncology 2020; 6:859-65.
59. Aribal E, Mora P, Chaturvedi AK, et al.: Improvement of early detection of breast cancer through collaborative multi-country efforts: Observational clinical study. European journal of radiology 2019; 115:31-8.
60. Martei YM, Pace LE, Brock JE, et al.: Breast Cancer in Low- and Middle-Income Countries: Why We Need Pathology Capability to Solve This Challenge. Clinics in laboratory medicine 2018; 38:161-73.
61. Pierz AJ, Randall TC, Castle PE, et al.: A scoping review: Facilitators and barriers of cervical cancer screening and early diagnosis of breast cancer in Sub-Saharan African health settings. Gynecologic oncology reports 2020; 33:100605-.
62. Murillo R, Robles C: Research Needs for Implementing Cancer Prevention and Early Detection in Developing Countries: From Scientists' to Implementers' Perspectives. Biomed Res Int 2019; 2019:9607803.

Figure 1: PRISMA flow diagram of study selection process

Figure 2: Map of geographical distribution of included studies

Figures 3a and 3b: Risk of bias assessment outcomes geographical distribution of the studies

Appendix 1: Search strategies

For Peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

FIGURE 1

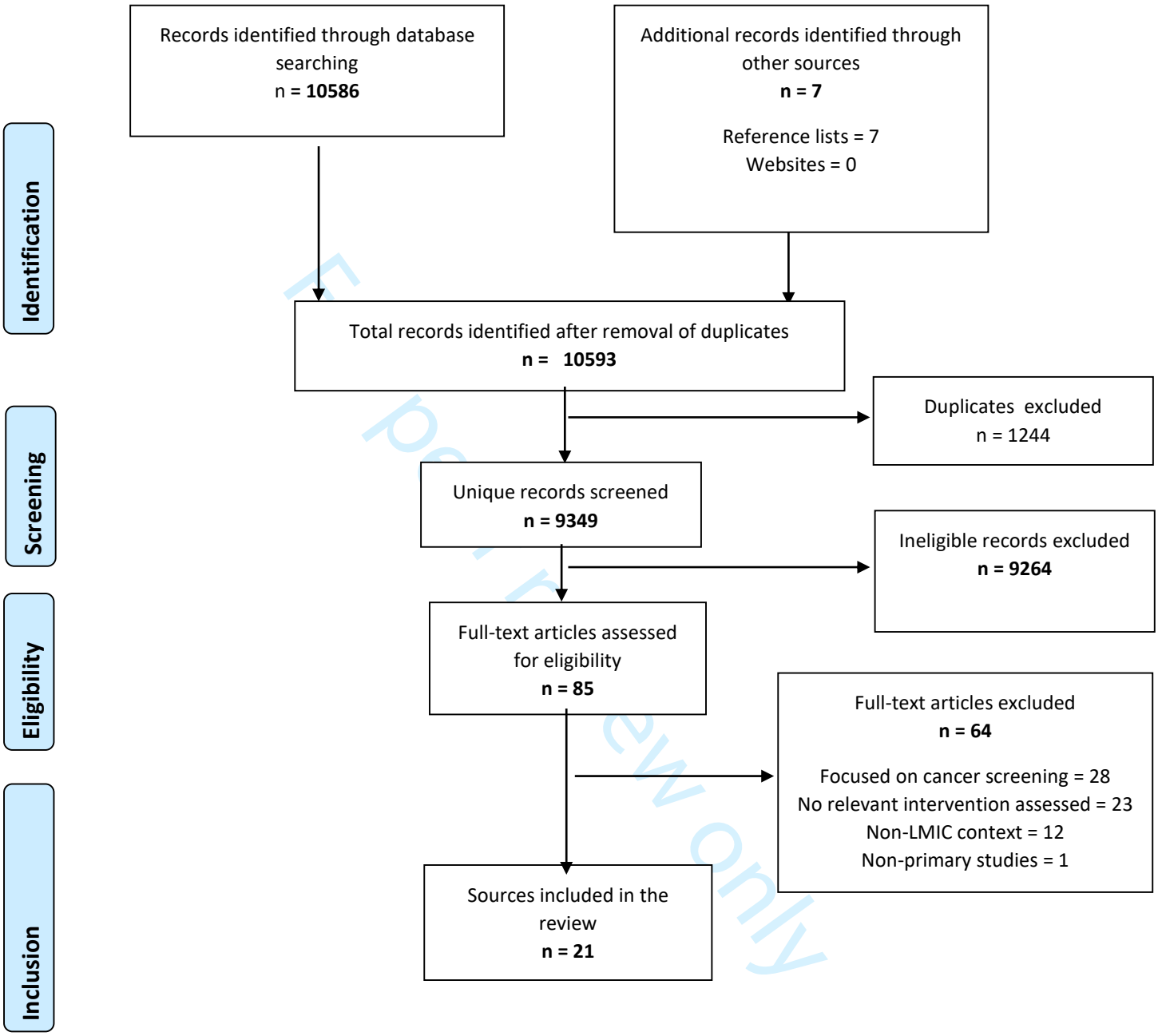


Figure 1: PRISMA flow diagram showing study selection process

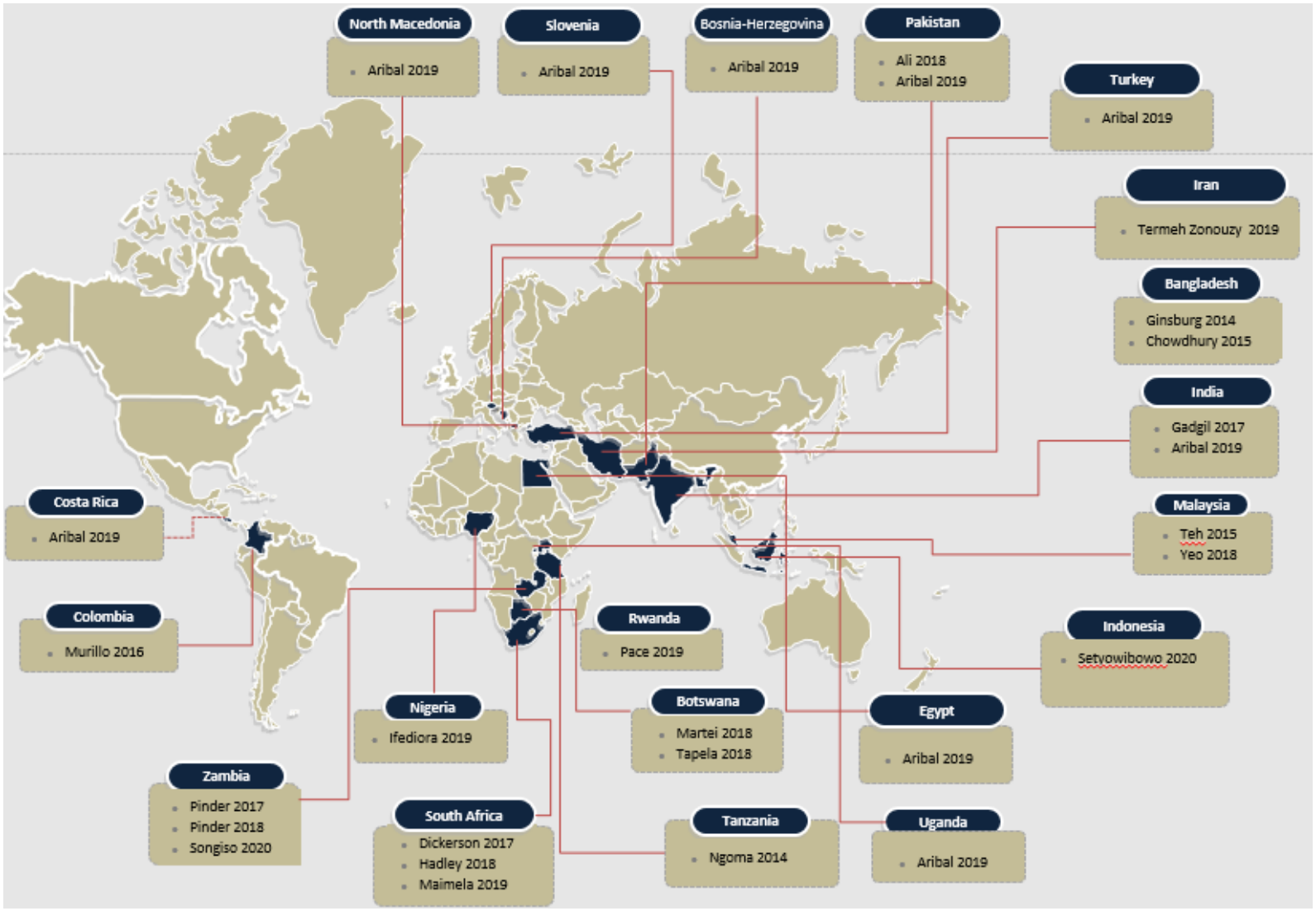


Figure 2: Geographical distribution of included studies (map created by authors using an open-source template sourced from https://yourfreetemplates.com/)

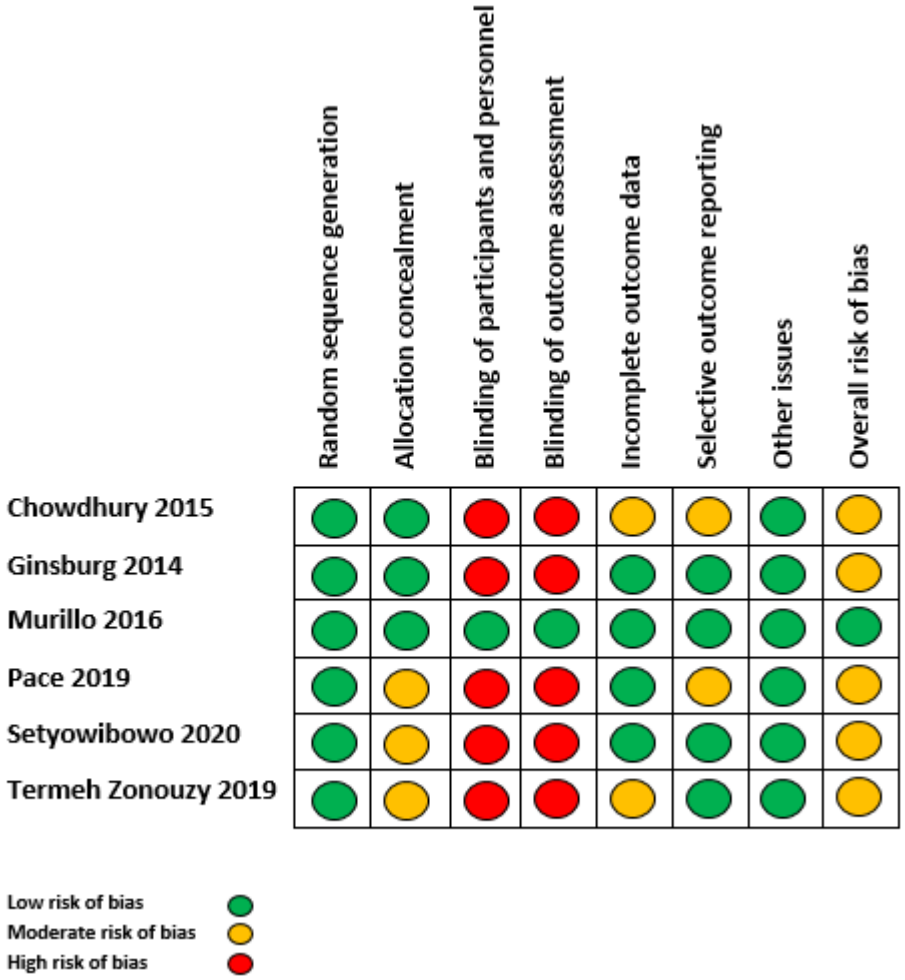


Figure 3a: Risk of bias assessment of randomized controlled trials

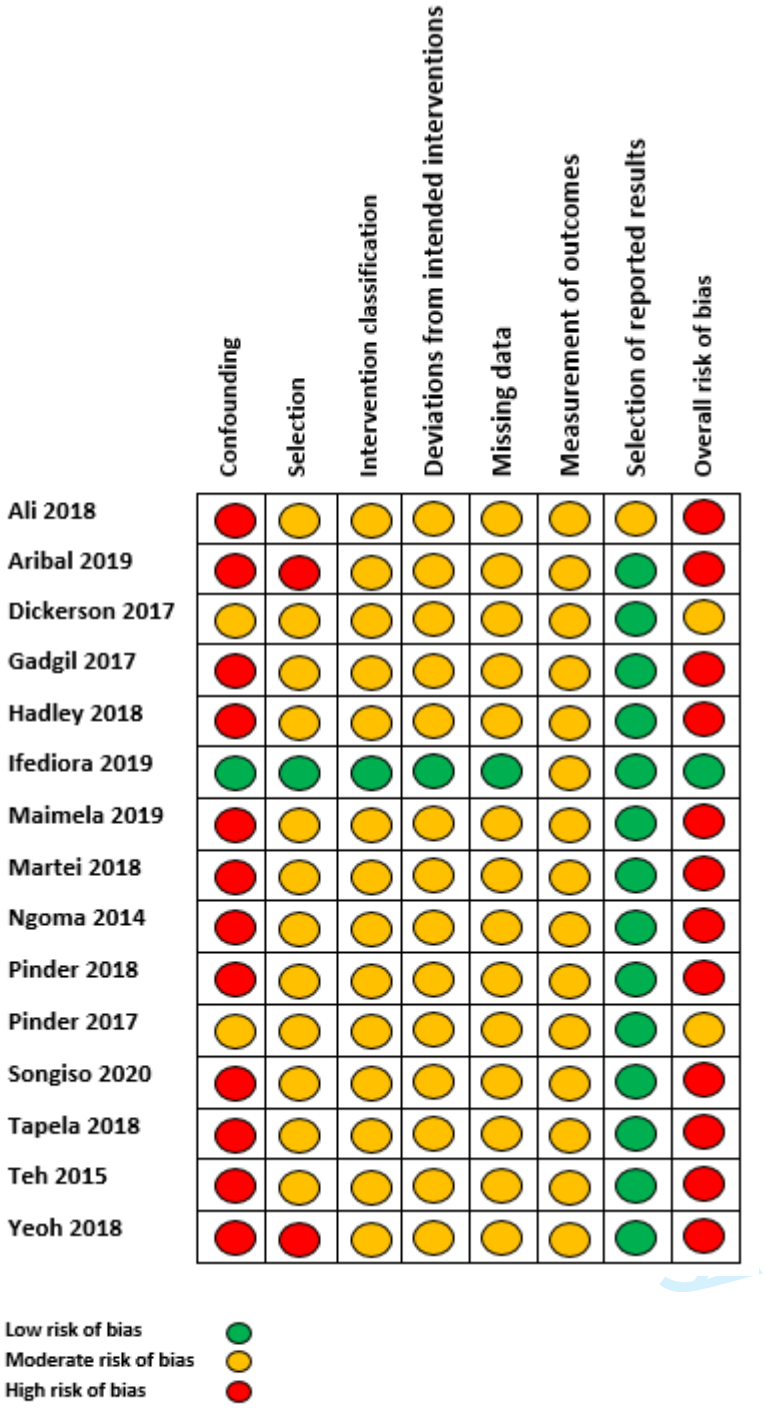


Figure 3b: Risk of bias assessment of observational studies

Appendix: Search strategies

PubMed/MEDLINE search strategy

Search #	Search Texts and Syntaxes
#1	Breast Neoplasms [Mesh] OR Breast cancer[Title/Abstract] OR breast[Title/Abstract] OR cervical cancer[Title/Abstract] OR cervix[Title/Abstract] OR cancer[Title/Abstract] OR malignant[Title/Abstract] OR neoplasm[Title/Abstract] OR neoplasia[Title/Abstract] OR malignancy[Title/Abstract] OR tumour[Title/Abstract]
#2	Diagnosis [Mesh] OR diagnostic[Title/Abstract] OR diagnosis[Title/Abstract] OR diagnosis[Title/Abstract] OR detection[Title/Abstract] OR discovery[Title/Abstract] OR Patient navigation[Title/Abstract] OR Patient pathway[Title/Abstract] OR care continuum[Title/Abstract]
#3	early[Title/Abstract] OR timely[Title/Abstract] OR time[Title/Abstract] OR late[Title/Abstract] OR delay[Title/Abstract]
#4	#2 AND #3
#5	Developing Countries OR Developing Country OR Developing Economies OR Developing Economy OR Developing Nation OR Developing Nations OR Developing Population OR Developing Populations OR Developing World OR LAMI Countries OR LAMI Country OR Less Developed Countries OR Less Developed Country OR Less Developed Economies OR Less Developed Nation OR Less Developed Nations OR Less Developed World OR Lesser Developed Countries OR Lesser Developed Nations OR LMIC OR LMICS OR Low GDP OR Low GNP OR Low Gross Domestic OR Low Gross National OR Low Income OR Lower GDP OR lower gross domestic OR Lower Income OR Middle Income OR Poor Countries OR Poor Country OR Poor Economies OR Poor Economy OR Poor Nation OR Poor Nations OR Poor Population OR Poor Populations OR poor world OR Poorer Countries OR Poorer Economies OR Poorer Economy OR Poorer Nations OR Poorer Population OR Poorer Populations OR Third World OR Transitional Countries OR Transitional Country OR Transitional Economies OR Transitional Economy OR Under Developed Countries OR Under Developed Country OR under developed nations OR Under Developed World OR Under Served Population OR Under Served Populations OR Underdeveloped Countries OR Underdeveloped Country OR underdeveloped economies OR underdeveloped nations OR underdeveloped population OR Underdeveloped World OR Underserved Countries OR Underserved Nations OR Underserved Population OR Underserved Populations OR Afghanistan OR Albania OR Algeria OR American Samoa OR Angola OR Armenia OR Azerbaijan OR Bangladesh OR Belarus OR Byelarus OR Belorussia OR Belize OR Benin OR Bhutan OR Bolivia OR Bosnia OR Botswana OR Brazil OR Bulgaria OR Burma OR Burkina Faso OR Burundi OR Cabo Verde OR Cape Verde OR Cambodia OR

For peer review only

	Cameroon OR Central African Republic OR Chad OR China OR Colombia OR Comoros OR Comores OR Comoro OR Congo OR Costa Rica OR Côte d'Ivoire OR Cuba OR Djibouti OR Dominica OR Dominican Republic OR Ecuador OR Egypt OR El Salvador OR Equatorial Guinea OR Eritrea OR Ethiopia OR Fiji OR Gabon OR Gambia OR Gaza OR Georgia OR Georgia Republic OR Ghana OR Grenada OR Grenadines OR Guatemala OR Guinea OR Guinea- Bissau OR Guyana OR Haiti OR Herzegovina OR Hercegovina OR Honduras OR India OR Indonesia OR Iran OR Iraq OR Ivory Coast OR Jamaica OR Jordan OR Kazakhstan OR Kenya OR Kiribati OR Democratic People's Republic of Korea OR Kosovo OR Kyrgyz OR Kirghizia OR Kirghiz OR Kyrgyzstan OR Lao PDR OR Laos OR Lebanon OR Lesotho OR Liberia OR Libya OR Macedonia OR Madagascar OR Malawi OR Malay OR Malaya OR Malaysia OR Maldives OR Mali OR Marshall Islands OR Mauritania OR Mauritius OR Mexico OR Micronesia OR Moldova OR Mongolia OR Montenegro OR Morocco OR Mozambique OR Myanmar OR Namibia OR Nepal OR Nicaragua OR Niger OR Nigeria OR Pakistan OR Palau OR Papua New Guinea OR Paraguay OR Peru OR Philippines OR Principe OR Romania OR Ruanda OR Rwanda OR Samoa OR Sao Tome OR Senegal OR Serbia OR Sierra Leone OR Solomon Islands OR Somalia OR South Africa OR South Sudan OR Sri Lanka OR St Lucia OR St Vincent OR Sudan OR Surinam OR Suriname OR Swaziland OR Syria OR Syrian Arab Republic OR Tajikistan OR Tadzhikistan OR Tajikistan OR Tadzhik OR Tanzania OR Thailand OR Timor OR Togo OR Tonga OR Tunisia OR Turkey OR Turkmen OR Turkmenistan OR Tuvalu OR Uganda OR Ukraine OR Uzbek OR Uzbekistan OR Vanuatu OR Venezuela OR Vietnam OR West Bank OR Yemen OR Zambia OR Zimbabwe
#6	#1 AND #4 AND #5

For peer review only

Scopus

Search #	Search Texts and Syntaxes
#1	TITLE-ABS-KEY (Breast cancer) OR breast OR (cervical cancer) OR cervix OR cancer OR malignant OR neoplasm OR neoplasia OR malignancy OR tumour
#2	TITLE-ABS-KEY Diagnosis OR diagnostic OR diagnosis OR diagnosis OR detection OR discovery OR (Patient navigation) OR (Patient pathway) OR (care continuum)
#3	Early OR timely OR time OR late OR delay
#4	#2 AND #3
#5	TITLE-ABS-KEY “Deprived Countries” OR “Deprived Population” OR “Deprived Populations” OR “Developing Countries” OR “Developing Country” OR “Developing Economies” OR “Developing Economy” OR “Developing Nation” OR “Developing Nations” OR “Developing Population” OR “Developing Populations” OR “Developing World” OR “LAMI Countries” OR “LAMI Country” OR “Less Developed Countries” OR “Less Developed Country” OR “Less Developed Economies” OR “Less Developed Nation” OR “Less Developed Nations” OR “Less Developed World” OR “Lesser Developed Countries” OR “Lesser Developed Nations” OR LMIC OR LMICS OR “Low GDP” OR “Low GNP” OR “Low Gross Domestic” OR “Low Gross National” OR “Low Income” OR “Lower income” OR “Lower GDP” OR “Lower Gross Domestic” OR “Middle Income” OR “Poor Countries” OR “Poor Country” OR “Poor Economies” OR “Poor Economy” OR “Poor Nation” OR “Poor Nations” OR “Poor Population” OR “Poor Populations” OR “poor world” OR “Poorer Countries” OR “Poorer Economies” OR “Poorer Economy” OR “Poorer Nations” OR “Poorer Population” OR “Poorer Populations” OR “Third World” OR “Transitional Countries” OR “Transitional Country” OR “Transitional Economies” OR “Transitional Economy” OR “Under Developed” OR “Under Served” OR “Underdeveloped Countries” OR “Underdeveloped Country” OR “underdeveloped economies” OR “underdeveloped nations” OR “underdeveloped population” OR “Underdeveloped World” OR “Underserved Countries” OR “Underserved Nations” OR “Underserved Population” OR “Underserved Populations” OR Afghanistan OR Albania OR Algeria OR “American Samoa” OR Angola OR Armenia OR Azerbaijan OR Bangladesh OR Belarus OR Byelarus OR Belorussia OR Belize OR Benin OR Bhutan OR Bolivia OR Bosnia OR Botswana OR Brazil OR Bulgaria OR Burma

For peer review only

	OR “Burkina Faso” OR Burundi OR “Cabo Verde” OR “Cape Verde” OR Cambodia OR Cameroon OR “Central African Republic” OR Chad OR China OR Colombia OR Comoros OR Comores OR Comoro OR Congo OR “Costa Rica” OR “Côte d'Ivoire” OR Cuba OR “Democratic People’s Republic of Korea” OR Djibouti OR Dominica OR “Dominican Republic” OR Ecuador OR Egypt OR “El Salvador” OR Eritrea OR Ethiopia OR “Equatorial Guinea” OR Fiji OR Gabon OR Gambia OR Gaza OR “Georgia Republic” OR Georgia OR Ghana OR Grenada OR Grenadines OR Guatemala OR Guinea OR “Guinea Bissau” OR Guyana OR Haiti OR Herzegovina OR Hercegovina OR Honduras OR India OR Indonesia OR Iran OR Iraq OR “Ivory Coast” OR Jamaica OR Jordan OR Kazakhstan OR Kenya OR Kiribati OR Korea OR Kosovo OR Kyrgyz OR Kirghizia OR Kirghiz OR Kyrgyzstan OR “Lao PDR” OR Laos OR Lebanon OR Lesotho OR Liberia OR Libya OR Macedonia OR Madagascar OR Malawi OR Malay OR Malaya OR Malaysia OR Maldives OR Mali OR “Marshall Islands” OR Mauritania OR Mauritius OR Mexico OR Micronesia OR Moldova OR Mongolia OR Montenegro OR Morocco OR Mozambique OR Myanmar OR Namibia OR Nepal OR Nicaragua OR Niger OR Nigeria OR Pakistan OR Palau OR Papua New Guinea OR Paraguay OR Peru OR Philippines OR Principe OR Romania OR Rwanda OR Ruanda OR Samoa OR “Sao Tome” OR Senegal OR Serbia OR “Sierra Leone” OR “Solomon Islands” OR Somalia OR “South Africa” OR “South Sudan” OR “Sri Lanka” OR “St Lucia” OR “St Vincent” OR Sudan OR Surinam OR Suriname OR Swaziland OR Syria OR “Syrian Arab Republic” OR Tajikistan OR Tadzhikistan OR Tajikistan OR Tadzhik OR Tanzania OR Thailand OR Timor OR Togo OR Tonga OR Tunisia OR Turkey OR Turkmen OR Turkmenistan OR Tuvalu OR Uganda OR Ukraine OR Uzbek OR Uzbekistan OR Vanuatu OR Venezuela OR Vietnam OR “West Bank” OR Yemen OR Zambia OR Zimbabwe
#6	#1 AND #4 AND #5

For peer review only

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7-8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7-8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	8

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	9
Study characteristics	17	Cite each included study and present its characteristics.	9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	10
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	10-18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	10-18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	18
	23b	Discuss any limitations of the evidence included in the review.	21
	23c	Discuss any limitations of the review processes used.	21
	23d	Discuss implications of the results for practice, policy, and future research.	21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	22
Competing interests	26	Declare any competing interests of review authors.	22
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	7

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi:



PRISMA 2020 Checklist

10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only